

Figure 11.13. Mixed and behaviorally mixed strategies.

four; we can call player 1's two strategies U and D and label player 2's as lL , lR , rL , and rR , where the first letter says what 2 does if 1 chooses U and the second says what 2 does if 1 chooses D . We can then translate this game into the corresponding normal form game shown in figure 11.12(b) and thereby define mixed strategies just as previously; player 1's mixed strategies are probability distributions over her pair of possible pure strategies, whereas 2's are probability distributions over his four strategies.

But there is another way to think about mixed strategies for player 2. Suppose that player 2 chooses to randomize only when necessary. If player 1 chooses U , then 2 randomizes between l and r , whereas if 1 chooses D , then 2 randomizes between L and R . That is to say, instead of randomizing over strategies player 2 randomizes over available actions at each information set.

Every probability distribution over 2's four pure strategies corresponds to this sort of randomization over actions at each information set. For example, suppose that 2 considers the mixed strategy: Play lL with probability $1/3$, play rR with probability $1/3$, and play rL with probability $1/3$. (So lR is played with probability zero.) This is the "same" as: If 1 plays U , play l with probability $1/3$ and r with probability $2/3$. And if 1 plays D , play L with probability $2/3$ and R with probability $1/3$. When we say that these two are the same, we mean: For any (mixed) strategy by player 1, whether we think of player 2 randomizing at the outset over his four strategies or we think of player 2 randomizing as needed at information sets, the given two mixed strategies for 2 (of the two different types) lead to the same distribution on outcomes of the game.

How do we know this is true? The "proof" is implicit in figure 11.13(a) and (b); we will talk our way through it, leaving the formal details to you. In figure 11.13(a), we show a joint probability table for player 2's actions at each of his information sets. For example, the lower left box corresponds to the pure strategy rL , and the number $1/3$ there is the $1/3$

probability that he chooses rL in the mixed strategy listed above. The numbers in the right and bottom margins of the table are the marginal probabilities; the $2/3$ along the right margin is the marginal probability that he will choose r . Hence you can see how we derived the "mix-as-you-go" strategy for 2 that corresponds to the listed mixed strategy. Now imagine that we were trying, for some mixed strategy by player 1, to work out the probability distribution over final outcomes. Specifically, suppose that player 1 chooses U with probability α . Then to work out what will happen, it is natural to write down the probability tree in figure 11.13(b), showing first the choice of player 1 and then the choice of player 2. Note well (this is the whole point): The outcome if player 1 chooses U and 2 chooses lL is the same as the outcome if 1 chooses U and 2 chooses lR ; if player 1 chooses U , all that matters is the marginal distribution of 2's choice of l versus r . Similarly, if player 1 chooses D , it is only the marginal distribution over L and R that is of consequence.

Suppose that instead of the mixed strategy in 11.13(a), player 2 adopted the mixed strategy shown in 11.13(c). Would it make a difference? Since the margins on l vs. r and on L vs. R in (a) and (c) are the same, it would not, at least insofar as all we care about is the distribution over outcomes, given player 1's choice of strategy. Many mixed strategies for player 2 will give the same mix-as-you-go strategy.

In general, for a given extensive form game, we could recover the strategy sets for the players and define mixed strategies as mixtures over the pure strategies. But, alternatively, we could consider mix-as-you-go strategies, which are more formally referred to as *behaviorally mixed strategies*. For each information set h , we let $\Delta(A(h))$ be the set of probability distributions over the set of actions $A(h)$ that are available at h . And for each player i , a behaviorally mixed strategy for i is an element of $\prod_{h \in H: i(h)=i} \Delta(A(h))$. The interpretation is that i , upon reaching an information set h with $i(h) = i$, chooses the action to take according to the h th component in this vector of probability distributions.

So we have a choice in how to construct mixed strategies for an extensive form game. Which should we use? At least for games with perfect recall, convenience should be our guide because the choice is without loss of generality. The following result is known as *Kuhn's theorem* (and is first proven in Kuhn [1953]).

Theorem. For an extensive form game with perfect recall, for every mixed strategy, there is a corresponding behaviorally mixed strategy which is "equivalent." And for every behaviorally mixed strategy, there is at least one (and often many)

"equivalent" mixed strategy. "Equivalent" here means "leads to the same distribution of outcomes," if we specify strategies for all the players.

This is not a trivial result. In particular, it doesn't hold for games without perfect recall. If you play contract bridge, you will be able to see this by doing problem 5.

We care about this because in different sorts of games the two different types of mixed strategies are of varying convenience. For simple games and for certain analytical results, it is often convenient to deal with strategies mixed in the normal form. For dealing with complex examples, it is usually more convenient to deal with behaviorally mixed strategies.^f So it is nice to know that the two are equivalent concepts (for games with perfect recall).

11.5. Bibliographic notes

I have tried to be relatively more formal and complete in this chapter than in others, because I don't know any "next level" textbooks to recommend. The serious reader should instead proceed to the original research papers on these subjects. References to these were provided in the course of the chapter. In the other direction, the reader interested in seeing a slower development of decision trees can consult Holloway (1979), while a somewhat less formal and anecdotal companion to most of what we will do on noncooperative game theory is Dixit and Nalebuff (forthcoming).

References

- Aumann, R. 1964. "Mixed and Behavior Strategies in Infinite Extensive Games." In *Advances in Game Theory*, M. Dresher, L. Shapley, and A. Tucker, eds., 627-50. Princeton, N.J.: Princeton University Press.
- Dalkey, N. 1953. "Equivalence of Information Patterns and Essentially Determinate Games." In *Contributions to the Theory of Games*, vol. 2, H. Kuhn and A. Tucker, eds., 217-45. Princeton, N.J.: Princeton University Press.
- Dixit, A., and B. Nalebuff. Forthcoming. *Thinking Strategically*. New York: W. W. Norton.
- Holloway, C. 1979. *Decision Making Under Uncertainty: Models and Choices*. Englewood Cliffs, N.J.: Prentice-Hall.

^f To prove existence of an equilibrium, the former is more convenient; see the next chapter. But when we analyze the centipede game with incomplete information in chapter 14, it is essential that we deal with behaviorally mixed strategies.

Kuhn, H. 1953. "Extensive Games and the Problem of Information." In *Contributions to the Theory of Games*, vol. 2, H. Kuhn and A. Tucker, eds., 193-216. Princeton, N.J.: Princeton University Press.

Thompson, F. 1952. "Equivalence of Games in Extensive Form." The Rand Corporation. Mimeo.

11.6. Problems

Be sure to do problem 4. It will return again and again in the sequel.

■ 1. Let us complicate the story of Jokx and Beljeau enormously by supposing that both Jokx and Beljeau must decide whether to commission market surveys that will reveal the size of the market. For Jokx, this survey will cost \$5,000; if they choose to spend this money, Jokx can learn for sure what the market size will be before they have to decide whether to proceed with Oligopoly. And Beljeau (which previously was assumed to have this information coming) now must decide whether to commission a survey; it will cost them \$5,000 as well. Both sides must decide whether to commission a survey and then whether to proceed with the respective products without learning what the other has done, both with regard to taking the survey and then to introducing their product.

- Draw decision trees for Jokx and Beljeau in this new situation.
- Draw a game tree representing the situation for the two. (This is a tedious exercise.)
- List all the strategies that each side has. (Depending on what you do with equivalent strategies, they have between six and sixteen apiece.)
- Give the reduced normal form of this game. (Again very tedious.)

■ 2. What are the extensive and strategic forms of the game if, beginning with the variation of problem 1, Beljeau must decide whether to become informed about the size of the market before learning what Jokx does, but if they decide to become informed, they learn whether Jokx chose to become informed or not?

■ 3. Two firms simultaneously produce quantities of a good, which costs them \$5 per unit to produce. They can produce any quantity between 0 units and 500 units. (To keep the game finite, assume they must select an integer number of units.) The units produced are then totalled, and a price for them is set that will clear markets where demand for the good is given by $P = \$1000 - 2Q$, where Q is the total quantity supplied. Each

firm gets revenue according to the equilibrium price and the amount that it sells. What does an extensive form of this game look like? What does the strategic form of this game look like? How would things change if one firm, say firm A, got to see how much firm B produced before launching production itself?

■ 4. Consider the following game, which we will call the “truth game.” There are two players, called 1 and 2, and a game-master. The game-master has a coin that is bent in such a way that, flipped randomly, the coin will come up “heads” 80% of the time. (The bias of this coin is known to both players.) The game-master flips this coin, and the outcome of the coin flip is shown to player 1. Player 1 then makes an announcement to player 2 about the results of the coin flip; player 1 is allowed to say either “heads” or “tails” (and nothing else). Player 2, having heard what player 1 says but not having seen the results of the coin flip then must guess what the result of the coin flip was — either “heads” or “tails.” That ends the game. Payoffs are made as follows. For player 2 things are quite simple; player 2 gets \$1 if his guess matches the actual results of the coin flip, and he gets \$0 otherwise. For player 1 things are more complex. She gets \$2 if player 2’s guess is that the coin came up “heads”, and \$0 if player 2 guesses “tails”, regardless of how the coin came up. In addition to this, player 1 gets \$1 (more) if what she (player 1) says to player 2 matches the results of the coin flip, while she gets \$0 more if her message to player 2 is different from the result of the coin flip.

Draw an extensive form representation of this game. Can you draw more than one extensive form representation? Then convert this extensive form representation to a normal form representation. (If you wish to test your intuition, answer: If you were playing this game as player 2, what strategy would you select? If you were player 1, what would you do? Record your reasons, and keep them available; later in the book we will analyze this game according to game-theoretic procedures.)

■ 5. If you read and understood sections 11.2 and 11.4 and if you know the rules of contract bridge, give the details for the following statements. Consider one hand in the game of contract bridge. Viewed as a four-player game, this game has perfect recall. But viewed as a two-player game, it does not. In this respect, viewed as a two-player game, Kuhn’s theorem is false. One way to see this is to note that it would be a huge advantage to a partnership that sometimes engaged in psych bidding if they could secretly signal to each other when they were doing so.

chapter twelve

Solution concepts for noncooperative games

Having modeled a situation with either an extensive form or a normal form game, our next objective is to analyze the model in order to make predictions about what will happen in the situation modeled. This chapter concerns the analysis and “solution” of noncooperative games. It is a long and complex chapter, and you should expect it to take a number of readings before it begins to make sense to you (if it ever does).

12.1. Opening remarks

We begin with two opening remarks and an exercise that sets the stage.

The subject population

Any predictions we make about how a game will be played must depend on the characteristics of the individuals playing the game. One would not expect to observe the same behavior from every group of players.

The theories we will present proceed on the presumption that all the players in the game are “rational” (in various senses that we will make more precise as we go along), that each credits their rivals with “rationality,” that each believes that all rivals credit their rivals with “rationality,” and so forth. Our objective is to predict how people act in gamelike settings, and so one must wonder how useful will be any theory based on such a strong presumption concerning the rationality of the subjects. Some players in some situations might fail to meet this assumption. If you are playing against one of these people, or you suspect that you may be, you would sometimes be foolish to presume that your rival is rational. We will see in chapter 13 an interesting variation on this: If your rival(s) suspect that you are not rational, or even if they suspect that you suspect that they suspect that you aren’t rational, then the “rational” actions for you

can be quite different than if you ignore this possibility. That is to say, irrational people are out there and their irrationality has a substantial impact in many competitive settings, including an impact on the actions of rational individuals. We would not be developing a useful theory, either positive or normative, if we didn't take the possibility of irrationality into account. We will do this in chapter 13, but for the time being (and with apologies for the delay), we will proceed on the assumption that there isn't significant probability that some player is irrational, or that players think that others might be irrational, and so on.

This is still weaker than the formal assumption upon which some of our arguments will be based, namely that each player is absolutely certain to be rational, each is absolutely certain that all others are rational, each is absolutely certain that all others are certain that all the players are rational, and so on. I would contend (without much argument, presumably) that the ideal conditions of absolute certainty of rationality are never met in the real world. So insofar as the theory we develop depends on this absolute certainty and isn't robust to an "almost certainty," we have a remarkably useless theory. We need to pay attention to how robust our theory is to the slight possibilities that some one or more of the players is irrational, or that one or more suspect that some other(s) might be irrational, and so on. We will have a few things to say along these lines in what follows, but this is a difficult subject that only recently has gotten very much attention.

Some ground rules and cheap-talk

The ground rules in noncooperative game theory are that the description of the game should include all relevant opportunities for the players. That is, if there is the possibility that players can negotiate and form binding agreements, or if there is the possibility that one player can precommit early on to a particular sequence of moves, then this should be reflected in the extensive form and in the list of possible strategies.

One of the hardest things to handle in this regard is the possibility that the players can communicate. We can imagine communication of two different sorts here. One possibility is that players can send signals that involve a significant expenditure of resources. For this sort of communication, we follow the policy of trying to build formally into the model the communication possibilities.¹

But there is also what is called "cheap-talk" — communication by one or both sides that is costless but which nonetheless might be useful in coordinating actions. Players can often communicate, and when players

¹ See especially chapter 17, which is largely given over to this subject.

can engage in cheap-talk, we could not hope to build a model with all the possibilities. How many speeches could one side make? How many replies could a second side make? How many iterations could there be in this speech/counterspeech regime? So it is typical in the analysis of noncooperative games to omit such communication possibilities from formal models. This doesn't mean that these possibilities won't affect the outcomes of games; they do so in some important ways in various situations. And we will want to note this sort of effect when we can. But once again, our remarks on this point will be very much less than complete. This is another subject which has only recently begun to receive attention, and there is no settled "conventional wisdom" about cheap-talk.

Exercise

The test of the theory we will develop must, in the end, be empirical. A fair bit of experimental work looks at how various subject populations act in different types of "toy games," and (rather less) empirical work has been done on how individuals act in the sorts of real life situations we hope to model and analyze with these tools.

We will not look at this empirical literature, relying instead on your (no doubt razor-sharp) intuition about how people act in competitive situations. To give your intuition a fair test (or, rather, to give the theory a fair test), consider now how you would play a few games. In figure 12.1 are thirteen games in normal form and two games in extensive form, and following in the text are two more games in "verbal form." They are all two-player games, which is something of a simplification, but they cover enough ground for most of the points we will make in this chapter.

In all cases, the rules are that you will play the game once and once only, with no enforceable agreements possible, and with no side-payments between the two players. The numerical units ought to be thought of as in utility terms for the players. But, if you prefer, you can think of these as being "small" monetary payoffs, in pennies or nickels, say. Your rival will be someone very much like yourself — a fellow student, say, or someone who is reading this book just as you are.

Before reading on in this chapter, consider each of these games in turn and answer the following questions:

- How would you play as player 1 and as player 2?
- What predictions do you make about the play of your opponent? Assess a probability distribution for your opponent's choice of strategy; e.g., in the game in figure 12.1(a), perhaps you assess that your opponent, if

		Player 2		
		t1	t2	t3
Player 1	s1	4,3	2,7	0,4
	s2	5,5	5,-1	-4,-2

(a)

		Player 2		
		t1	t2	t3
Player 1	s1	4,10	3,0	1,3
	s2	0,0	2,10	10,3

(b)

		Player 2	
		t1	t2
Player 1	s1	10,0	5,2
	s2	10,1	2,0

(c)

		Player 2		
		t1	t2	t3
Player 1	s1	10,0	5,1	4,-200
	s2	10,100	5,0	0,-100

(d)

		Player 2	
		t1	t2
Player 1	s1	10,10	0,0
	s2	0,0	1,1

(e)

		Player 2			
		t1	t2	t3	t4
Player 1	s1	0,0	0,0	100,100	0,0
	s2	0,0	0,0	0,0	99,99
	s3	100,100	0,0	0,0	0,0
	s4	0,0	100,100	0,0	0,0

(f)

		Player 2	
		t1	t2
Player 1	s1	0,0	90,91
	s2	91,90	0,0

(g)

		Player 2	
		t1	t2
Player 1	s1	1,-1	-1,1
	s2	-1,1	1,-1

(h)

		Player 2	
		t1	t2
Player 1	s1	3,1	1,3
	s2	0,5	4,2

(i)

Figure 12.1. Some games.

		Player 2	
		t1	t2
Player 1	s1	0,0	20,40
	s2	40,20	0,0

(j)

		Player 2	
		t1	t2
Player 1	s1	0,0	20,40
	s2	60,25	0,0

(k)

		Player 2			
		t1	t2	t3	t4
Player 1	s1	200,6	3,5	4,3	0,-1000
	s2	0,-10000	5,-1000	6,3	3,20

(l)

		Player 2	
		t1	t2
Player 1	s1	5,5	-3,8
	s2	8,-3	0,0

(m)

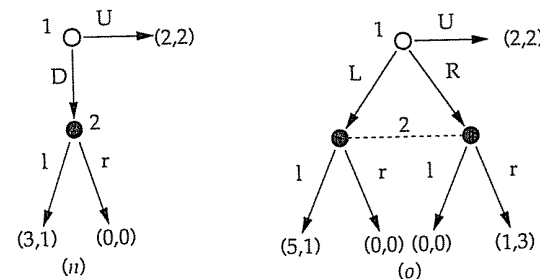


Figure 12.1, continued. Some more games.

choosing between s1 and s2, will choose s1 with probability .1 and s2 with probability .9, while if your opponent takes the role of player 2, he will play t1 with probability .6, t2 with probability .35, and t3 with probability .05. (This is just an example of the sort of assessment you are asked to make; my own assessment for this game would be quite different, as you will see when we get underway.) Of course, your answer to part (b) should have some connection with your answer to part (a). You might wish to write down your assessments and planned strategies for each game, with some notes to yourself about how you came to these assessments. We will discuss each of these games in turn as the chapter progresses, and you might wish to compare your assessments with what the theory will say you "ought" to have done.

As you give these assessments, please take special note of the following two sorts of situations:

Which individual strategies do you think are extremely unlikely for your opponent to choose?

In which games do you think there is an "obvious way to play," by which we mean an outcome (strategies for both players) that you think is very, very likely, and that you think any opponent you might encounter is likely to think very, very likely.

Note that I haven't tried to quantify "extremely" and "very, very" in these questions. That is for you to do.

Insofar as your predictions and/or your own choice of strategy depend on the specification we made concerning who is your rival, you might think through what would change if we changed the population from which your rival will be selected. The ground rules are also apt to affect your predictions. In particular, consider the following two variations for the games in figure 12.1:

(c) How do your predictions change if you can make a cheap-talk speech to your rival before playing (but he can't talk back to you)? How do your predictions change if the two of you can exchange cheap-talk?

The two games in verbal form are:

Divide the cities. Following is a list of eleven cities in the United States: Atlanta, Boston, Chicago, Dallas, Denver, Houston, Los Angeles, New York, Philadelphia, San Francisco, and Seattle. I have assigned to each city a point value from 1 to 100 according to the city's financial and commercial importance and its "quality of life." You will not be told this scale until the game is over, except that I tell you now that New York has the highest score, 100, and Seattle has the least, 1. I do think you will find that my scale is fair. I am going to have you play the following game against a randomly selected student of the Harvard Graduate School of Business Administration.² Each of you will be asked to list, simultaneously and without consultation, some subset of these eleven cities. Your list must contain San Francisco, and your opponent's must contain Boston. Then I will give each of you \$100 simply for playing the game. And I will add to/subtract from that amount as follows: For every city that appears on one list but not on the other, the person who lists the city will get as many

² If you are from the Harvard Business School, make it a randomly selected student from the Stanford Graduate School of Business.

dollars as that city has points on my scale. For every city that appears on both lists, I will take from each of you twice as many dollars as the city has points. Finally, if the two of you manage to partition the cities — if each city appears on one and only one list — I will triple your winnings. Which cities will you list?

Divide the letters. Following is a list of eleven letters: A, B, C, D, E, H, L, N, P, S, T. Each has been assigned a point value from 1 to 100 according to some random scheme that I will not tell you, except to say that N has been assigned the highest score of 100 points and T has been assigned 1. I am going to have you play the following game against a randomly selected student of the Harvard Business School. Each of you will be asked to list, simultaneously and without consultation, some subset of these eleven letters. Your list must contain S, and your opponent's must contain B. Then I will give each of you \$100 simply for playing the game. And I will add to/subtract from that amount as follows: For every letter that appears on one list but not on the other, the person who lists the letter will get as many dollars as that letter has points on my scale. For every letter that appears on both lists, I will take from each of you twice as many dollars as the letter has points. Finally, if the two of you manage to partition the letters — if each letter appears on one and only one list — I will triple your winnings. Which letters will you list?

12.2. Dominance and iterated dominance for normal form games

Let us begin by looking at the normal form of the Jokx-Beljeau game, which is reproduced here as figure 12.2. Does this representation suggest how the game will be played?

		Beljeau's strategy			
		t1	t2	t3	t4
Jokx' strategy	s1	-11,-19.4	5.2,4	35,-23.4	41.2,0
	s2	0,44.4	0,48	0,-3.6	0,0

Figure 12.2. The normal form representation of Jokx vs. Beljeau.

		Player 2		
		t1	t2	t3
Player 1	s1	4,3	2,7	0,4
	s2	5,5	5,-1	-4,-2

(a)

		Player 2		
		t1	t2	t3
Player 1	s1	4,10	3,0	1,3
	s2	0,0	2,10	10,3

(b)

Figure 12.3. Two normal form games that are dominance solvable.

The answer to this question is emphatically yes for Beljeau. If Beljeau anticipates that Jokx will follow strategy s1, then Beljeau's best response is t2. If Beljeau anticipates that Jokx will follow strategy s2, then Beljeau's best response is t2. No matter what Jokx does, *Beljeau's best response is t2*. It stands to reason, then, that we can anticipate t2 from Beljeau.³

You may be straining to come to the obvious corollary conclusion, namely that if Jokx understands all this (so Jokx knows that Beljeau will adopt its strategy t2), then Jokx will choose s1. But before doing so, let us analyze the logic being employed a step at a time. We have in this example, for Beljeau, a very strong criterion for assessing what it will do; viz., *its best response to Jokx is independent of what Jokx does*. Whenever a player in a game has a strategy that is so clearly optimal as this, we will assess very high probability that the player will use this strategy, assuming we have correctly modeled the player's situation.

Consider next the game depicted in figure 12.3(a), which is taken from figure 12.1(a). If player 1 plays s1, 2's best response is t2. If player 1 plays s2, 2's best response is t1. So we can't come to the strong sort of conclusion above for player 2. Neither can we come to this sort of conclusion about player 1: s2 is 1's best response to t1 and t2, but s1 is 1's best response to t3. Nonetheless, we can arrive at a "partial" conclusion about player 2: We can conclude that he won't choose t3. This is so because t2 is better than t3 no matter what player 1 does. If player 1 chooses s1, then player 2 gets 7 from t2 and only 4 from t3, while if player 1 chooses s2, player 2 gets -1 from t2 and -2 from t3. We can't say for sure what player 2 will do, but we can be pretty sure about something he *won't* do.

We are invoking here the criterion of *strict dominance*. One strategy for a player is said to dominate strictly another if, no matter what other players do, the player in question does strictly better with the first than with the second. When one strategy strictly dominates a second, we will

³ This should come as no surprise: You will recall that we reached this conclusion back when we drew the decision tree of Beljeau.

invariably conclude that the second strategy *will not* be played (or, if it is, we wonder why our model of the situation is incorrect).

Note that our first, very strong criterion (which worked for Beljeau) is an extreme case of strict dominance. It is the case where one strategy for a player strictly dominates all the others. Since by strict dominance we assess zero probability that any of the other strategies will be chosen, we are left concluding that the one strategy that dominates all the others will be chosen with probability one.

Now we take the next step in Jokx-Beljeau, a step we can take as well in figure 12.3(a). If we can predict that Beljeau will choose t2, then Jokx can as well, and Jokx should choose s1. Or in terms of the game in 12.3(a), if player 1 concludes that player 2 will not choose t3, then s2 is better for player 1 than is s1. The only reason player 1 would consider s1 is if she thought there was an appreciable chance that player 2 will play t3. Having concluded that this isn't the case, s2 dominates s1 for the strategies that are "left." And, to go yet another step, if player 2 anticipates that player 1 will anticipate that 2 will not play t3, and if 2 anticipates that player 1 will, therefore, select s2, player 2 then will choose t1 over t2. If we think players will think through the game in this fashion, we predict that the outcome will be s2 and t1. (How did you say you would play this game?)

We have "solved" both the Jokx-Beljeau game and the game in figure 12.3(a) by the criterion of *successive strict dominance*. We eliminate one strategy for one of the players on the basis of strict dominance. In the belief that a rational opponent will figure this out, this may allow us to rule out some of the opponent's strategies on the basis of "dominance," where we only require dominance against strategies not yet eliminated. And this could take us back to further elimination of strategies of the first player, and so on. Sometimes, this will end up with a single cell left, in which case we have figured out how rational opponents will play the game.^a

Be very careful in noting what is meant by "rational players" here. We presume, and this seems a reasonably safe prediction, that rational players will not play dominated strategies. But when we move from dominance to iterated dominance, we raise the stakes. In a second round of iterated dominance, we are assuming that players assume their opponents won't play dominated strategies. That is, players are presumed to believe that

^a You might wonder if this will be path dependent — if crossing out successively dominated strategies might lead to different results if crossed out in different orders. The answer is no, as long as crossing out is done on the basis of strict dominance, i.e., where you only cross out one strategy if some second strategy is strictly better than the first against every (remaining) choice by one's opponent(s).

their opponents are "rational." And in a third round, players presume that their opponents presume that all *their* opponents are rational. And so on. As the chain gets longer, the presumptions required grow more dubious; and we may feel less happy with predictions based on many rounds of iterated dominance.

Does this require players to be absolutely certain that their opponents are rational? (All we need to be fairly sure that players will not play a dominated strategy is a "fair certainty" that players are rational. If, in the game in figure 12.3(a), we assess probability .9 that player 2 is rational (will not play a dominated strategy), then we assess probability .9 or more that player 2 won't play t3. (This just repeats a definition.) But (when we move to iterated dominance, other considerations intrude. Suppose that the -4 in the s2-t3 cell of figure 12.3(a) was a -400. Then if player 1 assesses only probability .9 that player 2 is rational, she might choose to play s1 instead of s2. Or make this payoff a -40 and make the 3 in s1-t1 a -30. Suppose that player 2 is rational. Suppose that player 1 assesses probability .99 that player 2 is rational. But suppose that player 2 isn't so sure that player 1 has this high a regard for player 2's rationality. Then player 2 might well consider that player 1 might play s1 to be "safe"; at least player 2 might assess significant probability for this. And player 2 might, therefore, be led to t2 instead of t1.

For a prediction obtained by iterated dominance, there is some robustness in that prediction against the presumption that all players are certainly rational, all are certain that all are rational, and so on (for as many steps as there are steps in the chain of iterated dominations). That is, there is a probability less than one such that if every player believes that others are rational with at least this probability, each believes that each believes this with at least this probability, and so on; then we would still make the prediction that we obtained from iterated dominance. But (a) we might feel less happy with this sort of assessment if the number of iterations is large, and (b) the higher the cost of a "mistake" is, the closer to one must be this probability. We will return to these points near the end of this section.

Now look at the game depicted in figure 12.3(b), which is reproduced from 12.1(b). Can we do anything with dominance here? It would seem not; no one strategy strictly dominates another. But, under the assumption that player 2 is a subjective expected value maximizer, we can rule out t3. Why? Because no matter what probability distribution player 2 assesses for what player 1 will do, one of t1 or t2 (or both) is better than t3. Sometimes t1 is better than t3 (if s1 has probability greater than .3) and sometimes t2 is better (if s1 has probability less than .7). But t3 is never even tied for

best of all. And, if we accept this argument and cross out t3 (or, more to the point, if we think that player 1 accepts this argument and crosses out t3), then we can proceed to cross out s2, and thus settle on s1-t1 as the solution.

This is "strict dominance of one strategy by a combination of other strategies." It is more difficult to find in general than strict dominance of one strategy by a second, but it seems legitimate to apply whenever we can. Two remarks about this are in order: If the prizes in the game were in units of millions of dollars, and if player 2 is risk averse, player 2 might prefer the sure \$3 million to gambling. But we're assuming that either players are risk neutral or payoffs are in von Neumann-Morgenstern utility units, so this won't be a problem in theory. Secondly, we put the argument above as: Player 2, for any assessment about what player 1 might do, will choose either t1 or t2 as the strategy that maximizes expected utility. One could pose a different argument leading to the same conclusion: If player 2 randomizes by flipping a coin, playing t1 if heads and t2 if tails, then player 2 has a strategy that gives (an expected utility of) 5 if player 1 chooses s1 and 5 if 1 chooses s2, which strictly dominates t3. That is, t3 is dominated by a mixture of t1 and t2. I prefer the first line of argument, but the two come to the same thing.^b

Now look at the game in figure 12.4(a), reproduced from figure 12.1(c). No strategy strictly dominates another. But s1 *weakly dominates* s2; s1 is as good as s2 against t1 and is better against t2. In general, one strategy for a player weakly dominates a second if against any strategies by the other players the player does as well with the first strategy as with the second,

		Player 2	
		t1	t2
Player 1	s1	10,0	5,2
	s2	10,1	2,0

(a)

		Player 2		
		t1	t2	t3
Player 1	s1	10,0	5,1	4,-200
	s2	10,100	5,0	0,-100

(b)

Figure 12.4. Examples of weak dominance.

^b To be precise, in two-player games the two arguments certainly come to the same thing. In more-than-two-player games, they come to the same thing if a player is allowed to assess that the strategies chosen by his rivals might be correlated. If, however, players in a more-than-two-player game are presumed to assess that rivals' choices of strategy are independently distributed, then my preferred phrasing of the test is stronger. If you know about the separating hyperplane theorem, you can prove all this for yourself.

and against some combination of strategies by the other players, the player does strictly better with the first. Any player who maximizes expected utility and assigns strictly positive probability to every possible strategy combination by others will strictly prefer a weakly dominant strategy to the strategy that it dominates.

(Applying weak dominance successively can be problematic, however. Look at figure 12.4(b), reproduced from figure 12.1(d). While s_1 weakly dominates s_2 , it ceases to do so if the strictly dominated strategy t_3 is eliminated first; once t_3 is eliminated, player 1 is completely indifferent between s_1 and s_2 . Player 2 would certainly like to encourage player 1 to select s_2 , so perhaps billboards put up by player 2 saying that " s_2 is safe" would be in order.⁴

The use of (successive) weak dominance is not as noncontroversial as the use of (successive) strict dominance, in part because of examples such as 12.4(b).⁵ Despite such examples, economists tend to use successive weak dominance when they can. And when (as sometimes happens) all strategies except one for each player are eliminated by successive dominance, weak or strict, the game is said to be *dominance solvable*.⁶

How does weak dominance interact with less-than-complete certainty that the players are rational? — in two ways, depending on who is trying to guess whether his rivals are rational. Suppose that some given player, call him i' , has a given strategy s that is weakly dominated. Then if i' is less than completely certain that his opponents are rational, he may not wish to preclude them doing anything at all. That is, his assessment about his rivals' actions puts positive probability on every possibility. In this case s will never maximize i' 's expected utility, even weakly, and we have a good argument to conclude that i' won't use s . But this argument presupposes that i' is rational. And if some other player i'' is worried even a bit about the rationality of i' , then i'' might not want to count on i' avoiding s . Indeed, since s for i' is

⁴ It is also in player 1's interest that player 2 thinks that 1 will pick s_2 , so we might also imagine that player 1 would be renting space on available billboards, to convince player 2 that she will pick s_2 . But player 2, if he is sceptical, may wonder about this; it is in the interests of player 1 to mislead 2 into thinking that 1 will play s_2 even if player 1 fully intends to play s_1 , since by this means player 1 encourages the play of t_1 . How would you, as player 2, treat some pregame cheap-talk by player 1 that she fully intends to play s_2 ?

⁵ What does this example tell you about path independence and iterated weak dominance?

⁶ Remember to review as we go along what you had to say about these games when you did the exercise at the start of this chapter. My experience with student populations, albeit informally tabulated, is that in the game in figure 12.4(b), s_2 - t_1 is the modal response, despite the "logic" of weak domination. Admittedly, this is a very special game. But see below for a more robust example.

⁷ Some quite interesting games are dominance solvable. See, for example, the discussion of the Cournot game in section 12.8 and Rubinstein's bargaining game in chapter 15.

only weakly dominated, it seems more likely that a less than fully rational i' might not see this argument. Accordingly, if s must be deleted in some chain of iterated dominance and if others doubt the rationality of i' , the chain is more suspicious.

12.3. Backwards induction in games of complete and perfect information

So far we've discussed dominance and weak dominance in the context of normal form games. These criteria have important manifestations as well in extensive form games. Consider the extensive form game in figure 12.5(a), reproduced from figure 12.1(n). What predictions did you make about this game?

Most people predict that the outcome of this game will be that player 1 is very likely to choose D and player 2 is likely to respond with l . The argument supporting this prediction runs as follows: If player 1 chooses D , then player 2 is faced with a choice of l , which nets 1, or r , which nets 0. The dominating choice is l , and so we predict that if 1 chooses D , 2 (presuming 2 is rational) will choose l . And if player 1 believes that 2 is rational, 1 will replicate this logic, and she will choose D , which (she expects) will net 3, instead of the 2 she gets from U .

This, it turns out, is an argument equivalent to successive weak domination. In figure 12.5(b), we have the normal form game that corresponds to the extensive form game in figure 12.5(a). In this normal form, l for player 2 weakly dominates r . And once we eliminate r for player 2, D is the dominant choice for player 1. Note that the first step in this progression is weak domination only; if player 1 chooses U , l and r do equally well.

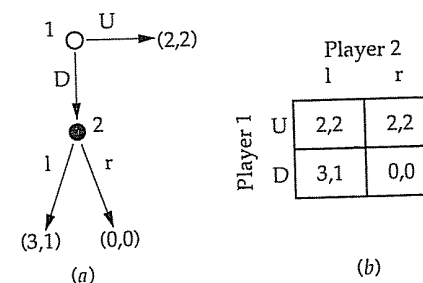


Figure 12.5. A simple extensive form game and its normal form counterpart.

It is sometimes maintained that the argument for the extensive form game is a bit more convincing than the corresponding argument of weak dominance for the normal form game. The logic for this assertion is that if we think of players 1 and 2 selecting strategies simultaneously and independently, then we might conceivably imagine player 2 choosing r , if he is absolutely certain that player 1 will choose U . But in the extensive form game, player 2 doesn't have this luxury. If he is given the move, player 2 knows that player 1 chose D , and he, therefore, knows that his choice will be of consequence.

Whether you find the two arguments equally compelling or you find the extensive form version more compelling, the connection should be obvious. And we can generalize this substantially. A *game of complete and perfect information* is an extensive form game in which every action node is, by itself, an information set. Think of any such game, and suppose that the tree has only a finite number of nodes. Then we can look at any "next-to-final" node — one whose successors are all terminal nodes. (Since the tree is finite, there must be at least one such. Provide a proof, if you wish.) Whoever has the move at that node, we can quickly decide how the player will act there; he will select whichever of the terminal node successors gives him the highest utility. (What if there is a tie among the terminal successors in terms of his utility? We will ignore this unhappy possibility, which complicates matters without adding much insight.) By weak domination, this player will never adopt a strategy that calls for him to take any other action at this next-to-final node. Now look for any node all of whose successors are either terminal nodes or the next-to-final node we just analyzed. Again, there must be one such. We can now solve the game from this point on (again assuming there are no ties), and so on. Calling any node that we have solved a solved node, as long as there are unsolved nodes, at least one unsolved node has successors that are all either solved or terminal, and we can proceed to solve that node next. Eventually every node in the game will be solved, and we will have a predicted solution for the game.

If the foregoing verbal description was less than completely transparent, you will do well to create and solve an example. Remember the ground rules: Every action node must be, on its own, an information set; there must be (only) finitely many nodes; and ties in payoffs of players are to be avoided. (In case you've forgotten, remember as well that game trees may not have cycles.)

And note how this is just iterated weak dominance, strengthened (if you find it more compelling) because when a player is called upon to move at a given node, he knows that his choice will make a difference.

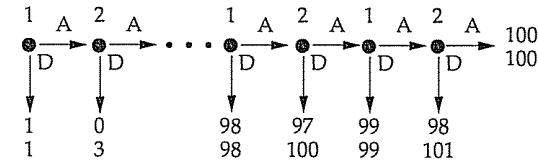


Figure 12.6. Rosenthal's centipede game.

The procedure sketched above is sometimes called backwards induction. One of the first results in game theory was, essentially, that finite games of complete and perfect information are dominance solvable by this backwards induction procedure, at least as long as ties are ruled out.⁷ An application of this result will indicate some of the limitations of non-cooperative game theory. The rules of chess make it a finite game of complete and perfect information.⁸ Hence we conclude that chess is dominance solvable. Either white has a forced win, or black does, or the game, played optimally, must end in a tie. The game tree for chess is, however, a bit too large for practical application of the backwards induction procedure. So despite the fact that chess must have a "solution," it remains an interesting puzzle how to play the game.⁹

Before departing this subject, one further example may be of interest — the so-called centipede game, popularized by Rosenthal (1981) and depicted in figure 12.6. If we apply backwards induction to this game, we see that at the next-to-final node player 2 will surely choose D , taking 101 instead of 100 and leaving player 1 with 98 instead of with 100. Hence at the next-to-next-to-final node player 1 will choose D , taking 99 instead of the 98 she would get by choosing A and letting 2 choose D . This leaves 2 with 99, and so 2 a node earlier chooses D and takes 100. And so on — going back up the tree, players 1 and 2 always take D instead of A , winding up with 1 choosing D at the first node in the game for payoffs of one apiece. (Game theorists, like economists, are not the best of linguists. Moving *up* the tree usually means moving in the direction of the tree's roots. Sorry about that.)

Since there are such lovely payoffs out at the end of the tree, this outcome seems rather a shame. More to the point, based on irregularly gathered evidence, it is a poor prediction. Asked to play this game, one

⁷ The original proof of this result was for two-player zero sum games, where the sum of the two players' payoffs at each terminal node is zero. In this case, ties are irrelevant.

⁸ If a particular position recurs three times, the game is drawn.

⁹ We will return to the example of chess at the very end of the book.

typically finds even rather sophisticated players moving a fair ways out to the end before one player or the other chooses *D*.

There are two points to make in this regard. First, the prediction that player 1 will choose *D* to start the game is based on 200 rounds of iterated dominance. It is rather hard to believe that players are so sure of their opponent's rationality and their opponent's certainty of theirs, and so on, that we can trust to a prediction so based. Second, suppose that you, as player 2, are absolutely certain that your opponent is rational, and that your opponent is certain that you are, and so on, for as many iterations as this takes. So you are certain that your opponent will begin the game with *D*. And then, *despite all your certainties*, your opponent chooses to start the game with *A*. What do you think then? You have pretty good evidence that your initial hypothesis was wrong. But to justify that your follow-on choice should be *D*, you have to believe ex post that your theory still holds (with very high probability). After all, if your opponent isn't rational, or doesn't think you are, or isn't sure that you think he is, ... then maybe it is better to try *A* and see how far out the tree you can get. And now put yourself in the position of player 1. Just think what havoc you can play in the mind of your opponent by choosing *A*. And look at all those big payoffs (for both of you!) out at the end of the tree. Shouldn't you give *A* a shot?

This is an example to which we will return several times, so for the time being, simply note that many rounds of iterated dominance is sometimes a bit too much to believe. And iterated dominance can be especially unbelievable in extensive form games, where a "wrong prediction" early in the game may lead players to reexamine their faith in the maintained hypothesis that all the players (and especially the one who did the "wrong thing") are rational.

12.4. Nash equilibrium

In the previous two sections, we attempted to predict the outcomes of games by asking the question: What actions by the players can we *rule out*? This may seem a rather indirect way to get at the question of what players will do, and it may leave us well short of a positive prediction. Nonetheless, that question is a model of straightforward logic when compared with what we take on for the next four sections.

Consider the games in figure 12.7, which are reproduced from figure 12.1(e,f,g). Recall the second subsidiary question that you were asked in the exercise at the start of this chapter: In these games, is there an "obvious way to play the game," an outcome that you think is very, very likely, and

12.4. Nash equilibrium

		Player 2	
		t1	t2
Player 1	s1	10,10	0,0
	s2	0,0	1,1

(a)

		Player 2	
		t1	t2
Player 1	s1	0,0	90,91
	s2	91,90	0,0

(b)

		Player 2			
		t1	t2	t3	t4
Player 1	s1	0,0	0,0	100,100	0,0
	s2	0,0	0,0	0,0	99,99
	s3	100,100	0,0	0,0	0,0
	s4	0,0	100,100	0,0	0,0

(c)

Figure 12.7. Three normal form games.

that (you would expect, with high probability) your rivals would think is very likely?

The typical responses are a definite yes for the game in 12.7(a), namely the outcome s1-t1, a definite no for the game in 12.7(b), and a conditional maybe for the game in 12.7(c). Regarding the last game, the answer I have heard typically runs as follows: If the players are unable to communicate, then s2-t4 seems very likely, although perhaps it is not very, very likely. The reason for this is that players will recognize that it is in their mutual interest to coordinate their actions, and while s2-t4 is not the absolutely best outcome for the two players, it seems the "safest" or "clearest" way to coordinate. If the players can engage in cheap-talk before the game is played, on the other hand, then one of s1-t3 or s3-t1 or s4-t2 is very, very likely to be the outcome, although there is no way to know which until after the cheap-talk is heard. That is, the players will certainly coordinate their actions; their interests are not in the least opposed. And they will use the opportunity of cheap-talk to coordinate on one of the three "best" outcomes.

Looking at the game in figure 12.7(a), and perhaps as well at the game in figure 12.7(c), we are using our intuition to say that there is an

"obvious way to play this particular game." On a case-by-case basis, we can explain why these outcomes are "obvious," although what is obvious may depend on context — on whether players can engage in cheap-talk, for example. Even though our intuition may only be engaged on a game-by-game basis, there is a general formal test that we can pose for any candidate "obvious way to play": If this strategy profile is obvious, so that (we expect, with high probability) all the players will figure this out, then it ought to be that players, in their own best interests, carry out their part of this profile. Put differently, what each player is meant to do ought to be a best response to what all the other players are meant to do. This suggests the following formal definition. We pose the definition for the context of a finite player normal form game, where the players are indexed by $i = 1, \dots, I$, their (respective) available strategy sets are denoted by S_i , and for $s = (s_1, \dots, s_I) \in \prod_{j=1}^I S_j$, $u_i(s)$ is i 's payoff if the strategy profile s is played.

Definition. A strategy profile $\hat{s} = (\hat{s}_1, \dots, \hat{s}_I)$ is a Nash equilibrium if for each player i and $s_i \in S_i$,

$$u_i(\hat{s}) \geq u_i(\hat{s}_1, \dots, \hat{s}_{i-1}, s_i, \hat{s}_{i+1}, \dots, \hat{s}_I).$$

Or, in words, a Nash equilibrium is a strategy profile in which each player's part is as good a response to what the others are meant to do as any other strategy available to that player.

This definition is given for normal form games, but the same basic concept applies to extensive form games — the easiest way to make the translation is to say that a strategy profile for an extensive form game is a Nash equilibrium if it corresponds to a Nash equilibrium as defined above in the corresponding normal form game. This translation hides some interesting facets of equilibria in extensive form games, which we get to only in section 12.7. Until then we will stick for the most part to normal form games.

The contention is that being a Nash equilibrium is a *necessary condition* for an obvious way to play the game, if an obvious way to play the game exists. And, indeed, in the games in figure 12.7(a,c), our candidate "obvious ways to play" are all Nash equilibria. But note carefully that

(a) Being a Nash equilibrium is certainly not sufficient for a strategy profile to be the obvious way to play a given game. In the game in figure 12.7(a), s_2 - t_2 is another Nash equilibrium, and we would hardly call it the obvious way to play this game.

(b) There is no reason to suppose that every game admits an obvious way to play. For example, if players 1 and 2 are unable to communicate, the game in figure 12.7(b) possesses no obvious solution, although it has two Nash equilibria, namely s_2 - t_1 and s_1 - t_2 .¹⁰

In the great majority of the applications of noncooperative game theory to economics, the mode of analysis is equilibrium analysis. And in many of those analyses, the analyst identifies a Nash equilibrium (and sometimes more than one) and proclaims it (them?) as "the solution." I wish to stress that this practice is sloppy at best and probably a good deal worse.¹¹ We advance the formal concept of a Nash equilibrium in a much narrower spirit: This formal concept is *an* answer to the question: If there is an obvious way to play the game, what properties must that "solution" possess? Given our loose definition of "an obvious way to play the game" — it is something all the players can figure out, and all expect the others can figure out — it is hard to see how some outcome that is not a Nash equilibrium could ever qualify.

But this is a very weak question, and it is clear that having the answer "Nash equilibrium" is pretty thin gruel if what we are after is a way to solve games. All we have is a test of solutions derived by some other means. Some obvious questions arise.

(a) If "Nash equilibrium" is only one answer to this question, what more can be said? Can we sharpen this necessary condition? In this regard, note that a Nash equilibrium can involve weakly dominated strategies. For example, in the game in figure 12.5(b), U - r is a Nash equilibrium, even though l weakly dominates r . Insofar as we believe that players will avoid weakly dominated strategies, we could add to our necessary condition that the "solution" should be a Nash equilibrium in strategies that are undominated, even weakly. Accepting this (although wait until section 12.7 before doing so), is there still more to say?

(b) What are the means by which we are to identify "obvious ways to play a (given) game?"

(c) What would lead one to imagine that a given game admits an "obvious way to play?" This question is closely related to (b) in that if we know

¹⁰ In fact, as we will discover momentarily, this game has a third Nash equilibrium, which is also not an obvious way to play.

¹¹ And, I'm sorry to say, in later chapters I will be somewhat sloppy in just this fashion.

why this sort of solution might exist, we would probably have a good clue to what it might be.

(d) What can one say about games that do not admit a "solution?" And what, if anything, does the notion of Nash equilibrium say concerning such games?

Make no mistake — there are games that, in certain contexts, do not possess "solutions." The game in figure 12.1(j) is a good case in point, at least if played under conditions of no communication. The game in figure 12.1(k) is another. And the game in figure 12.1(l) is especially interesting with regard to question (d) above. In discussing this game with students, I have found that most, cast in the role of player 2, choose t_3 , and most expect that player 2 will choose t_3 . Accordingly, most students cast in the role of player 1 choose s_2 , and most students cast in the role of player 2 expect player 1 to choose s_2 . But because of those large negative payoffs for player 2 in all the other columns, player 2 must be *very* sure what player 1 is going to do before moving from t_3 . That is to say, the standards of "very, very sure" are raised when the consequences of being wrong are dire. While I would guess that in eighty percent of the times I have had students play this game, s_2 - t_3 was the outcome, it is still not quite "obvious" enough to qualify as an "obvious way to play the game," given the consequences of being wrong.¹² This game does not admit an obvious way to play.

The point is that this game has Nash equilibria, three of them in fact. One is s_1 - t_1 , and a second is s_2 - t_4 . The third is an equilibrium in mixed strategies, something which we will introduce in the next section, but for the time being take my word for it: In no Nash equilibrium of this game is t_3 played. Indeed, t_3 is the only strategy that is unplayed in some Nash equilibrium. So in answer to question (d) above: When the game doesn't admit an "obvious way to play," looking at its Nash equilibria can give precisely the wrong answer. The concept of Nash equilibrium should have no command on our attention when the game doesn't admit of a "solution."¹³

This leaves us with questions (a), (b), and (c). We will get to those questions in a bit, but before doing so, we chase down a bit of esoterica concerning the notion of a Nash equilibrium.

¹² And, indeed, my very casual empirical sense is that there is enough play of s_1 by player 1 (between ten and twenty percent) to make t_3 the optimal strategy for player 2. It is certainly what I would do, playing this game.

¹³ For another example, see the end of section 12.6 for a game with a unique Nash equilibrium which, nonetheless, is not what one normally sees when the game is played.

12.5. Equilibria in mixed strategies

Some games don't have Nash equilibria. At least, they don't have equilibria if we stick to the definition given (loosely) above. Consider the games in figure 12.8 reproduced from figure 12.1(h) and (i).¹⁴ In both these games s_1 - t_1 isn't a Nash equilibrium, because if player 1 is choosing s_1 , then player 2 would wish to move to t_2 . At s_1 - t_2 , player 1 wishes to move to s_2 . At s_2 - t_2 , player 2 takes us back to t_1 . And at s_2 - t_1 , player 1 wishes to shift to s_1 , which exhausts all the possibilities for a Nash equilibrium in these games.

Despite what was just said, these two games do have Nash equilibria. What was just shown is that no pure strategy profile is a Nash equilibrium. But consider the game of figure 12.8(b), and imagine that player 1 plays s_1 with probability .6 and s_2 with probability .4. Then player 2, by playing t_1 , will net expected value 2.6. And playing t_2 will net 2.6 for player 2. And playing any mixed strategy will net expected value 2.6. So every strategy by player 2, mixed and pure, is a best response to this strategy by player 1. In particular, one best response is the mixed strategy where player 2 plays t_1 with probability .5 and t_2 with probability .5. And this strategy causes player 1 to net expected value 2 from s_1 and 2 from s_2 . So, in particular, playing s_1 with probability .6 and s_2 with probability .4 is a best response to player 2's strategy. We have a mixed strategy profile (that is, a mixed strategy for each player) such that each player's strategy is a best response to what the other is playing. Hence we have a Nash equilibrium. Similarly in the game of figure 12.8(a), it is an equilibrium for both players to mix with probability 1/2 for each of their respective strategies.

In these examples, players randomize among all their strategies. So you don't get the wrong idea about this, consider the game in figure 12.1(l).

		Player 2	
		t_1	t_2
Player 1	s_1	1,-1	-1,1
	s_2	-1,1	1,-1

(a)

		Player 2	
		t_1	t_2
Player 1	s_1	3,1	1,3
	s_2	0,5	4,2

(b)

Figure 12.8. Two games with only mixed strategy equilibria.

¹⁴ The game in figure 12.8(a) is called "matching pennies," after the children's game.

We have already remarked that this game has two Nash equilibria, s1-t1 and s2-t4. Because these equilibria do not involve any randomization, they are called *pure strategy equilibria*. Now consider the following pair of strategies. Player 1 plays s1 with probability 9000/9001 and s2 with probability 1/9001, and Player 2 plays t1 with probability 1/101 and t2 with probability 100/101. Given player 2's strategy, player 1 nets:

$$(200)(1/101) + (3)(100/101) + (4)(0) + (0)(0) = 500/101 \text{ from s1; and}$$

$$(0)(1/101) + (5)(100/101) + (6)(0) + (3)(0) = 500/101 \text{ from s2.}$$

So player 1 is content to play the mixed strategy. And player 2, against player 1's strategy, nets:

$$(6)(9000/9001) + (-10000)(1/9001) = 44000/9001 \text{ from t1;}$$

$$(5)(9000/9001) + (-1000)(1/9001) = 44000/9001 \text{ from t2;}$$

$$(3)(9000/9001) + (3)(1/9001) = 27003/9001 \text{ from t3; and}$$

$$(-1000)(9000/9001) + (20)(1/9001) = -8,999,980/9001 \text{ from t4.}$$

So player 2 is indifferent between strategies t1 and t2, and strictly prefers these two to t3 and t4. The range of best responses for player 2 (to player 1's strategy), then, is any mixed strategy that puts zero probability on t3 and t4. And our strategy profile has player 2 playing precisely this sort of mixed strategy.

Note that in this sort of equilibrium, the probabilities used by the players to randomize are determined not by their own payoffs but by the payoffs of the other players. If, for example, in figure 12.8(b) we changed the 1 that player 2 gets from s1-t1 to a 2, then the Nash equilibrium would be for player 2 to stick with the .5-.5 randomization and for player 1 to change to playing s1 with probability .75 and s2 with probability .25. The point of randomizing is to keep the other player(s) just indifferent between the strategies that the other player is randomizing among. One randomizes to keep one's rival guessing and not because of any direct benefit to oneself. Put another way, in a mixed strategy Nash equilibrium each player has no *positive* incentive to randomize according to the called-for mixing probabilities; as long as the other player is fulfilling his part of the equilibrium, there are many best responses for the first.

For this reason, many people find the idea of a mixed strategy Nash equilibrium incredible. Are we to believe that individuals faced with real economic decisions decide what to do by flipping a coin? And if we believe

that, are we further to believe that individuals will bother to perform just the right randomization, when doing so is not particularly to their advantage? Nothing said so far should be taken to imply that the answer is yes. Remember that being a Nash equilibrium is necessary for "the obvious way to play the game," if an obvious way to play exists. It is in the eyes of the players (or the analyst) whether mixed strategies ever can constitute the obvious way to play the game.

In some cases, it doesn't seem unreasonable to suspect that players use randomized strategies. When playing poker, for example, good players will randomize (bluff) to avoid becoming predictable.¹⁵ In games where one player checks on the other — in cases of auditing or quality inspection, to take two examples — a policy of random checking is often employed.

Still, if you conclude for yourself that mixed strategy equilibria are often not very intuitive and (therefore) not candidates for the "solution" of a given game, you won't get too much of an argument here.

But, in any case, having the notion of mixed equilibrium around is useful in one respect. It gives a rather nice mathematical result:

Proposition. *Every finite player, finite strategy game has at least one Nash equilibrium, if we admit mixed strategy equilibria as well as pure.*

This result can be generalized to classes of games with infinitely many players and/or infinitely many strategies under certain conditions. We won't be concerned with such extensions here.

The proposition is rather easy to prove if you recall Kakutani's fixed point theorem from chapter 6. Let $i = 1, \dots, I$ index the players, let S_i be the (pure) strategy space for player i , and let Σ_i be the space of mixed strategies for player i (i.e., Σ_i is the space of probability distributions on S_i). Write $\Sigma = \prod_{i=1}^I \Sigma_i$; that is, Σ is the space of mixed strategy profiles.

For each $\sigma = (\sigma_1, \dots, \sigma_I) \in \Sigma$, define

$$\phi(\sigma) = \{\hat{\sigma} = (\hat{\sigma}_1, \dots, \hat{\sigma}_I) \in \Sigma \mid \hat{\sigma}_i \text{ is a}$$

best response to $(\sigma_1, \dots, \sigma_{i-1}, \sigma_{i+1}, \dots, \sigma_I)$ for each $i\}$.

That is, $\phi(\sigma)$ gives the vector correspondence of best responses by the players to σ . We leave as an exercise the proof that ϕ is upper semi-continuous and convex-valued. Then by Kakutani's fixed point theorem, there is some σ

¹⁵ There is a counterargument to this: Good players randomly decide when to bluff in order to be unpredictable *in the future*. This is different from the argument in a randomized strategy, where a player randomizes for current purposes only. Put differently, in the last hand of poker you will ever play, will you choose what to do based on the flip of a coin?

such that $\sigma \in \phi(\sigma)$. Which then, by the definition of ϕ , makes σ a Nash equilibrium.

The use of mixed strategies is troubling to much of the laity and even to some game theorists; do we really think that players will choose their actions by rolling dice? Owing to this, there have been attempts to domesticate the concept. The most prevalent is an argument that mixed strategies are an artifice of an unnaturally coarse model of reality. That is, in any real application, each player has many items of personal preference known to himself and unknown by others — things like what side of the bed he got out of, his exact tolerance for risk, etc. One would not wish to build that much detail into a model; the model would become intractable. But that detail is “really” there. And the detail is what determines how an individual chooses between strategies. What maintains the equilibrium is that each player, trying to decide what the other will do, feels that he faces the equilibrium probability distribution over the opponent’s (or opponents’) strategies. The individual can make a definite choice between the strategies over which he is supposed to randomize based on his personal factors. He can’t predict what the other side will do, because he doesn’t know the actual values of the other’s personal factors. The mathematical results that support this philosophical position are known as *purification theorems*. A classic reference on this subject is Harsanyi (1973); Aumann et al. (1983) is also recommended to the interested reader.

12.6. Why might there be an obvious way to play a given game?

We’ve said that if there is an obvious way to play the game, then it is *necessary* that this is a Nash equilibrium. But we’ve also said that the only claim that the concept of Nash equilibrium has on our attention is predicated on there being an obvious way to play the game. And we’ve seen several examples of games without an obvious way to play. So if we think that there is some reason to pay attention to the notion of a Nash equilibrium, we had better give reasons why some games might seem (to the players and to the analyst) to be endowed with a “solution.”

We have, of course, seen examples of games that seem to have this sort of solution. The Jokx-Beljeau game and the games in figure 12.7(a) and (c) would be two such, in the case of 12.7(c) depending on what the players can say to one another. Games that are dominance solvable would be another class of games with obvious solutions, at least if the number of rounds of iterated dominance required is not too large: The games in figure 12.3(a) and 12.5 would seem to qualify, although the game in figure 12.6 would not. What is it about these games and others like them?

There is no single answer to this question, and in many cases the answer, such as it is, can only be given informally. This is why the formal notion of a Nash equilibrium is useful; as a necessary condition for any

12.6. Why might there be an obvious way to play a given game?

candidate solution, the concept applies no matter what story is believed to be behind the existence of the solution. The formal notion provides a useful test for solutions that others may suggest to you, and it gives us something of a formal language with which to discuss matters. But this utility of the formal concept doesn’t mean that one can do without the story about why a game in question does have a solution. So we turn now to some of the stories that are told in the literature.

Preplay negotiation

One means to the existence of an “obvious way to play” is (explicit negotiation or cheap-talk among the players, conducted prior to play of the game. If preplay negotiations take place, we cannot guarantee that the players will come to an agreement, nor can we say what agreement will be reached. But, if the players do negotiate and they do reach an agreement, then we would expect the agreed-to action to be “self-enforcing” in the sense that each individual player, expecting the others to carry out their part of the agreement, does not have an incentive to deviate. That is, we will want the agreement to be a Nash equilibrium.

There is an important caveat to add here. If players can engage in preplay negotiation, they might be able to move beyond the possibilities suggested by the formal rules of the game, especially if doing so will make everyone better off. We could model this possibility by putting within the formal rules of the game all these possibilities. But some sorts of possibilities are sometimes modeled “indirectly”; the possibilities are not included in the formal model of the game, but the solution concept is changed to reflect their possibility. The most famous of these is Aumann’s correlated equilibrium. Aumann (1974) considers what will happen if the players are able to build various sorts of mechanical coordinating devices or employ a disinterested party who will act as a coordinator. For example, consider the two-player game depicted in figure 12.9. This game has a single mixed strategy Nash equilibrium, in which both players have expected payoff 3.75. Now imagine that, via preplay negotiation, the players agree to the following scheme. Some third party will roll a six-sided die. If the die comes up with one or two spots

		Player 2	
		t1	t2
Player 1	s1	4,4	3,5
	s2	5,3	0,0

Figure 12.9. A game with a correlated equilibrium.

up, player 1 is told to play s_2 . If it comes up three through six, this third party tells player 1 to play s_1 . At the same time, if the die comes up one through four, player 2 is told to play t_1 , while five or six causes player 2 to be instructed to play t_2 . Each player is told only what to play and not how many spots came up (precisely). This arrangement is "self-enforcing": If player 1 is told to play s_2 , she knows that player 2 was told to play t_1 , and s_2 is a best response to t_1 . While if player 1 is told to play s_1 , she assesses probability $1/2$ that player 2 was instructed to play t_1 and $1/2$ that player 2 was instructed to play t_2 . Assuming 2 carries out his instructions, s_1 is a best response. A similar argument applies for 2. The point is that the roll of the die and the "intervention" of this third party allow 1 and 2 to correlate their actions in a self-enforcing manner. And in this correlated equilibrium, each player has an expected payoff of 4, which is more than either gets in the unique Nash equilibrium of this game. Can we maintain that the two players can engage in preplay negotiation but they cannot obtain the services of a third party for this purpose? If they can correlate their actions in this fashion, then the range of self-enforcing agreements is larger than the set of Nash equilibria, which presumes independent selection of actions. On all these points, see Aumann (1974). And for even fancier devices of this sort, see Forges (1986) and Myerson (1986).

Convention

Consider the game in figure 12.7(b), played under the following conditions: There is a large population of players who are matched randomly at random times and who at those times must play this game. (Moreover, players are randomly assigned "positions" in the game; a given individual will sometimes be put in the position of player 1 and sometimes in the position of player 2.) The players, when matched, cannot communicate. In this case, it is not hard to imagine that some convention will be developed concerning how to play this game — say, always play s_2 - t_1 . This convention is no more likely, on the face of things, than the reverse. By the same token (taking a slightly more complex example), there is no reason why red means stop and green means go on traffic lights. This is convention, and a lot of "good outcomes" on the roads result.

Interesting equilibria "by convention" sometimes are tied to seemingly irrelevant signals or historical summaries. By "seemingly irrelevant" I mean: These things do not affect the range of strategies a player has, or players' payoffs from various strategy profiles. But these things become relevant to the play of the game because convention makes them relevant. Traffic lights are a classic example. Or consider playing the game in figure 12.1(j) in a large population, with random matchings of players, but where the distinguishing labels of player 1 and player 2 are absent, so players cannot use those labels to coordinate. Imagine that each player carries with himself a summary of how well he has fared in the past —

say the sum total of all his winnings in these games. And imagine two different "societies," distinguished by the following two conventions: In the first society, when two players meet, the one who has fared less well in the past is given the edge in this encounter. That is, if player 1 has fared less well in the past, we play s_2 - t_1 . (If the players have done precisely as well in the past, convention is silent as to what happens.) In the second society, instead of *noblesse oblige*, the reverse holds: Whichever party has fared better in the past is given the edge. One might argue which is the better society/convention. But both conventions serve at least the purpose of getting players to a Nash equilibrium. (For analysis of this sort of thing, see Rosenthal and Landau [1979].)

Learned behavior

A variation on the convention stories concerns learned behavior. Go back to the game in figure 12.7(b), and consider a pair of players who play this game a number of times. We might not expect them to be able to coordinate their actions on the first round of play or on the second, but it is not hard to imagine that after a few interactions some *modus vivendi* will spring up between them. (Suppose that the players have constant roles. That is, one individual is always player 1 and the other is player 2. It is then interesting to ask: Given that the difference in payoffs is only 1, versus a loss of 90 if coordination fails, will players tend to stick with one coordinated solution, or will they learn to alternate? Or, put another way, if the 91s were both 1s, we might expect players to learn to alternate. But what about intermediate cases, such as the game in figure 12.1(j)? And what will happen in the game in figure 12.1(k)?)

An early manifestation of learned behavior stories were models of "fictitious play." Imagine two players playing a normal form game with the following decision rules. Begin with any strategy at all. Then, in each round of play, choose whatever strategy is best against the "average play" of your rival in previous rounds. That is, if in one hundred rounds of play your rival has chosen s_1 39 times, s_2 22 times, s_3 16 times, and s_4 23 times, then you play whatever is best against the mixed strategy (by your opponent) of (.39, .22, .16, .23). If both players act and "learn" in this fashion, will they settle down to some equilibrium? (If they do, it is clear that this will have to be a Nash equilibrium.) It turns out that this sort of behavior does not necessarily lead to an equilibrium; a famous example of Shapley (1964) establishes this. But one can still wonder to what extent learning by the players can lead them to situations in which each is fairly sure what the others are doing and acting in his own best interests in response reinforces what others think they know about the actions of the

first. For one development of this sort of story, especially in the context of extensive form games, see Fudenberg and Kreps (1989).

Although it will make more sense after chapter 14, note that a "problem" with the learned behavior story, insofar as it is meant to apply to a pair of players who play the same game repeatedly, is that the repeated play by itself can enlarge the set of possible Nash equilibria. (If you don't understand this remark, don't panic; wait for chapter 14.) To assume that players play the same game with the same opponents over and over and they thereby "learn" some static Nash equilibrium of the game may require that players are myopic to a certain extent. The story works a bit better if there is a group of individuals who are randomly matched, but then we come to a gray area between learned behavior and convention.

Focal points and principles

The final story we tell about where "solutions" to given games might come from is related to the stories in the previous two subsections. We can introduce this story by returning to the games in figure 12.7(a) and (c). In the game in 12.7(a), it seems clear why s_1-t_1 is the "solution," even though there is another Nash equilibrium; s_1-t_1 gives both players the most they can get in this game, and the players can see that this doesn't put their interests in opposition. Even if the players have never spoken and are not allowed to speak, it seems pretty clear that each will realize this (at least, with high probability) and act "accordingly." Note well the quotation marks around accordingly, since it is only accordingly if we make the leap for the two players simultaneously, or rather if we believe that each player, making the leap, simultaneously believes that his opponent is making the corresponding leap.

In 12.7(c), if the players cannot communicate, it is not the optimality of s_2-t_4 that commends itself, but the "uniqueness" of this (mode of) coordinating. It just seems likely that each player, looking at this game and thinking about how coordination might be achieved, will see this uniqueness as his best hope (and expect that the other side will see this as well).

Now consider the two verbal games from the first section. Take the game with the eleven cities first. This game has an enormous number of Nash equilibria, even if we restrict attention to pure strategies. Since San Francisco and Boston are preassigned, it has 512 pure strategy equilibria (the 512 partitions of the other nine cities). Moreover, each of these is a strict equilibrium; the players have positive incentive not to deviate in the slightest. Accordingly, you might think that playing this game leads to massive confusion. But it hasn't done so when I've had it played by students from Stanford University. Over eighty percent of the students (who

are U.S. nationals or otherwise long-time residents of the U.S.) asked have given the same list of cities, and rarely has anyone's list deviated from this by more than one. For Stanford students, this list is Dallas, Denver, Houston, Los Angeles, Seattle, and San Francisco. It is evident what is going on here; students use an east-west division of the United States to divide the cities. There are eleven cities, so students typically expect that they should get five or six. New York is said to have the highest value and Seattle the lowest, and it is clear to whom these two cities "belong," so it is typically felt that equity considerations should lead the western list to have six. And, rather nicely, the east-west principle leads to just a five-six division.

This doesn't always work. By changing the list of cities, one can make things much less nice. For one thing, when one of the participants in this game (when I've played it with students) is from outside the United States and one is from the U.S., the U.S. resident has worried about how much geography the other student knows. (These worries are only allowed to be expressed after the game is played, since a worry, properly phrased, gives the game away: One student once wondered aloud beforehand whether, since geography was such a tricky thing, whether his rival knew the order of the English alphabet.) Or if one removes Atlanta and Philadelphia from the list and adds Phoenix and San Diego, play becomes much less predictable, even for U.S. residents.

And, of course, the fact that these are cities is crucial. If the game involves not cities but a list of letters, then alphabetical order is the overwhelming choice for division: The list with T always takes S, P, and N, very often L, and almost never H. Again, one can confuse matters easily. Make N (the 100 point letter) the sixth letter in order, and things get touchy. Have one list begin with A and the second with C, and confusion results. Whatever is going on here only goes on in some cases. But, crucially, when it does, players tend to know that it is doing so.

The point is that in many games an equilibrium suggests itself to the players for reasons of symmetry or optimality or some other (qualitative) principle like those two. Context, the identity of the players, their backgrounds, and other such fuzzy factors play a huge role in this. This is the sense in which this is akin to convention or learned behavior: Although it is learning about "how people think" in general, rather than learning about "how people act conventionally" in a specific situation or "how person X acts in this situation," it is still a (manifestation of) socially driven expectations and behavior. The equilibria that people sometimes just naturally focus upon are called *focal points* or *focal equilibria*, and we will refer here to the organizing principle (equity, uniqueness, geography, alphabetical order) as the *focal principle*.

Game theorists have not done a good job of formalizing what constitutes a good focal principle, and these ideas remain informal and imprecise. The best (perhaps the only really good) discussion of these ideas can be found in Schelling (1980).

A nonstory

There are stories then, about why a game might possess a "solution" in the sense used here. There could be preplay negotiation; it could be the result of convention or learned behavior; and it could simply be that the solution is focal. At the start of this subsection we added, the game might be dominance solvable, which is a good story, as long as one doesn't need to invoke too many rounds of iterated dominance.

We close this section with a different story, one that is sometimes told and which when heard should be regarded with scepticism. This is the story that strategy profile X is the *only* Nash equilibrium, and so it must be the obvious way to play the game.¹⁶ I contend, however, that uniqueness of the equilibrium alone is not a convincing story.

To see why, consider the following variation on the game in figure 12.7(a). First, change the two 10s both to 100s. Now it seems even more obvious that the "solution" is $s1-t1$. But add the following complication: Player 1 chooses both a row (either $s1$ or $s2$) and a nonnegative integer. That is, player 1's strategy space is $\{s1, s2\} \times \{0, 1, 2, \dots\}$. Player 2 chooses a column (either $t1$ or $t2$) and a nonnegative integer. If player 1 chooses $s2$ or player 2 chooses $t2$, payoffs are just as before. But if player 1 chooses $s1$ and player 2 chooses $t1$, then whoever named the greater integer gets 101 and the other gets 100. If the integers named are identical, then each gets 99. If you think about this game, you will see that it has only one Nash equilibrium, $s2-t2$. And even with all this greatest integer stuff, I rather doubt that $s2-t2$ is the "solution." I would predict, with a fair degree of assurance, that the outcome of the game will be partially described by $s1-t1$; I have no particular prediction to make about how the greatest integer stuff will resolve. This, then, is a game in which there is no "solution," but where there is a very good and reliable "partial solution," at least to my mind. And this "partial solution" involves outcomes that are part of no Nash equilibrium.

You may be unhappy with this example because of the very strange nature of the game; "name a large integer" is not the stuff of many realistic

¹⁶ You will also hear that in a given game profile X is the only refined equilibrium for one of the refinements that will be given in the next section, and hence X must be the obvious way to play the game. Absent any supporting arguments, this is if anything less convincing than the story for plain Nash equilibria.

applications, and the one penny differences are not significant. We can think of this as really two games combined: The first is the game in figure 12.7(a) but with 99s instead of 10s. This game has two Nash equilibria and one quite obvious way to play. And the second is a strange game that is played only if we get the 99s equilibrium in the first game, a game where the stakes are one and two pennies, which has no Nash equilibrium at all, and which clearly has no obvious way to play. If we "decompose" the original game into these two pieces, then there is no reason to think that the presence of this strange second piece affects the reasonableness of playing the 99 equilibrium in the first piece.

Having set up this straw argument, let us demolish it: This is precisely the point. The decomposition of this game into two "pieces" is not something done formally; it is something we did based on intuition of how the game will be played. When formal criteria such as the uniqueness of equilibria support one's intuition, or even where the uniqueness of the equilibrium seems intuitively to be a focal principle that is likely to appeal to the particular players in the particular situation, then well and good. The formal criterion *by itself*, however, is inadequate.

In case you are still unhappy, we can make a slightly weaker point with the centipede game of figure 12.6. This game has more than one Nash equilibrium. But every Nash equilibrium of this game leads to the same outcome, namely player 1 playing D at the initial node of the game (proving this is a nice exercise). So if one is inclined to believe that any game with a unique Nash equilibrium *outcome* must have that outcome as obvious, then one believes that the obvious outcome to the centipede game is that player 1 plays D to begin, which we asserted before and continue to assert is not so obvious at all.

12.7. Refinements of Nash equilibrium

If being a Nash equilibrium is a necessary condition for some strategy profile to be the "obvious way to play" a particular game, perhaps there are other, stronger tests that ought to be passed. For example, we have already remarked on one weakness of the Nash equilibrium concept in section 12.3, namely that a Nash equilibrium can have a player using a weakly dominated strategy. One might wish to say, therefore, that a better necessary condition for a "solution" is that it be a Nash equilibrium with the additional *refinement* that no weakly dominated strategy is played with positive probability. (We will comment on this particular refinement of Nash equilibrium in a moment. For now consider it an example of what we might try to do.)

The general program of requiring more of a "solution" than just that it be a Nash equilibrium is known in the literature as the program of *perfecting or refining* Nash equilibrium. The literature on this subject runs in two veins; one can base refinements on arguments that are derived from considerations of the normal form of the game, as in the case of weak dominance, or one can base refinements on extensive form considerations. We have already noted a connection between weak dominance in the normal form and "backwards induction" in extensive form games of complete and perfect information, so you can expect a close connection between these two veins. In this section, we will move back and forth between the two veins, although emphasis will be on the extensive form, on grounds that extensive form considerations are more intuitive.¹⁷ Throughout, when we look at extensive form games, we will consider only games with perfect recall, so that Kuhn's theorem applies.

This is a very long and involved section. I have divided it into numbered subsections, so that it seems less of an endless mass. Certainly try to get through subsections 12.7.1 and 12.7.2. The remainder of this section is increasingly "optional," although some things discussed later in the book (especially in chapters 13 and 17) depend on these later subsections.

I give two final warnings: First, there is no settled conventional wisdom about refinements of Nash equilibrium. Indeed, there is much controversy about which refinements are reasonable and which not. What follows inevitably reflects my personal biases and my own way of viewing Nash equilibrium and noncooperative game theory, a fact that I will attempt to make clear by extensive use of the pronoun *I*. Of course, you get a strong dose of my personal philosophy throughout this book, anyway, but this particular subject is one in which opinions are rather broadly distributed, and my opinions are a bit extreme. Second, to develop fully my philosophy on these things, I need some tools that will be given in the next chapter. Hence in this section I ask a number of questions to which I can, for now, give only cryptic answers. The final section of chapter 13 will provide more satisfactory answers to the questions that are asked but not really answered here.

12.7.1. Weak dominance

We already sketched our first refinement: For games in normal form, look only at Nash equilibria that do not involve the play of weakly dominated strategies. I find this refinement fairly noncontroversial except in

¹⁷ My assertion that extensive form considerations are more intuitive is *not* unanimously held.

		Player 2	
		t1	t2
Player 1	s1	10,0	5,2
	s2	10,11	2,0

Figure 12.10. Refining with weak dominance.

very special and extreme examples (such as the game in figure 12.1[d]), and I would be suspicious of any "solution" that involved the play of a weakly dominated strategy.^d

Since I view this refinement as noncontroversial, it makes a good place to begin to discuss the root of all controversy concerning refinements. Consider the game in figure 12.10. This is a variation on the game in figure 12.1(c), but with player 2's payoff improved in the s2-t1 cell. Just as in the game from 12.1(c), s1 weakly dominates s2, and so the solution by iterated weak dominance is s1-t2. Note that this is a Nash equilibrium. So is s2-t1. But s2-t1 involves the play of a weakly dominated strategy, so we would "refine" it away. Of course, the problem with this is that s2-t1 is better for both players than is s1-t2.

Now with reference to this particular game, I am quite willing to entertain the hypothesis that there is no "solution," especially if the players can't communicate. But suppose the players can engage in preplay negotiation. Shouldn't we then expect them to settle on s2-t1? And doesn't this mean that the weak dominance criterion isn't a good refinement?

The argument against player 1 playing the weakly dominated strategy s2 is that player 1, if she isn't sure that player 2 will play t1, doesn't optimize with s2. If player 1 assesses any probability at all, however small, that player 2 will choose t2, then s1 gives a higher expected utility. But the point of preplay negotiation is that player 2 lays to rest the fears of player 1 on this score. Player 1 is given assurances about player 2's behavior.¹⁸ If there is no preplay communication, one can mount a similar

^d My suspicions would *not* be provoked by equilibria in weakly dominated strategies for certain games with infinitely many strategies. In very natural games with infinitely many strategies, all the Nash equilibria involve weakly dominated strategies. See, for example, the discussion of Bertrand equilibrium in section 12.8. This is something of a mathematical artifact of games with infinitely many strategies; hence it is not cause for serious concern.

¹⁸ But the argument, even with preplay communication, is not as simple as this. Player 2 is as much at risk (or more) at the profile s2-t1: What if player 1 plays s1? So we must also consider whether player 2's fears are laid to rest by preplay communication. To repeat from footnote 4, even if player 1 has every intention of playing s1, she also has every incentive

argument: If s_2-t_1 is advanced as the “solution,” then player 1 is supposed to think that player 2 recognizes this. And if player 2 recognizes this, then player 2 will play t_1 . Which leads player 1 to dismiss both her fears and the argument against playing s_2 . And once that argument is dismissed, there is nothing wrong with s_2-t_1 being the solution.

The problem here is one of complete confidence in a “solution” versus being very, very sure, but not certain, that players will see how to play the game and act accordingly. If players are completely certain, then nothing is wrong with playing a weakly dominated strategy. If they are only very, very sure, then they would rationally avoid weakly dominated strategies. So the question comes down to: Is Nash equilibrium appropriate for cases where players are very nearly certain or must it be reserved for cases of complete certainty? If the latter, then the weak-dominance refinement is on weak ground. And even if the former, this refinement is on weak ground in those (perhaps rare) instances with complete certainty.

I prefer to think that there is never complete certainty in the sorts of situations one models with games, and so the appropriate theory is one that involves nearly certain players who still don't rule out entirely any possible action by their opponents. Given this, weak dominance seems a useful refinement of Nash equilibrium.

And, I hasten to add, in the particular example of figure 12.10, my position would be that there is no “obvious solution”; changing the payoffs from figure 12.1(c) in this manner gives us a game that is hard to play, despite the fact that it has a unique equilibrium in undominated strategies. The usual refrain applies: Nash equilibrium, now refined by weak dominance, is a necessary condition for a “solution” *when one exists*.

The tension just observed is the hallmark of all the literature on refinements. For normal form games, one is concerned with how strategies perform if one's opponents take actions that have “zero” probability in the equilibrium; one has to reason about the *relative* likelihood of things that one feels are not going to happen with any significant probability. In terms of extensive form games, this tension takes the following form: Most refinements that are based on extensive form considerations are keyed to what will happen “out of equilibrium” — that is, at points in the game tree that will not be reached if the equilibrium is played. But how can we reason about what will happen conditional on things that are not meant to happen in the first place? If this question seems cryptic now, keep it

to convince player 2 that she will play s_2 (so player 2 will respond with t_1). If player 2 is intelligently sceptical about what he hears, player 1 may have a hard time reassuring him that she really, truly intends to play s_2 .

firmly in mind as we continue through this section. It will be a constantly nagging counterpoint to everything that follows.

12.7.2. Subgame perfection (and iterated weak dominance)

Now we move to extensive games. Recall first the game in figure 12.1(n), reproduced (and turned on its side) in figure 12.11(a). In this game are two Nash equilibria: $U-r$ and $D-l$. Let us be precise about this. First, why is $U-r$ an equilibrium? Because if player 2 is choosing r , then player 1 prefers U to D . And given that player 1 is choosing U , player 2 doesn't care which action he takes. They both give payoff 2, since player 2 won't get the opportunity to move. The profile $D-l$ is also an equilibrium because, if player 2 is choosing l , then player 1 prefers D to U , while if player 1 is choosing D , then player 2 prefers l to r .

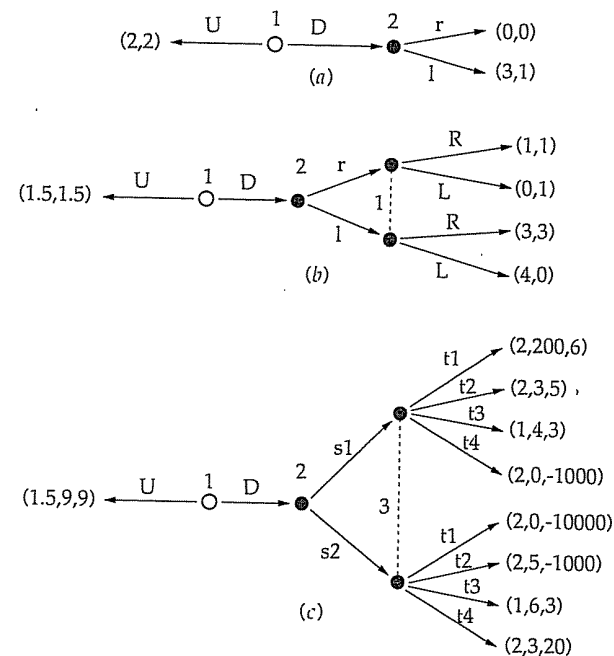


Figure 12.11. Subgame perfection.

This is a good time to make an important observation and set some terminology about Nash equilibria in extensive form games, based on the example of U-r. If player 1 chooses U, then player 2's information set (which is a single node) is not reached. We say that player 2's information set is *out-of-equilibrium* or *off the path of play*. In this case, what player 2 does there is inconsequential to player 2's expected payoff, since there is no chance (given the strategy of the other player) that player 2's choice will matter. To be a bit imprecise, the choice of action of a player at an information set that is off the path of play is unrestricted by the criterion of Nash equilibrium, since that choice cannot affect the payoff of the player who moves there.⁶ But the strategy of a player at an out-of-equilibrium information set can affect what other players choose at *in-equilibrium* information sets, since those players, in evaluating the value to them of other strategies they might employ, have to take into account what will happen if they cause a previously out-of-equilibrium information set to be reached. In the U-r equilibrium, player 1 doesn't choose D because of what player 2 does at player 2's out-of-equilibrium information set.

While both U-r and D-l are Nash equilibria, we argued before that U-r is not a very sensible prediction. If player 2 is given the move, it seems reasonable to suppose that 2 will choose l. We saw before that this argument could be cast as a special case of weak dominance, but now let us pose a different refinement for extensive form games that also captures this argument as a special case.

To do so, consider the games in figures 12.11(b) and (c). Note that in the game in figure 12.11(b), U-r-R is a Nash equilibrium. Given that player 1 chooses U, the action of player 2 at his information set is unconstrained. And given that player 2 is choosing r, U and R are best choices for player 1. And in the game in figure 12.11(c), U-s2-t3 is a Nash equilibrium. The actions of players 2 and 3 are unconstrained; given player 1 chooses U, their information sets are off the path of play. And given that they choose s2 and t3, respectively, player 1 prefers U.

I wish to focus on the fact that in each of these games what follows (from the decision node of player 2) is a game in its own right. That is to say, take the node of player 2, which we call t , and all its successor nodes, the set $S(t)$. First, node t is an information set as a singleton set. Second, every node that follows t is in an information set that involves only nodes that are successors to t . So if any successor node to t is ever

⁶ To be precise, this is true of any information set that is off the path of play even if the player who moves there does everything possible to get play to go there. That is, the information set must be off the path of play by virtue of what others are doing and not simply because the player in question, at an earlier move, is moving away from that information set.

reached, the player who moves at that node will know that node t had been reached earlier. We make the following definition. Recall that for any given node t , $h(t)$ denotes the information set that contains t .

Definition. A *proper subgame* of an extensive form game is a node t and all its successors $S(t)$ that have the properties: $h(t) = \{t\}$; and for all $t' \in S(t)$, $h(t') \subseteq S(t)$.

Suppose we have a "solution" to an extensive form game. We will think of this as a behavior strategy — a prescription of how each player will act at each information set in the game. As always, a "solution" is meant to be a prescription for actions such that every player is very nearly certain that this is how others will act, and every player is nearly certain that others are nearly certain about this. And suppose this game has a proper subgame. Then if the subgame is reached in the course of play, players will be very, very sure what their opponents are doing in the subgame. Because this is a proper subgame, players will be able to evaluate how well they do in the game for the various actions they might take, conditional on the subgame being reached. And it wouldn't be sensible for them to take actions in the subgame which, given what others are doing, give them less expected utility in the subgame than the best they can do. Before moving to examples, we give the formal definition.

Definition. A *subgame perfect Nash equilibrium* for an extensive form game is a Nash equilibrium for the game that, moreover, gives a Nash equilibrium in every proper subgame of the game.

As examples, consider the games in figure 12.11. Take 12.11(a) first and the Nash equilibrium U-r. This is not subgame perfect, because in the proper subgame that begins with player 2's node, the choice of r is not "an equilibrium." That is, starting from his node, player 2 gets 1 from l and 0 from r, and so (conditional upon getting to this node, 2 prefers l.

Take the game in 12.11(b) and the Nash equilibrium U-r-R. In the subgame that begins with the choice of D by 1, r-R is not a Nash equilibrium. Player 1's choice of R is a best response in the subgame, if we assume that 2 is playing r. But against R, in the subgame player 2 prefers l to r. Note well: Player 2 doesn't care about l versus r in the entire game if player 1 begins with U, since then player 2's choice is irrelevant. But if player 2 is put on the move, player 2 knows that his choice is in fact relevant. And given that player 2 is expecting player 1 to play R later on, the better choice is now l. Since r-R is not a Nash equilibrium in the proper subgame beginning with 2's node, U-r-R is not a subgame perfect Nash equilibrium.

And take the game in 12.11(c) and the Nash equilibrium U-s2-t3. In the subgame beginning with player 2's node, s2-t3 is not a Nash equilibrium. And so U-s2-t3 is not subgame perfect.

In games with complete and perfect information, the notion of subgame perfection applies with a vengeance: Since every node is, as a singleton set, an information set, every node begins a proper subgame. It is straightforward to see, for example, that in a finite game of complete and perfect information, if the rollback procedure (i.e., iterated weak dominance) generates no ties, then the strategy profile obtained by rolling back is also the unique subgame perfect equilibrium.

In most applications of game theory to economics, subgame perfection is accepted as an entirely reasonable refinement of Nash equilibrium. More than that, when a given game has a unique subgame perfect equilibrium, it is usually held that this uniqueness is a convincing reason to conclude that this equilibrium is the "solution" of the game. The reader will realize that I most certainly do not ascribe to the second assertion; an example has already been given of a game with a unique Nash equilibrium that, I contend, would almost never be played. And we have an example of a game of complete and perfect information, namely the centipede game, in which I don't find the unique subgame perfect equilibrium to be the obvious solution. But, beyond this, I would contend that there are games for which the outcome is fairly clear, and that outcome, while an equilibrium outcome, is not part of a subgame perfect equilibrium.

The game in figure 12.11(c) is such a game. In case you don't recognize it, the game between 2 and 3 beginning at 2's node is the same as the normal form game in figure 12.1(l). When we examined that normal form game, we concluded (or, rather, I asserted) that it admits no obvious way to play where the standards for an obvious way to play are necessarily very high, since the penalty for being wrong (for player 3 in the game as depicted here) is enormous. My experience with this game, though, leads me to assess a very high probability that player 3 will pick t3 and so, if I were player 1, I would find the choice of U to be fairly obvious. And I would expect most reasonable people put in the role of player 1 would on similar grounds pick U. I comfortably predict that the outcome of this game will be that U is chosen and the game ends.

Notwithstanding this, as we remarked earlier (and you are asked to verify in problem 3), the only Nash equilibria for the subgame involve the play of t1, t2, and t4. If 3 is playing t3 with zero probability, as he must in any equilibrium in the subgame, then 1 will play D. So in every subgame perfect equilibrium of this game (and there are three of them), player 1 chooses D.

What is going on here? The short answer is that application of subgame perfection requires two things: (1) The game must have a "solution" in all parts of the game tree, including those parts of the game that this solution says are unlikely to be reached. For the game in figure 12.11(c), the outcome U isn't a full solution in this sense, since this outcome doesn't say what will happen if 2 is given the move. (And as I maintained before, I don't find any readily apparent solution to the subgame that starts with player 2's node.) (2) One must assume that players act in a way that maximizes the payoffs ascribed to them by the model given their conjectures about the actions of their opponents, even in parts of the game that are unlikely to be reached if the "solution" is correct. In out-of-equilibrium subgames, the "solution" is clearly disproved. Yet in applying subgame perfection we are meant to maintain faith in the validity of the "solution" in whatever remains. In games such as the centipede game, this is a bit hard to accept.

Or to give a still shorter answer, with subgame perfection one assumes that we can reason with confidence about things that will happen conditional on things that "shouldn't" have happened. This assumption should at least be cause for concern.

These short answers don't nearly do justice to this basic point of the philosophy of refinements. To give a better answer, though, requires a substantial digression, especially for the reader for whom this is the first pass through the common types of Nash equilibrium refinements. Hence we will leave matters of this sort here, to be picked up in the last section of chapter 13.

12.7.3. Sequential equilibrium

Throughout this subsection, we continue to work with extensive form games with perfect recall.

Consider the game depicted in figure 12.12(a).¹⁹ One Nash equilibrium of the game is player 1 chooses D; player 2 chooses a; player 3 chooses L. Note why this is a Nash equilibrium. Player 1's choice is optimal given player 2's choice of a and 3's choice of L. Player 3, given 1's choice of D, concludes (when given the move) that she is at the left-hand node in her information set, from which L is optimal. And player 2's choice is optimal because it is irrelevant — given that player 1 is choosing D, player 2's information set (which consists of a single node) is off the path of play.

As we saw above, one has to wonder about choices made off the

¹⁹ This example is originally due to Selten (1975).

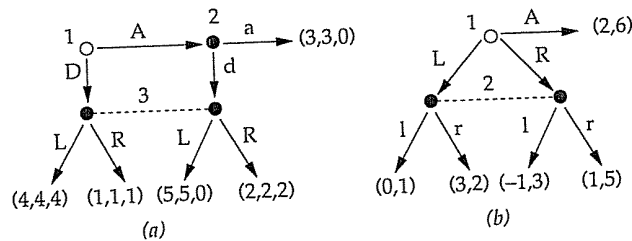


Figure 12.12. Two extensive form games.

path of play. And there is reason to wonder about player 2's choice in this instance. Player 2, if he is given the move and if he maintains the hypothesis that player 3 will choose L, "ought" to choose d. Choosing action a nets 3, while d nets 5 if player 3 chooses L. But if 2 is choosing d instead of a, then player 1 would do better with A than with D. And if player 1 is supposed to choose A and 2 to choose d, then player 3, put on the move, should conclude that she is at the right-hand node in her information set, which leads her to choose R, which then causes 2 to go back to a. (Although now, if 3 is choosing R and 2 is choosing a, player 1 should choose A, and we have a second Nash equilibrium.)

As before, in the equilibrium D-a-L the off-the-path play of a is suspicious. Accordingly, we might try to pose a refinement of Nash equilibrium that eliminates this equilibrium as a potential solution. Subgame perfection as a formal criterion doesn't help us in this game, because the game has no proper subgames. Except for the initial node, only player 2's node is a single-node information set, and one of its successors is joined in an information set to a node that isn't one of its successors. But while subgame perfection doesn't apply, the spirit of subgame perfection seems clear enough here: A player, given the move and assessing that other players will (subsequently) conform to the equilibrium prescription, should choose optimally, even if this player is given the move unexpectedly.

Before trying to formalize this into a new refinement of Nash equilibrium, consider the game in figure 12.12(b). Here we have a Nash equilibrium where player 1 chooses A and 2 chooses l. This is an equilibrium: Given 2 is choosing l, 1 prefers A to both L and R, whereas if 1 is choosing A, 2's choice is off the path of play and so unrestricted. But is 2's choice "reasonable?" Once again, the formal criterion of subgame perfection is no help. Since 2's information set contains two nodes, there are no proper subgames with which to work. And there is a technical difficulty in applying the spirit of subgame perfection to 2's choice: We would like to

evaluate the ex post expected payoff to 2 of choosing r or l, given that 2 is given the opportunity to move. But we can't evaluate 2's expected payoff ex post because we don't know what odds 2 would assess for being at the left-hand or right-hand node in his information set. We can't use Bayes' rule and player 1's equilibrium strategy to get that ex post assessment, because player 2 assesses zero probability ex ante that he will be given the move by player 1. But surely this shouldn't matter: *No matter what assessment player 2 makes concerning which node he is at in his information set, r is better than l.* (And if player 2 chooses r, then player 1 should choose L.)

The notion of a sequential equilibrium is meant to cover these two examples. A sequential equilibrium consists of two pieces. The first piece is, as in any Nash equilibrium, a strategy profile, which (thinking in terms of behavior strategies) prescribes for every information set a probability distribution over the actions available. We will use π to denote a typical strategy profile, with π^i denoting the behavior strategy of player i .

The second piece of a sequential equilibrium is a system of beliefs, denoted by μ . Beliefs assign to every information set h in the game a probability distribution over the nodes $x \in h$, with the interpretation that these are the beliefs of player $i(h)$ about where in his information set he is, given that information set h has been reached. Formally, μ is a function from the set of decision nodes X of the game tree to $[0, 1]$, such that for every information set $h \in H$, $\sum_{x \in h} \mu(x) = 1$.

To take an example, the game in figure 12.12(a) has three information sets. The first, belonging to player 1, has a single node; so 1's beliefs are necessarily that she is at this node with probability one, given that she is given the move. The second, belonging to player 2, also has a single node, so again the only possible beliefs put probability one on that single node. But the third, belonging to player 3, has two nodes, and beliefs for this information set would prescribe the probabilities that player 3 attaches to these two nodes. Or, for the game in figure 12.12(b), it is the two-node information set of player 2 for which beliefs are nontrivial.

Roughly speaking, a sequential equilibrium is a (profile of) strategies π and beliefs μ such that starting from every information set h in the game tree player $i(h)$ plays optimally from then on, given that what has transpired previously is given by $\mu(h)$ and what will transpire at subsequent nodes belonging to other players is given by π . This condition is called sequential rationality.

To see sequential rationality in action, consider first the game in figure 12.12(a), the strategy profile where player 1 chooses D, 2 chooses a, and 3 chooses L, and the beliefs where player 3 at her information set assesses probability 1 that she is at the left-hand node. (As noted above, the

beliefs of players 1 and 2 are trivial.) Player 1's choice is optimal given her trivial beliefs. Player 3's choice is optimal given her beliefs; if she assesses probability one that she is at the left-hand node, then L nets her 4 units while R nets her 1. (Note that 3's choice of L is optimal given any beliefs that assign probability $2/5$ or more to her being at the left-hand node.) But player 2's strategy is not sequentially rational; his beliefs are trivially given, and if he maintains the hypothesis that 3 will play L, then he obtains 5 from choosing d and only 3 from choosing a.

And consider the game in figure 12.12(b), the strategy profile where player 1 plays A and player 2 plays l, and beliefs by player 2 that the left-hand node in his information set has (conditional) probability α . It doesn't matter what α is; player 2's choice of l is not sequentially rational. Given α , he nets expected utility $\alpha + 3(1 - \alpha)$ by playing l and $2\alpha + 5(1 - \alpha)$ by playing r. Since the second amount exceeds the first for any α , there is no way that a choice of l by player 2 can be sequentially rational, and so A-l is not the strategy profile part of any sequential equilibrium.

We are looking here at some very simple examples; do not be misled by their simplicity. In some games there are information sets with more than one node and with actions by other players following; in these games both beliefs (about what has happened so far) and the strategy profile (about what will happen in the future) come into play in checking for sequential rationality. For example, in the game in figure 12.13, suppose we specified that player 1 plays L, player 2 plays d, and player 3 plays D. And for player 2 (who is the only player with a nontrivial information set), we specify beliefs that put probability .3 on the topmost node in his information set. Then d is not sequentially rational for 2: If he chooses d, he nets $(.3)(4) + (.7)(2) = 2.6$, while u nets $(.3)[(0)(2) + (1)(8)] + (.7)(1) = 3.1$. Note well how player 3's choice of D affects the second computation; if player 3's part of the strategy profile were to play D with probability .5 and U with probability .5, we would calculate that the consequences of u for player 2 are

$$(.3)[(.5)(2) + (.5)(8)] + (.7)(1) = 2.2,$$

which is less than what 2 expects to get from d, given his beliefs. Thus, with this change in player 3's strategy and holding 2's beliefs fixed, 2's action becomes sequentially rational. On the other hand, if we maintain this change in player 3's strategy and modify 2's beliefs so that he assesses, say, probability .7 that he is at the topmost node in his information set, then (fixing 3's strategy) his choice of d is once again not sequentially rational. (Work this out!)

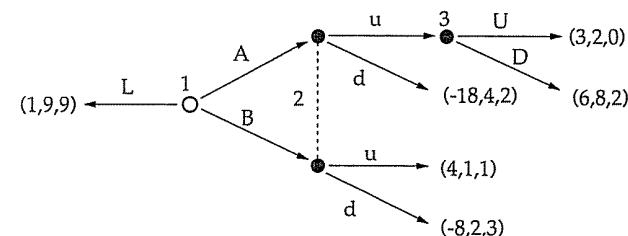


Figure 12.13. An extensive form game.

Sequential rationality is one part of the definition of a sequential equilibrium. In addition, we want beliefs and strategies to make sense together. For example, suppose in the game in figure 12.12(a), player 1 chooses A, player 2 chooses d, and player 3 chooses L. Moreover, player 3's beliefs at her information set are that she is at the left-hand node with probability .95. You can easily check that each player's part of the strategy is sequentially rational. Yet this isn't even a *Nash* equilibrium, let alone any sort of refinement of Nash equilibrium. The problem is that player 3's beliefs are crazy given player 1's and 2's supposed strategies. If the game begins with the play of A followed by d, then player 3 ought to believe given the move that she is at the right-hand node in her information set with probability one. Bayes' rule would certainly tell her this.

At a minimum, we would want to insist that the strategy profile π and the beliefs μ are consistent at the level of Bayes' rule in the following sense: Given the strategy profile π , for any information set h that will be reached with positive probability (if players use the strategies given by π) beliefs at h are computed from the strategies via Bayes' rule.

(In case you are a bit rusty concerning applications of Bayes' rule, let me work an example. Consider the game in figure 12.12(a) and suppose player 1 chooses A with probability $1/3$ and D with probability $2/3$, and player 2 chooses a with probability $1/4$ and d with probability $3/4$. Then, in the course of play, we will come to the left-hand node of player 3's information set with (prior) probability $2/3$ and the right-hand node with (prior) probability $(1/3)(3/4) = 1/4$. That is, there is prior probability $11/12$ that player 3 will be given the opportunity to move (and $1/12$ that the game will end without 3's information set being reached). By applying Bayes' rule, therefore, player 3's beliefs upon being given the move should be that she is at the left-hand node with probability $(2/3)/(11/12) = 8/11$ and at the right-hand node with probability $(1/4)/(11/12) = 3/11$.)

But Bayes' rule will not apply to information sets that are not reached with positive probability in the course of play. And it is precisely at such information sets that beliefs are important.²⁰ So we might want to insist on rather more consistency than just consistency with Bayes' rule when it applies. Suppose, for example, that the game in figure 12.12(a) is modified as follows: Player 4 begins the game, choosing either to end the game (action E) or give player 1 the move (action G), after which the game is just as in 12.12(a). For the strategy profile where 4 chooses E, 1 chooses D, 2 chooses a and 3 chooses R, player 3's information set is not reached with positive probability; player 4, at the start of the game, sees to that. But should we then allow player 3 to have any beliefs at all? Or should 3, given the move, believe that while it is patent that 4 must have chosen G, player 1 is still likely to have picked D, and so her (player 3's) beliefs should put probability one on the left-hand node in her information set?

Let us set some terminology. A behavior strategy profile π is said to be *strictly mixed* if every action at every information set is taken with positive probability. Given a strictly mixed strategy profile π , every information set h will be reached with positive probability, and so Bayes' rule can be used to define a unique (strictly positive) system of beliefs μ that is Bayes' consistent with π .

Definitions. A strategy profile π and beliefs μ are said to be *consistent* in the sense of sequential equilibrium if there is a sequence of strictly mixed strategy profiles $\{\pi_n\}$ such that, for μ_n the strictly positive beliefs that are Bayes' consistent with π_n , the limit of the vector (π_n, μ_n) in n is (π, μ) .²¹

A *sequential equilibrium* is a strategy profile π and a system of beliefs μ that are consistent with each other in the sense above and that satisfy sequential rationality at every information set.

The notion of consistency that is given above may look innocuous, if it doesn't look completely mysterious, but it isn't innocuous at all. (It may well be completely mysterious.) Rather a lot of bodies are buried in this definition of consistency. One can try to unearth the bodies, to see what the definition entails, and in the problem set, I give you a few exercises that will help you do this. (See also the discussion at the end of chapter 13.)

The restrictions implicit in this notion of consistency are held by many to be too severe. Insofar as this is so, one can keep sequential rationality and try to find a less restrictive way of tying together strategies and

²⁰ See problem 7.

²¹ To those of you who don't know what a limit is, I apologize, but there is no other way to do this.

beliefs.^f In particular, you will find in the literature an alternative equilibrium notion called *perfect Bayes equilibrium*, which is the conjunction of sequential rationality (for some beliefs) and weaker notions of consistency.^g

Sequential equilibria are well-behaved:

Proposition. Every sequential equilibrium is a subgame perfect equilibrium.

You will be asked to prove this in the problems.

Clearly, since I attacked the "credibility" of subgame perfection in the previous subsection, I must be willing to attack the credibility of sequential equilibrium as well. And indeed I am, although I will hold off on my attack until the next chapter, noting here that the attack when it comes will be an attack on *both* consistency and sequential rationality.

One reason for attacking the refinements of sequential and subgame perfect equilibrium is easy to give here. Suppose we modify the game in figure 12.12(b) so that player 1's choice of action is done in two stages; first she chooses between A and not A, and if she chooses not A, then she chooses between L and R. This is depicted in figure 12.14. Note that this extensive form is strategically equivalent to the extensive form in figure 12.12(b), in the sense that the two generate the same reduced normal form game. But in this second extensive form, player 1's second decision node does begin a proper subgame. And then, in any subgame perfect equilibrium, player 1 will choose L and player 2 will choose r. That is, in this form of the game, not A-L-r is the unique subgame perfect equilibrium. The same sort of thing is true about sequential equilibrium. If we change the payoff for player 2 in the event of R-r in figure 12.12(b) from 5 to 1, then A-L is a sequential equilibrium

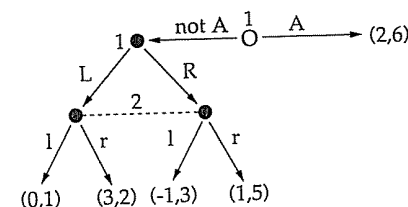


Figure 12.14. A strategically equivalent representation of the game in figure 12.12(b).

^f However some of the reasons for doubting this notion of consistency are also reasons for doubting sequential rationality. For an argument along these lines, see Kreps and Ramey (1987).

^g The notion of perfect Bayes equilibrium was developed contemporaneously with sequential equilibrium, but it has only recently been formalized; see Fudenberg and Tirole (1988).

strategy in the game as depicted in figure 12.12(b), but it would not be part of a sequential equilibrium if we broke player 1's choice among A, L, and R into two pieces as in figure 12.14. (Show this.)

It is not considered desirable to be able to change the set of refined equilibria by changing from one extensive form game to another that is strategically equivalent; this is construed as a weakness of the subgame perfection and of sequential equilibrium refinements. But in their defense remember that these refinements are advanced as necessary conditions for a "solution" to the game in question. Sufficiency is not implied by meeting these refined tests, so that meeting a test in one game form but not another (if the games are strategically equivalent) is likely to be an indication that the equilibrium in question is not a good candidate for the "solution" of the given game.

12.7.4. Restrictions on out-of-equilibrium beliefs

Roughly put, a sequential equilibrium adds to the notion of a Nash equilibrium the requirement that actions taken off the path of play must be rationalized by some beliefs about what has happened up to the point where the actions must be taken. This is only a rough paraphrase, because the consistency of beliefs with the strategy profile does restrict somewhat what range of beliefs can be held. But the rough paraphrase captures the "original intent" of this refinement.

What the bare notion of a sequential equilibrium does not capture, however, are various considerations concerning which beliefs are more or less reasonable. The game in figure 12.15, reproduced (and turned on its side) from figure 12.1(o), provides one example of this.²³ Consider the Nash equilibrium where player 1 chooses U and player 2 chooses r. (Check that this is a Nash equilibrium.) We have to wonder whether 2's choice of r is reasonable; after all, this is a choice of action that is off the path of play if player 1 is meant to choose U. Is this choice of action sequentially rational? Yes, if player 2 holds beliefs that he is equally likely to be at either of the two nodes in his information set, then r is better than l.

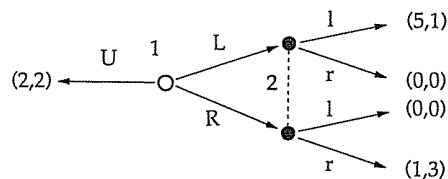


Figure 12.15. An extensive form game.

²³ This game is due to Kohlberg and Mertens (1986).

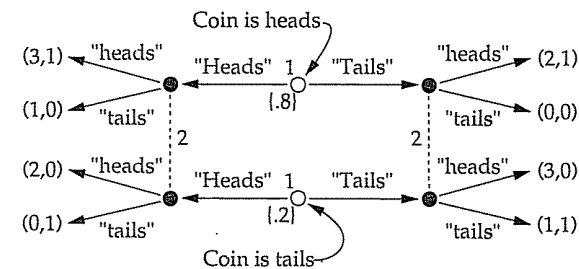


Figure 12.16. The truth game in extensive form.

So we come down to the question: Is it reasonable to suppose that 2 will hold beliefs that make r better than l, given that 2 expects that 1 will choose U? The argument against this is quite simple: Player 1 by choosing U can guarantee herself 2. If 1 chooses R, then the best she can do is 1. If she chooses L, on the other hand, she has a chance of obtaining 5. Shouldn't 2 conclude on this basis, if he is given the move, that player 1 must have chosen L, in an attempt to get the 5? If he concludes this, then he should believe that he is at the topmost node of his information set with probability (close to) one, which would mandate a choice of l instead of r. (And if he is going to choose l, then player 1 will shift from U to L.)

What we have done in this example is to reason what a player's beliefs should be at an information set that is off the path of play. In so doing, we see that a particular sequential equilibrium can only be sustained with beliefs that are not so intuitively sensible. The argument in this example can be made pretty sharply. In the game, U strictly dominates R. So it seems ridiculous for player 1 to be supposed to be choosing R, even as an out-of-equilibrium choice.²⁴ Even if 2 supposes at first that U will be selected, faced with the irrefutable evidence that L or R was selected (namely, being asked to move), 2 ought to conclude that it was L that was chosen after all, which then informs 2's beliefs at his information set.

A somewhat trickier example is given in figure 12.16. This is the "truth game" of problem 4 in chapter 11. (If you did that problem, you can check your answer now.) Consider, in this game, the following Nash equilibrium: Regardless of the result of the coin flip, player 1 says it was "Tails." Player 2 responds "heads" if player 1 says "Tails" and player 2 responds "tails" if player 1 says "Heads."

²⁴ But see subsection 12.7.6 in which a counterargument will be given.

Let's first check that this is a Nash equilibrium. For player 1, given that player 2 will respond to "Tails" with "heads" and to "Heads" with "tails," it is more important to say "Tails" and get the response "heads" than (if the coin flip is in fact Heads) to tell the truth. And for player 2, if player 1 is meant to say "Tails" no matter what the results of the coin flip are, then hearing "Tails" signifies nothing; the chances that the flip was in fact Heads is the same .8 as in the prior. Thus "heads" in response to "Tails" is optimal. While "tails" in response to "Heads" is a best response in the sense that any action taken off the path of play is a best response; since player 1 is meant to say "Tails" there is zero prior probability that player 2 will have to respond to "Heads," and thus all responses to "Heads" are equally good.

As you can imagine by now, we want to think hard about 2's off-the-path response to "Heads"; is it a sensible response? We can ask, for example, whether player 2 might hold any beliefs having heard "Heads" that would make a response of "tails" sequentially rational. That is, have we described the strategy profile part of a sequential equilibrium? The answers to these questions are yes. If player 2, having heard "Heads," believes that the chances the coin was in fact Heads are below .5, then "tails" is the sequentially rational response.²⁵

So we are led to ask, Are these beliefs sensible? If player 2 expects player 1 to say "Tails" and player 1 unexpectedly says "Heads," is it sensible for player 2 to infer that there is a 50% or greater chance that the coin flip came up Tails? An argument to say that this is not sensible runs: Player 1 has more incentive to say "Heads" when the coin flip is Heads than when it is Tails, since player 1 gets a unit of payoff for telling the truth. Whatever conjecture player 1 might make about player 2's responses to "Heads" and to "Tails," player 1 seems no less apt to say "Heads" when the flip is Heads than when it is Tails, or to say "Tails" when the flip is Tails instead of when it is Heads. Hence if player 2 hears "Heads," it seems sensible to conclude that it is no less likely that the result of the flip was Heads than in the prior assessment of .8 for Heads. But if 2's beliefs reflect this, if 2 assesses probability .8 or more that the coin was Heads if she hears "Heads" from 1, then we can't sustain a response by him of "tails" to "Heads," and our equilibrium is destroyed.

In both these examples, the mode of analysis takes the following form: An equilibrium is given, with at least some actions taken at information sets that are not reached if the equilibrium is followed. Hence, in the usual fashion, those actions are not constrained by the criterion of Nash

²⁵ And these beliefs are consistent in the sense given in the previous subsection.

equilibrium. Moreover, we can sustain or rationalize those actions for some beliefs by the player who takes the action. But the beliefs necessary to sustain the actions seem unintuitive or unreasonable. One would wish to formalize what makes a system of beliefs unreasonable; certainly one would wish to be more formal than we have. But this can be done. For example,²⁶

We will restrict attention to a particular class of extensive form games, known as *signaling games*. These games have two players called 1 and 2. Nature moves first, selecting one of a number T of possible initial nodes according to a strictly positive probability distribution ρ . Player 1 is informed of nature's choice and sends a signal to player 2 about this; the set of signals available to player 1 is denoted by S . Player 2 then takes an action a in response to this signal, not knowing which $t \in T$ was chosen by nature, drawn from some set A . Player 1's utility is given by $u(t, s, a)$, and player 2's is given by $v(t, s, a)$, both as functions of nature's choice t , 1's signal s , and 2's response a . Note that the truth game is an example of a signaling game, where $T = \{\text{Heads, Tails}\}$, $S = \{\text{"Heads," "Tails"}\}$, and $A = \{\text{"heads," "tails"}\}$.

An equilibrium for a general signaling game consists of (behavior) strategies for players 1 and 2. We use π_1 to denote 1's strategy, where $\pi_1(s; t)$ is interpreted as the probability that player 1 sends message s if nature chooses t . We require that $\sum_{s \in S} \pi_1(s; t) = 1$ for each t . And we use π_2 to denote 2's strategy; $\pi_2(a; s)$ gives the probability that 2 chooses action a having heard signal s , where $\sum_{a \in A} \pi_2(a; s) = 1$ for each s is required.

A signal s is "along the path of play" if $\sum_{t \in T} \rho(t) \pi_1(s; t) > 0$. That is, there is some circumstance in which s is sent by 1. For any such signal s we can use Bayes' rule to compute 2's beliefs about t given 2 hears s . She assesses probability

$$\mu(t|s) = \frac{\rho(t) \pi_1(s; t)}{\sum_{t' \in T} \rho(t') \pi_1(s; t')}.$$

And a pair of strategies (π_1, π_2) is a Nash equilibrium if

- (a) For each t , if $\pi_1(s; t) > 0$, then s maximizes $\sum_{a \in A} u(t, s', a) \pi_2(a; s')$ (in s')
- (b) For each s that is along the path of play, $\pi_2(a; s) > 0$ implies that a maximizes $\sum_{t \in T} v(t, s, a') \mu(t|s)$ in a' , where μ is computed as above from π_1 by Bayes' rule

Moreover, π_1 and π_2 are part of a sequential equilibrium if (a) and (b) above hold and if, moreover, for any s that is not along the path of play, there

²⁶ Read on at your own risk. What follows will be referred to only in optional reading in chapter 17. But if you plan to be an economic theorist of virtually any stripe, you will have some need to be conversant with what follows, at least until fashions in economic theory change.

are some beliefs $\mu(t|s)$ such that (b) holds. Of course, these beliefs must agree with those that are computed by Bayes' rule and are the beliefs part of the sequential equilibrium. (One can verify that the restrictions of consistency do not restrict the off-the-path beliefs.)

Now suppose we have sequential equilibrium strategies π_1 and π_2 . (That is, some beliefs off the path rationalize the off-the-path prescriptions of π_2 .) We will examine whether the beliefs accompanying these strategies are intuitively sensible.

Suppose for some $t \in T$ and s and $s' \in S$,

$$\min_{a \in A} u(t, s, a) > \max_{a \in A} u(t, s', a).$$

Then we would know that any strategy that involved 1 sending s' if nature picks t is strictly dominated by a strategy where 1 sends s instead. It doesn't matter how 2 responds to s or s' ; whatever his (contingent) response, 1 does strictly better with s than with s' . Accordingly, we would wish to restrict 2's beliefs so in the event of s' being sent he doesn't put any weight on t being nature's choice. Why (this logic goes) would 1 send s' if nature picks t , when the strictly dominant s is available? Formally,

Sequential equilibrium strategies π_1 and π_2 fail the test of dominated messages if for some t , s , and s' as above, it is impossible to sustain 2's equilibrium response to s' with beliefs that put zero conditional probability on t given s' .

(Note that since s' is dominated given t , it can't be that $\pi_1(s'; t) > 0$. Hence if s' is along the path of play, beliefs conditional on s' computed by Bayes' rule will put zero probability on t to begin with. This test will be more restrictive than the simple sequential equilibrium test only if s' is off the path.)

This test does not apply to the truth game, since no message is dominated by another, no matter what nature does to start the game.⁹ But we can formulate a stronger test that does work for the truth game. Fixing equilibrium strategies π_1 and π_2 , for each $t \in T$ let $u^*(t) = \max_{s \in S} \sum_{a \in A} u(t, s, a) \pi_2(a; s)$; that is, $u^*(t)$ gives player 1's conditional expected payoff in the equilibrium, conditional on nature picking t to start the game. Then say that s' is *equilibrium dominated at t* if $u^*(t) > \max_{a \in A} u(t, s', a)$. That is, maybe no alternative signal s dominates s' as before, but the best that can be had by sending s' is dominated by what 1 gets in the equilibrium, conditional on t .

Sequential equilibrium strategies π_1 and π_2 fail the test of equilibrium domination if for some t and s' that is equilibrium dominated at t , it is impossible to sustain 2's equilibrium response to s' with beliefs that put zero conditional probability on t given s' .

The equilibrium where 1 sends the signal "Tails" no matter what she learns about the coin flip fails this test. To see this, note that in this equilibrium, if the coin flip comes up tails, then player 1 will get a payoff of 3 in the equilibrium;

⁹ However see problem 12.

she says "Tails," which is the truth, and 2 responds with "heads." The best 1 can do if she says "Heads" is 2. So, at this equilibrium, $s = \text{"Heads"}$ is equilibrium dominated at $t = \text{tails}$. Hence to pass the test, we have to be able to sustain 2's reaction to $s = \text{"Heads"}$ with beliefs that put zero weight on $t = \text{tails}$ given "Heads." And this, as we noted in our informal arguments, we cannot do.

For comparison with the literature, you should know that the test of equilibrium domination is referred to in the literature as "the intuitive criterion" on grounds that it is such an intuitive test that it surely must be passed by any reasonable solution. Notwithstanding this bit of editorializing by its inventors, the justification for using the equilibrium domination test in applications is deservedly quite controversial. I will not try to recount the arguments for and against here; instead I give references at the end of this chapter. However, so that you can better consume the literature, it is useful to know at least one further thing (which is also covered in the references given below): The test of equilibrium domination is often further strengthened; when you find mention of *divine* or *universally divine* equilibria, the author is invoking one of these strengthenings.

12.7.5. Trembling-hand perfection

In the previous few subsections we worked with extensive form games. Now we turn back to considerations based on the normal form.

We have stated numerous times that a "solution" to a given game is advanced as a strategy profile in which we and, more to the point, all the players have a lot of confidence as a prediction about how everyone will play. What we haven't done, in any complete or rigorous sense, is to say what explanation might be offered for some player deviating from this solution. One could tell many stories (and we will see a few of them at the end of the next chapter), but the simplest is that players sometimes make mistakes. A player faced with a choice of several actions, one of which he means to play, might through inattention or a slip of the hand (or the pen, or the tongue) take another.

Now if the probability of these mistakes is significant, we wouldn't have that much faith in our "solution" as a nearly certain prediction. But since mistakes do happen, we would probably wish to dismiss predictions that aren't "safe" against the risk of a mistake.

The standard example of an equilibrium in weakly dominated strategies provides the simplest example. Take the game in figure 12.1(c). The profile s_2-t_1 is a Nash equilibrium; player 1 is happy with s_2 as long as there is no chance of t_2 . But if there is any chance of t_2 , and in particular if there is some chance, however small, of player 2 playing t_2 by mistake, then s_2 is not optimal. It may be nearly optimal, but a truly cautious

player 1, worried that player 2 might slip, would opt for s1. And player 2, understanding this, responds with t2.

We capture this idea with the following formal definition. Fix an I player normal form game with finitely many strategies for each player. Let S_i be the set of strategies available to player i , for $i = 1, \dots, I$. Let Σ_i denote the set of mixed strategies available to player i , with typical element σ_i . Let $\sigma = (\sigma_1, \dots, \sigma_I)$ denote a typical mixed strategy profile. Let $\bar{\Sigma}^i$ denote the set of *strictly mixed* strategies for n — each $\sigma^i \in \bar{\Sigma}^i$ is a mixed strategy that attaches positive probability to each pure strategy in S^i .

Definition. A Nash equilibrium profile $\sigma = (\sigma^1, \dots, \sigma^I)$ is **(trembling-hand) perfect (in the Normal form)** if there is for each player i a sequence of strictly mixed strategies $\{\sigma_n^i\}$, such that

(a) $\lim_{n \rightarrow \infty} \sigma_n^i = \sigma^i$ for each i , and

(b) for each i and $n = 1, 2, \dots$, σ^i is a best response against the profile of strategies

$$(\sigma_n^1, \dots, \sigma_n^{i-1}, \sigma_n^{i+1}, \dots, \sigma_n^I),$$

which we hereafter write as σ_n^{-i} .

The key to this definition is that the σ_n^j must all be strictly mixed. Player i intends to play σ^i , and he assumes that his fellow players intend to play σ^{-i} . But his fellow players, or so the story goes, are susceptible to making mistakes. If they intend to play σ^{-i} , in fact they play σ_n^{-i} for “large” n , a profile in which any combination of actions by i ’s opponents is possible, but in which i ’s opponents are “nearly” sure to be playing according to their intentions. (The last assertion follows from [a].) And the equilibrium condition is that what i intends to do (if he doesn’t slip) must be a best response against what his opponents actually do (which is [b]), and not simply against what they intend to do (which is the Nash criterion).

We can say lots of things about this definition. First, why is there this limit in n ? If players make mistakes, it would seem that we should have functions $\phi^j : \Sigma^j \rightarrow \bar{\Sigma}^j$, which give as a function of the intentions of j the actual (mistakes-included) play of j . We want j to be “local” — it shouldn’t move too far in translating j ’s intentions into j ’s actual actions. But players play against a definite, nonvanishing chance of errors; not against this fancy vanishing sequence. One is queasy, however, to fix ad hoc functions ϕ^j . We are looking for necessary conditions, remember. If

in the context of a particular game, we knew something about the precise “technology of errors,” then we could sharpen our necessary condition by building into our model this technology. (For an example, read on.) This definition imposes the *minimal* requirement (for our story) that our “solution” can be rationalized by *some* technology of errors, no matter how small we think the possibility of errors is.

Second, we require that each σ^i is optimal against σ_n^{-i} for each n . Although this wasn’t made explicit, it is implicit that σ_n^{-i} is made up of independent mixed strategies by i ’s opponents. This means, in particular, that the chances (and type) of an error by one of i ’s opponents is independent of errors by other opponents. The assumption of independent errors is conventionally made, but depending on what one thinks of as the reason errors are made, one might wish to allow correlation in errors. Fancy versions of trembling-hand perfection allow for this.

Third, it is implicit in this definition that all the players have the same notions about how their fellow players slip, when those fellow players do slip. That is to say, for each player j a single sequence of strictly mixed strategies σ_n^j is used to form the “test” for the intended σ^j , for each j . One can, once again, think of modifications that would remove this implicit assumption.

Fourth, the definition refers to this as (trembling-hand) perfection. Note the parenthetical modifier. The originator of this concept, Selten (1975), uses the simple name “perfection” without the modifier. But to distinguish from Selten’s earlier (1965) concept of subgame perfection, convention has been to use the modifier “trembling-hand,” which refers to slips of the hands that act.

Fifth, we motivated this notion by an example of an equilibrium in a weakly dominated strategy. And you should find it easy to prove, for two-player games, that no weakly dominated strategy is ever part of a trembling-hand perfect equilibrium.^h But trembling-hand perfection doesn’t extend to exclude strategies that are excluded by iterated weak dominance. For an example, take the normal form game depicted in figure 12.17(a). Note that Ba for player 1 is weakly dominated by Bb. Having removed Ba, Y weakly dominates X. And if player 2 is playing Y, then player 1 chooses Bb. Nonetheless, A-X is a trembling-hand perfect Nash equilibrium: There is no question that if player 2 is playing a strategy “close” to choosing X for sure, player 1 is doing the best that she can

^h There is an analogous statement for games with more than two players that you can try to dope out on your own. For a hint, go back to the early sections of this chapter and find references to separating hyperplanes.

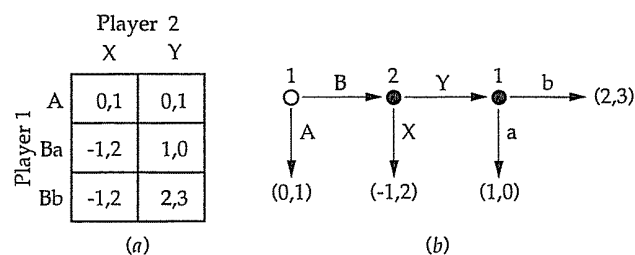


Figure 12.17. A game in normal and extensive form.

with A. And against the strictly mixed strategy where 1 plays A with probability $1 - 2/n$, Ba with probability $1/n$, and Bb with probability $1/n$, X for player 2 nets for himself a payoff of $(1)(1 - 2/n) + (2)(1/n) + (2)(1/n) = 1 + 2/n$, while Y nets $(1)(1 - 2/n) + (0)(1/n) + (3)(1/n) = 1 + 1/n$. So X is better. Letting n go to infinity, we pass the test of trembling-hand perfection.

Sixth (and finally), the definition refers to (trembling-hand) perfection (in the normal form), where now you should note the second parenthetical modifier. Consider the extensive form game in figure 12.17(b). This game is an extensive form version of the game in 12.17(a). Note that it is a game of complete and perfect information, so it has a solution by backwards induction, which (of course) is its solution by iterated weak dominance: Bb-Y. This is, of course, the unique subgame perfect equilibrium. Hence we conclude from the previous paragraph that trembling-hand perfection does not imply subgame perfection.

However, think a moment about those mistakes that players are meant to be making. Player 1 has two opportunities to make a mistake, at least in the extensive form game. If either by intent or by mistake player 1 chooses B and 2 replies with Y, then it is clearly a *further* mistake for player 1 to choose a. If we imagine that, whatever player 1 does at the start of the game, she is much more likely to choose b than a if the game gets out that far, then 2 should intend to choose Y, and 1 should intend to choose B. What happened to our previous argument that A-X is trembling-hand perfect? To make that argument, we needed the chances of Ba by mistake to be of the same order of magnitude as Bb by mistake. But if we think that Ba compounds two mistakes and Bb only one, and if (in consequence) we think that Ba must be much less likely than Bb, even if both are very unlikely (because 1 intends to choose A), then we can't sustain the choice of X by 2.

Proceed formally as follows. For any extensive form game, think of constructing a new extensive form game in which every information set of every player in the game belongs to a distinct player; that is, there are as many players in this new normal form game as there are information sets. Letting h denote a typical information set and thus a player in this new extensive form game, in this new extensive form game we let h 's payoff function be precisely the payoffs of $i(h)$ in the original game. Think of the following story: Player i may "control" more than one information set, but he assigns his decision-making authority at each one to a different agent. These agents of i all have the same payoffs as i , so they act in his interests. But we regard each as an individual player. And then we pass to the normal form of this new extensive form game, which is called the *agent normal form* of the original extensive form game.

And now look for trembling-hand perfect equilibria of the agent normal form game. Since we implicitly assumed in the definition of trembling-hand perfection that players tremble independently, we would, by applying that definition to the agent normal form, be assuming that the various "agents" of a given player tremble independently; and thus in terms of the original extensive form, we are assuming that a mistake at one information set by i does not affect the chances of a mistake at any other. In particular, in terms of the example in figure 12.17(b), even if 1's first agent intends A, Ba by 1's two agents must be much less likely than Bb, because b by 1's second agent is a second mistake.

Selten (1975) reserves the term (trembling-hand) perfect equilibrium for this version of the definition — perfection in the agent-normal form. We follow Selten in this regard and use the second modifier "in the normal form" to denote perfection where one doesn't use the agent-normal form of an extensive form game. You are asked to prove the following in problem 13:

Proposition. Every (trembling-hand) perfect Nash equilibrium of a given extensive form game is a sequential equilibrium, and hence it is subgame perfect.

One can also show that every (finite) extensive form game has a trembling-hand perfect equilibrium; indeed, every finite normal form game has an equilibrium that is trembling-hand perfect in the normal form. This is a bit harder to show.ⁱ

ⁱ And still harder to show, but demonstrating the close connection between normal form and extensive form refinements, for "most" extensive form games "most" of the sequential equilibria are trembling-hand perfect. You'll need to consult Kreps and Wilson (1982) to see what this means precisely.

12.7.6. Proper equilibria and stable sets of equilibria

Reconsider (once again) the game from figure 12.1(o), reproduced in this section as figure 12.15. The equilibrium U-r is, as we noted previously, sequential; it is also trembling-hand perfect. We argued earlier that as a sequential equilibrium it was not too sensible; to support the play of r by 2, 2 must hold beliefs that he is at the node that follows R with probability $1/4$ or more. But R is dominated by U for 1 (and L is not). Hence it didn't seem reasonable for 2 to hold these out-of-equilibrium beliefs.

Now that we have a story concerning why one might see off-the-path play, namely such play results from mistakes, we can further investigate this argument. Myerson (1978) presents a strengthening of perfection, called *properness*, which throws doubt on this argument.

Myerson reasons as follows: If player 2 is playing r, then player 1 nets 0 from L and 1 from R. Of course, playing U and getting a payoff of 2 is best of all for 1. But given 2's supposed play of r, the play of L is a "bigger" mistake for player 1 than the play of R, in the sense that it is more costly. If we assume that players are more likely to make smaller mistakes than bigger, it might be reasonable to assume that player 2, put on the move, thinks it at least as likely that 1 erred with R as that 1 erred with L. And, in this case, player 2 optimally responds with r.

I will not give here the precise definition of Myerson's properness. Roughly, it is like trembling-hand perfection, with the addition that one assumes that if one strategy by a given player is a bigger mistake than a second (as measured by expected payoffs at the supposed "solution"), then the first of these strategies is much less likely to be played by mistake. Properness, then, takes the basic notion of trembling-hand perfection and adds to it a notion of which mistakes are more or less likely to be made. Proper equilibria are defined for the normal form of a given game (although one could adapt the definition for the agent normal form). As we see in the example of figure 12.15, there are games solvable by iterated dominance, where equilibria other than the solution so derived are proper.

The mathematically most complex weapon in the arsenal of Nash refinements is Kohlberg and Mertens' (1986) concept of *stability*. Kohlberg and Mertens, noting earlier efforts, including those sketched above, construct the following "wish list" for a solution concept:

- (1) Every (finite) game should have a solution.
- (2) The solution should be trembling-hand perfect.
- (3) The solution should depend only on the reduced normal form of the game. Changing the extensive form should not matter, if no "strategic" changes are thereby entailed.

- (4) Players should never be playing (weakly) dominated strategies.
- (5) Any solution should survive iterated application of dominance.

They show by example that these requirements cannot possibly be met by any solution concept that prescribes a single strategy profile, and they argue that the appropriate response to this difficulty is to move to a "set-valued" solution concept; that is, a set of strategy profiles comprises a solution. This leads to modification of (2) and (5); (2) is written: Every solution should contain a perfect equilibrium. And (5) becomes: The solution of a game should contain the solution of a game derived by the deletion of dominated strategies in the first game. And they add a sixth requirement:

- (6) The solution (which is now a set of strategy profiles) should be topologically connected.

Perfection, as we noted above, fails on requirements (3) and (5). Properness (defined for the normal form) fails (5). Kohlberg and Mertens, by moving to a set-valued solution concept, provide a number of (quite complex) formal definitions that meet various subsets of these desiderata; *stability* is used to denote a concept that meets requirements (1), (3), (4), and (5). Very roughly put, the nature of these definitions is that a "solution" should be stable against a very large class of perturbations to the game; that is, a solution is a set of strategy profiles so that, if one perturbs the game a bit (so that, say, all the players are slightly susceptible to trembling in some manner), then the perturbed game has an equilibrium that is close to some profile in the solution.

I will not attempt to give even a summary formal treatment of this. Consult either the paper by Kohlberg and Mertens (1986) or the book of van Damme (1987). In addition, you can consult Banks and Sobel (1987) or Cho and Kreps (1987) for connections between stability and the sort of refinement posed in 12.7.4, based on restrictions on out-of-equilibrium beliefs.

12.8. Reprise: Classic duopoly

Now that we know all about game theory and Nash equilibria, we can complete our treatment of the classic models of duopoly. We left things in a state of bewilderment concerning the origins of all those different sorts of conjectures and with a suggestion that the "rules" of competition might help us resolve that bewilderment. More specifically, we suggested that, for example, Cournot conjectures made sense in the following sort of situation:

Two producers work in isolation preparing the quantity they bring to market. These quantities are decided upon with a foreknowledge of this market structure and with knowledge of the characteristics of demand, but neither side gets to see how much the other is producing. Each producer brings the quantity it produced to a central marketplace, where it is handed over to a "state sales commission"

that sets price so that the total amount brought to market is just demanded at the set price.

The point we made before is that if competition has this sort of structure, then it is sensible to hold to the (Cournot) conjecture that one producer will react to a change in the level of output of its rival by holding fixed its quantity, since the first producer must fix its quantity without seeing precisely what its rival has done.

On the other hand, the following structure of competition was meant to justify von Stackelberg conjectures:

We have the same two producers as above, and the same state sales commission, but one producer is able to produce its quantity first. That is, this producer brings to the market a quantity x_1 , which the second producer observes. And then the second producer decides on an amount x_2 to produce, knowing that price will be set so that the market clears at total quantity $x_1 + x_2$.

Now the structure of competition makes it reasonable to suppose that the second producer will take as given the quantity decision of the first producer, since this quantity decision is in fact fixed, and optimize. And the first producer, realizing this, picks its quantity accordingly.

And, on a third hand, we considered the following competitive structure as justifying Bertrand conjectures:

Things work much differently in this case. The two producers, independently, call the state sales commission and give a price at which each is willing to sell. The state sales commission investigates market demand at those prices and then calls the two producers, telling them how much they are obligated to bring to the market.

Now Bertrand conjectures are sensible, because firms must fix their prices simultaneously, without any opportunity to react to what their opponents do.

Of course, each of these three "stories" describes an extensive form game between the two firms. In the first, the first producer selects a quantity, and "then" the second producer selects a quantity without knowing what the first producer has done. We have the extensive form game depicted in figure 12.18(a). (A complication here and in the rest of this section is that each firm has infinitely many actions it can take; infinitely many quantities can be selected in this case. But one can be careful about the formalities and nothing untoward will happen.) The von Stackelberg game is just like the Cournot game, except that the second firm sees the action of the first before picking its quantity; we have the game depicted

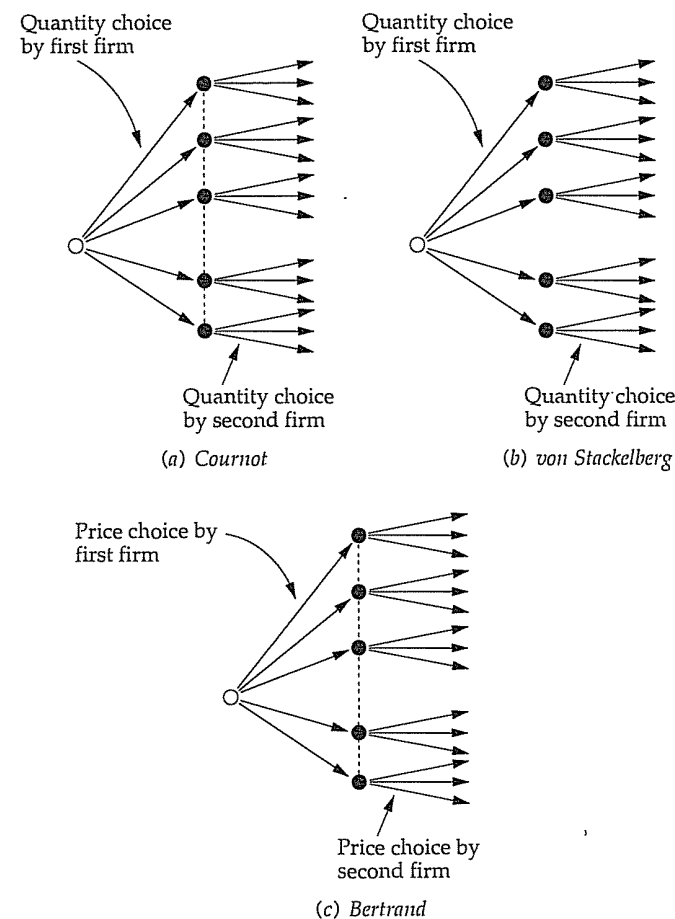


Figure 12.18. The Cournot, von Stackelberg, and Bertrand extensive form games.

in figure 12.18(b). And the Bertrand game is just like the Cournot game, except now price is the action chosen by each firm.

Actually, there is one complication in the Bertrand game. In our analysis in chapter 10, we didn't say precisely how demand would be allocated

between the two firms if they charged the same price. To make up a game, we need to specify payoffs for every terminal node, and the firms' payoffs will, in places, depend on how they divide the market when they charge the same price. It is customary, and we will follow custom in this, to assume that the market divides symmetrically — half of demand going to each firm.

Somewhat naturally, we find that the Cournot, Bertrand, and von Stackelberg equilibria (in conjectures) are Nash equilibria of the respective games. In fact, holding to the parameterization of linear demand $P = A - X$ and constant unit costs k , we can say the following:

Concerning Cournot, the Cournot equilibrium is the unique Nash equilibrium, and the game is even dominance solvable. To see this, note that no firm would ever produce more than $A - k$, because to do so would drive prices below k and profits below zero for sure. Indeed, firms would never produce more than $(A - k)/2$, because whatever (nonnegative) amount your rival produces, you are better off with $(A - k)/2$ than with some higher amount. But if your rival will never go above $(A - k)/2$, then you would never produce less than $(A - k)/4$, nor would your rival, since (against any amount less or equal to $(A - k)/2$, $(A - k)/4$ is better than anything less. And if your rival won't go less than $(A - k)/4$, you will never go more than $3(A - k)/8$. (Check my math here.) The same is true for your rival, and if your rival won't go above $3(A - k)/8$, you will never go less than $5(A - k)/16$. And so on, until everything is eliminated except $(A - k)/3$ for both of you. This takes *infinitely* many iterations of dominance, because there are infinitely many other strategies to get rid of. But in the end, only the Cournot quantities of $(A - k)/3$ for each of you survive.²⁷

Concerning Bertrand, again we have a unique Nash equilibrium (where each firm charges $P = k$). The argument we gave in chapter 10 is easily adapted to show this. One notices in this regard an extraordinary thing: The strategy of naming the price k is weakly dominated by a strategy of naming any other higher price. Note why this is so. Naming $P = k$ means profits of zero no matter what your opponent does. Naming, say, $P = k + 1$ could mean positive profits, if your opponent names this price or one higher, and $P = k + 1$ can never lead to negative profits. (Of course, it leads to zero profits if your opponent follows the equilibrium strategy of naming $P = k$.) The unique Nash equilibrium involves weakly

²⁷ This argument transcends the specific parameterization we've used. But pressing this argument to its utmost generality is not to my mind very interesting; for reasons to be given in just a bit, I don't find the Cournot game all that interesting.

dominated strategies. This sort of thing sometimes happens with games that have infinitely many strategies, and, if you are unduly concerned about it (now that I've mentioned it), it would be a good idea to prove for yourself that it won't happen if the range of available prices that can be charged is finite — say prices must be in integer multiples of one penny.

Concerning von Stackelberg, there are many Nash equilibria in this game. For example, the second producer could adopt the strategy: Produce $A - k - x_1$ units if $x_1 \neq 0$, and produce $(A - k)/2$ if $x_1 = 0$. One best response for the first producer to this is to set $x_1 = 0$: The first producer makes zero profits, but given the second producer's strategy, it can't do any better. And given that the first producer sets $x_1 = 0$, the second producer's strategy is a best response; what it does for any x_1 that isn't zero is out-of-equilibrium, and it is doing the optimal thing against the chosen x_1 . But, if you followed section 12.7, you will see immediately that this Nash equilibrium isn't subgame perfect. In any subgame that begins with a choice of x_1 different from zero, the second producer doesn't optimize by choosing $x_2 = A - k - x_1$. Indeed, the von Stackelberg game is a game of complete and perfect information, and we can use backwards induction to show that the unique equilibrium so obtained (and that is subgame perfect) is the von Stackelberg equilibrium we computed in chapter 10. You should be clear on why this is: In our analysis of von Stackelberg, we assumed that the second firm would react "optimally" to firm 1's choice of quantity, no matter what that quantity was. But this is just an expression of backwards induction (or subgame perfection).

The three "games" given above don't come close to exhausting the possibilities. For example,

Back to our two producers who produce quantities independently. But now they call the state marketing commission in advance, saying how much they plan to bring. And if one says it will bring more than the other, the second is given permission by the state marketing commission to bring up to the amount being brought by the first.

Or how about the following very complex situation.

The two producers must first install capacity for production. Capacity is costly, coming at a cost of $k_1 > 0$ per unit capacity installed. Capacity is installed by the two simultaneously and independently. Then each sees how much capacity the other installed. And then each calls the state marketing commission and quotes a price at which it is willing to sell. Production is possible up to the level of capacity, at an additional marginal cost of $k_2 \geq 0$ per unit. No production

beyond the level of capacity is possible. If a capacity constraint does bind when prices are named, the state marketing commission rations demand by some system or other. This good is resalable among consumers, and all consumers are aware of market demand conditions.

If you like a challenge, try to find all the subgame perfect equilibria for these two games.²⁸

And we can go on from here. We can consider all sorts of variations of these stories including: choices of technologies prior to selection of quantities or of prices; installation of capacity when capacity can only be installed at a finite rate; considerations of how much information to reveal to one's opponent about demand conditions or one's own cost function, if there is initial uncertainty about these things, and so on, and on. Starting in the late 1970s, there was an enormous amount of work of this sort, where the objective was to use these sorts of oligopoly "games" to chase down how competitors would fare in equilibrium under different structural conditions. In all of these, the rivals eventually had to compete, and the model of competition more often than not was one of the three given above. So when you see a paper that says "the firms compete Cournot style," the author usually means that the firms select quantities simultaneously and independently, and price is set so that markets clear. When "firms compete Bertrand style," they name prices simultaneously. When the competition is von Stackelberg, with one leader and (possibly many) followers, the leader chooses quantity first (usually), and then the others react with optimal quantity responses (usually simultaneously if there is more than one follower, although sometimes you'll see one firm pick its quantity, then a second, then a third, and so on). In the problems, you will get a taste of these sorts of game-theoretic models of oligopoly.

So finally we come to the big question of chapter 10: Which of the classic models of duopoly is "correct?" The obvious response at this point is it depends on the details of competition. For real-life oligopolists, therefore, the answer seems to be that none of the models for the structure of competition given above is very realistic, so the classic models explained in this fashion don't give us much guidance. For one thing, firms don't name quantity or price once and for all; they name price or set quantity one day and can reset either the next, reacting perhaps to what their rivals did the previous day. This dynamic character of competition in oligopoly makes an enormous difference to the set of possible equilibria, whether

²⁸ Be forewarned that the second game does give a very nice answer: The unique equilibrium is the Cournot equilibrium for costs $k_1 + k_2$. But showing this is quite difficult. See Kreps and Scheinkman (1983) for the details.

the oligopolists set quantities on a given day, or name prices on a given day, or do some of each. It is to this sort of complication that we turn in chapter 14.

Before leaving this topic again, let me anticipate chapter 14 and offer a word in defense of the classic models of duopoly. The "games" described above are not necessary for the predictions of the classic models to be valid. It suffices for those predictions that rivals have the right sorts of conjectures about their opponents and that an equilibrium is reached. These game-theoretic models say why those sorts of conjectures might be reasonable, but there is nothing to prevent players from holding particular conjectures even if those conjectures are unjustified by the structure of competition. Indeed, we'll see in chapter 14 that adding dynamic considerations to the structure of competition leads us back to having to wave our hands about the origins of the conjectures that players hold. In the end, the test of the classic models must be empirical; this sort of theory may be useful for refining the empirical predictions that we will test, but one shouldn't reject a prediction on grounds that there is reason to doubt the "best" theoretical model one can think of to justify the prediction.

12.9. Bibliographic notes

Many references have been scattered throughout this chapter that I will not repeat here. Some of the basic ideas in this chapter such as dominance, dominance solvability, and Nash equilibrium are covered in many venerated textbooks, and I will not attempt to provide references to those. Other topics that have been covered here are the subject of ongoing research, so references I might provide are apt to become dated quite quickly. With those caveats in place, I would recommend the following partial reading list to anyone interested in beginning to pursue these ideas.

For the use of dominance and iterated dominance criteria, begin with Bernheim (1984) and Pearce (1984).

Correlated equilibrium was only mentioned in passing, but the serious student should certainly consult Aumann (1974).

The various rationales for Nash equilibrium are scattered in many places. But Schelling (1980) is an absolute must-read for anyone interested in noncooperative game theory. Schelling is more verbal and less mathematical than most of what you will see in the literature of game theory, but his book bursts with ideas that await higher levels of formalization. As for models of "learned" behavior, my favorite (predictably) is Fudenberg and Kreps (1989).

Concerning refinements of Nash equilibrium, van Damme (1987) presents a unified and up-to-date treatment. The subject was pioneered by Selten (1965, 1975), who defined subgame perfection and (trembling-hand) perfection. Myerson's properness (1978) came next. Sequential equilibrium is analyzed in Kreps and Wilson (1982), stability is developed by Kohlberg and Mertens (1986), and perfect Bayes equilibrium is defined in Fudenberg and Tirole (1988). The literature on formal restrictions on out-of-equilibrium beliefs was begun by McLennan (1985); Banks and Sobel (1987) and Cho and Kreps (1987) present analyses along the lines sketched in 12.7.4. Objections to refinements based on the equilibrium domination are given in Okuno-Fujiwara and Postlewaite (1986).

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12.10. Problems

- 1. Prove that in any game of complete and perfect information with finitely many nodes, there is always at least one decision node whose immediate successors all are terminal nodes.
- 2. Prove that if a game is dominance solvable, then the "solution" so determined is a Nash equilibrium. Prove that if a game is dominance

solvable using only strict dominance at each step, this solution is the only Nash equilibrium that the game possesses.

■ 3. For the game in figure 12.1(l), we gave in the text two pure and one mixed strategy equilibria. Show that this game possesses no other equilibria, either mixed or pure.

■ 4. Prove that in the centipede game (figure 12.6), every Nash equilibrium begins with player 1 playing D with probability one.

■ 5. A correlated equilibrium for a two-player normal form game can be thought of as a probability distribution over the space of strategy profiles such that for each player and for each of the player's pure strategies (that have positive probability) the strategy is a best-response given the player's opponent chooses according to the conditional distribution on the opponent's strategies, conditional on the strategy for the first player. Find all the correlated equilibria for the game in figure 12.9.

■ 6. (a) For the game in figure 12.13, let α be the probability assessed by 2 that he is at the topmost node in his information set, given that he is given the move, and let β be the probability he assesses that player 3 chooses U. For which pairs of α and β is the choice of u by player 2 sequentially rational?

(b) For this game, what is the set of all sequential equilibria?

■ 7. Consider an extensive form game in which no player ever moves more than once along any path through the game tree. (If you wish, take the simple case of a game in which each player has a single information set.) Suppose we have behavior strategies π and beliefs μ that are Bayes' consistent in the sense: Given the strategy profile π , for any information set h that will be reached with positive probability (if players use the strategies given by π) beliefs at h are computed from the strategies via Bayes' rule. Prove: If π is sequentially rational given μ at every information set that is reached with positive probability given π , then π is a Nash equilibrium for the game. Is the requirement that no player ever moves more than once necessary? (Hint: the answer to the last question depends on your precise definition of sequential rationality.)

For the converse, suppose we have a Nash equilibrium π for a given extensive form game. Compute beliefs at information sets that are reached with positive probability given π by Bayes' rule. Show that at these information sets π and the beliefs computed are sequentially rational.

Problems 8 and 9 concern the notion of consistency in sequential equilibria. They are for the pathologically mathematically addicted only.

■ 8. Fix a strategy π for an extensive form game and beliefs μ that are consistent with π in the sense of sequential equilibrium.

(a) In any proper subgame of the game, we can speak of applying Bayes' rule within the subgame at information sets h in the subgame that are reached with positive probability, conditional on the subgame being reached. Prove that μ is consistent with Bayes' rule in every proper subgame, in just this fashion.

(b) Define an *almost-proper subgame* to be an information set h and all the successor nodes to nodes in h , which will be denoted $S(h)$, with the property that if $x \in S(h)$ and $x' \in h(x)$, then $x' \in S(h)$. For an almost-proper subgame whose "root" is h , given a specification of beliefs over nodes in h and a strategy profile π , one can use Bayes' rule to compute beliefs on information sets in $S(h)$ that are reached if π is played, conditional on h being reached according to beliefs μ on h . Show that μ is consistent with Bayes' rule in every almost-proper subgame in this sense.

■ 9. Fix a (finite) extensive form game and a behavior strategy π for the game. For simplicity, consider games without moves by nature. Define a π -labeling of the game as a pair of functions $\kappa : A \rightarrow \{0, 1, \dots\}$ and $\zeta : A \rightarrow (0, \infty)$, where A is the set of actions for the game, such that (a) $\kappa(a) = 0$ if and only if $\pi(a) > 0$ and (b) $\zeta(a) = \pi(a)$ if $\kappa(a) = 0$. (Follow the convention that every action is taken at a single information set in the tree.) Given a π -labeling, define two new functions $\lambda : X \rightarrow \{0, 1, \dots\}$ and $\xi : X \rightarrow (0, \infty)$, where X is the set of decision nodes, as follows: If x is the initial node, $\lambda(x) = 0$ and $\xi(x) = 1$. (Recall that there is a single initial node, since we've assumed no moves by nature.) Otherwise, $\lambda(x)$ is the sum of all $\kappa(a)$ for actions a that lead from the initial node to x , and $\xi(x)$ is the product of the $\zeta(a)$ for the same sequence of actions. For a given information set h , let $\lambda(h) = \min_{\{x \in h\}} \lambda(x)$. And for a node $x \in h$, let $\mu(x) = 0$ if $\lambda(x) > \lambda(h)$ and let $\mu(x)$ be given by

$$\frac{\xi(x)}{\sum_{\{x' \in h: \lambda(x') = \lambda(h)\}} \xi(x')}$$

if $\lambda(x) = \lambda(h)$.

(a) Prove that for any π -labeling, if we define μ in this fashion, μ is consistent with π in the sense of sequential equilibrium.

(b) (Very hard!) Prove that if μ and π are consistent in the sense of sequential equilibrium, then there is a π -labeling such that μ is obtained by the procedure above from this labeling.

In other words, this labeling procedure characterizes all the beliefs that are consistent with a given π . (If you are very ambitious, you might try to modify things so that moves by nature are allowed.)

■ 10. Prove that every sequential equilibrium is a subgame perfect Nash equilibrium. (If you prove that it is a Nash equilibrium, you are 90% of the way there.)

■ 11. Consider the class of signaling games given in subsection 12.7.4. Prove that any beliefs that are Bayes consistent with a given strategy profile are consistent in the sense of sequential equilibrium. (Don't use problem 9 unless you've proven it.)

■ 12. Consider the truth game of figure 12.16.

(a) The sequential equilibrium in which player 1 always says "Tails" fails the test of equilibrium domination. Find an equilibrium that passes this test.

(b) Write out this game in normal form. Show that the game is dominance solvable, if we allow dominance in mixed strategies. Can you modify the payoffs of the game so that it is not dominance solvable but still equilibrium domination is effective in eliminating a sequential equilibrium?

■ 13. Prove that every trembling-hand perfect equilibrium (for the agent normal form) is sequential. Show by example that the converse is false.

The remaining problems in this chapter concern game-theoretic variations on the classic models of oligopoly. They are primarily about those models and so relate more to material in other chapters, but they could not have been posed until we had covered Nash equilibria and subgame perfection.

■ 14. Prove that in the Bertrand game, if prices must be charged in integer multiples of a penny, then there is always at least one Nash equilibrium in which players do not use weakly dominated strategies.

■ 15. Mr. X and Ms. Y have entered into the following contractual arrangement with D. M. Kreps. Between now and the end of next week, each will install a certain amount of capacity for the manufacture of square based tri-prong slot poiuyts. Then Kreps will assume control of the capacity provided to him and produce and sell poiuyts. Demand for poiuyts is given

by the demand function $D = 10 - P$. (To keep matters simple, assume that production takes place for a single period.) The marginal cost of production of poiuyts is zero up to the total level of capacity that Mr. X and Ms. Y install. That is, if Mr. X installs capacity k_x and Ms. Y installs capacity k_y , then Kreps can produce up to $k_x + k_y$ poiuyts and sell them, at zero additional cost. Kreps can, however, produce less if he so chooses.

The contract signed by X, Y, and Kreps calls for Kreps to act in the best interests of X and Y. (He receives a small fixed fee for doing so, which you may ignore.) So that there is no ambiguity about what are the best interests of X and Y, the contract calls for Kreps to set output so that gross profits (= revenue) are maximized. (Kreps, who knows a lot about linear demand systems and zero marginal cost production functions, can be counted on to do this.) These gross profits are split between X and Y in proportion to the capacities they build. That is, if $k_x = 2k_y$, then X receives two-thirds of the gross profits and Y receives one-third.

Capacity is costly for X and Y. Precisely, there is a constant marginal cost of 2 per unit of installed capacity. Hence X's net from the entire arrangement, if he installs k_x , Y installs k_y , and Kreps produces $Q \leq k_x + k_y$, is

$$\frac{k_x}{k_x + k_y}(10 - Q)Q - 2k_x,$$

and similarly for Y.

(a) Suppose that X and Y collude in terms of the capacities they install. That is, k_x and k_y are set to maximize the sum of the net profits of X and Y from this arrangement. (Assume, for the sake of definiteness, that $k_x = k_y$.) What capacities will X and Y install?

(b) (Good luck!) Unhappily, X and Y cannot (or do not) collude. They install capacity levels k_x and k_y "Cournot" style, choosing their capacity levels simultaneously and independently. What is (or are) the equilibrium (or equilibria) in this instance?

(c) If demand is $P = A - Q$ and the marginal cost of capacity is r , can you find any values of A and r such that the total amount of capacity installed in an equilibrium exceeds A ?

■ 16. Since you know all about perfect competition, monopoly, and oligopoly, we can find out how various types of firms might feel about uncertainty concerning the prices of its factors of production and output. Consider a profit-maximizing

firm that produces a single good from several factors. The firm is characterized by a production function $y = f(x_1, \dots, x_n)$, where y is the level of output obtainable from factor inputs $x_1, \dots, x_n$. We will use p to denote the price of the output good, and w_i to denote the price of factor input i . When there is uncertainty a priori about these prices, the firm is allowed to choose its production plan *after* any uncertainty in prices resolves.

(a) Suppose that the firm is competitive in all its markets. That is, it takes prices $p, w_1, \dots, w_n$ as given. Consider two possible price vectors, $(p, w_1, \dots, w_n)$ and $(p', w'_1, \dots, w'_n)$, and let $(p'', w''_1, \dots, w''_n)$ be the simple average of the two: $p'' = (p + p')/2$, $w''_i = (w_i + w'_i)/2$, etc. Let π be the firm's level of profits in case of the prices $p, w_1, \dots$; π' be the firm's profits in case of $p', w'_1, \dots$; and π'' be the firm's profits in case of $p'', w''_1, \dots$. Show that $\pi'' \leq (\pi + \pi')/2$. (That is, the firm, in terms of expected profits, is risk seeking in the prices it faces.)

(b) Now we suppose that the firm is a monopolist in its product market, but is competitive in all its factor markets. Hence the firm itself determines p, p' , etc. Suppose that it makes profits π when facing factor prices $(w_1, \dots, w_n)$, π' when facing $(w'_1, \dots, w'_n)$, and π'' when facing $(w''_1, \dots, w''_n)$, where $w''_i = (w_i + w'_i)/2$. Can we say that $\pi'' \leq (\pi + \pi')/2$? Provide a proof, if so. If you need to make a further assumption to get this result, state the assumption and say why you need it. If there is no hope of getting this result, indicate why that is.

(c) Finally, consider this question in the context of a von Stackelberg duopoly. Two firms produce an undifferentiated commodity for which demand is given by $P = A - X$, where P is price and X is total supply. Demand is unchanging. Each firm has production technology with a fixed cost F for producing anything at all, and a (constant) variable cost c for producing every unit. Note well that the fixed cost F can be avoided by producing zero. Firm 1 moves first, deciding whether to produce anything and, if it does produce, the amount x_1 that it will supply. Firm 2 sees the decisions of firm 1, and then it decides whether to produce anything at all and, if so, how much. (In cases where firm 2 is indifferent between two courses of action, you may assume that it acts in a way that maximizes the profits of firm 1.) The firms regard the "factor costs" F and c as given to them; their actions do not affect those costs.

Is the result analogous to the result in part (a) of this problem (and in part (b), if it is correct there) correct in this case for firm 1? That is, for all (F, c) and (F', c') , is the firm's average of profits given those two levels of costs at least as large as its level of profits when costs are $((F + F')/2, (c + c')/2)$?

If so, can you conjecture how general the general proposition at work here is? If not, can you succinctly state what went wrong? (One way to do this is to give the "general proposition" in a fashion that applies to those cases [(a) and, perhaps, (b)] where the result is true, and fails to apply in this case.)

■ 17. Part (a) should be easy. Part (b) will introduce you to how uncertainty might be handled in the classic models. Part (c) foreshadows some developments to come, and you might well miss the point of it now.

(a) Consider a duopoly where the two firms in question produce an undifferentiated product. Demand for the product is given by $P = A - X$, where X is the total output of the two firms. Each firm has a simple cost function: $TC_i(x_i) = c_i x_i$ where the subscript i refers to the firm, x_i is the firm's output, and c_i is a constant with $0 < c_i < A$. The firms compete Cournot style. What is the equilibrium? Give equilibrium price, levels of output, and profits. (You may assume that $A + c_1 > 2c_2$ and $A + c_2 > 2c_1$ throughout this problem, although if you want a real challenge, try doing the entire problem without this assumption.)

(b) Now suppose that while the two firms are described as above, firm 2 is uncertain about the cost function of firm 1. Specifically, c_1 takes on one of a finite number of values, namely $c_{11}, c_{12}, \dots, c_{1n}$ with probabilities $p_1, \dots, p_n$. Firm 2 holds this probability assessment when it chooses its level of output x_2 ; firm 1 when it chooses x_1 knows the value of c_1 (and is allowed to condition its level of x_1 on c_1). Otherwise, the two engage in Cournot competition. By this we mean: An equilibrium is a quantity $\hat{x}_2$ output by firm 2 and, for each level of firm 1's costs, a quantity $\hat{x}_1(c_1)$, such that firm 2, believing that firm 1's output will be given by the function $\hat{x}_1$, maximizes its *expected profits* with output level $\hat{x}_2$, and firm 1, believing that firm 2 will have output $\hat{x}_2$ and knowing that its own costs are given by c_1 , would choose to produce $\hat{x}_1(c_1)$.

(c) Suppose the setting is as in (b), but prior to the selection of x_1 and x_2 , firm 1 is able to provide irrefutable evidence to firm 2 concerning its cost structure if it chooses to do so. What do you think will happen in this case?

■ 18. Now reconsider problem 17, assuming that the two duopolists produce differentiated commodities. (It will be good enough to assume that demand is given by the demand system in problem 20.) Consider also what happens if the two engage in Bertrand competition in this case.

■ 19. Xyllip is a controlled substance that, by law, can be produced by only

two companies, ABC Chemical and XYZ Pharmaceutical. The production of Xyllip for each firm involves combinations of kapitose and legume according to the production function

$$X = k^{1/2}l^{1/2},$$

where X is the level of Xyllip output, k is the level of kapitose input, and l is the level of legume input. (It will be okay to assume that these two firms produce only Xyllip; this isn't really true, but taking into account their other products wouldn't change the answers you'd get.) The two firms are price takers in the markets for kapitose and legume, each of which sell for \$1 for purposes of this problem. There are no fixed costs in the production of Xyllip. Demand for Xyllip is given by the demand function $P = 14 - X$.

(a) As a warm-up exercise: ABC and XYZ act as Cournot duopolists in the sale of Xyllip. That is, you can assume that they simultaneously and independently name quantities they will supply to the market, and price is set so that markets clear. What is the equilibrium in the Xyllip market in this case?

A third firm, Kinetiks, has found a way to synthesize a chemical known as Legusude. Kinetiks holds a patent on this process, and it is the only firm capable of producing Legusude. Legusude has only one possible use; it can be used in place of legume in the production of Xyllip. If one uses Legusude instead of legume, the production function for Xyllip is given by

$$X = k^{1/2}L^{1/2},$$

where now L is Legusude input to the production process. We will discuss fixed costs in the manufacture of Xyllip from Legusude (and kapitose) in a moment. The production of Legusude is (for simplicity) very simple; Kinetiks can make as much Legusude as it wishes at zero marginal cost. (There are some fixed costs in the manufacture of Legusude, but you should ignore them throughout.)

Kinetiks has announced that they are willing to supply Legusude to ABC and to XYZ. Kinetiks is not allowed to produce Xyllip, so its only market for selling Legusude is to ABC and XYZ.

To produce Xyllip from Legusude instead of from legume, ABC and/or XYZ must convert their production processes. This is a straightforward

operation, but involves a fixed cost incurred by any firm that converts; the fixed cost is \$1. If a firm (or both) converts so that it can use Legusude, it retains the option of using legume if the price of legume is below that of Legusude. But even if the firm that has converted uses legume, it must pay the fixed cost.

Imagine the following sequence of events, after Kinetiks has announced its ability to produce Legusude. ABC and XYZ must simultaneously and independently decide whether to convert their production processes so that they can utilize Legusude. This is an irrevocable decision: If a firm decides to convert, it thereafter incurs the fixed cost of \$1. The conversion decisions of the two firms are observed by all participants, and then (if either of the firms has decided to convert), Kinetiks announces a price at which it is willing to sell Legusude. There can be no bargaining over price between Kinetiks and ABC and/or XYZ. But, at the same time, Kinetiks is required by law to set a single price and sell as much Legusude at that price as anyone wishes to buy. The price being charged by Kinetiks is observed by all participants, and then ABC and XYZ simultaneously and independently engage in production of Xyllip; that is, once production conditions are known for the two firms, they compete Cournot style.

(b) Give an extensive form "representation" of this game, using the sort of symbolic pictures of figure 12.18.

(c) We are interested in knowing what the subgame perfect equilibria (or, if there is only one, equilibrium) of this game are. To this end, suppose that in stage 1, ABC and XYZ both convert and then Kinetiks announces price q for Legusude. What is the outcome in the final stage of Cournot competition between ABC and XYZ? Given this, what price q should Kinetiks name for Legusude, if both firms have converted?

(d) Repeat step (c) for the case where the game begins with ABC deciding to convert and XYZ deciding not to convert.

(e) What are the subgame perfect equilibria (or equilibrium) of this extensive form game?

(f) Suppose that you are the CEO of Kinetiks. What might you do to change the "rules" of this strategic interaction so that you do better than you do in (e)? You should not answer this question by saying that you should go into the production of Xyllip; that is forbidden. But you should try to invent plausible changes in the rules of the "game" and then investigate their consequences for the profits of Kinetiks.

■ 20. Consider a two-firm industry where each firm has a single output, and the outputs of the two firms are reasonably close substitutes for one another. More precisely, if x_1 is the level of output of firm 1 and x_2 the level of output of firm 2, then the price that firm 1 gets in the market is $p_1 = 10 - x_1 - .5x_2$, while firm 2 gets price $p_2 = 10 - x_2 - .5x_1$.

Firm 2 has a very simple technology in which every unit produced has constant marginal cost 2. Firm 1 will soon have a constant marginal cost technology, but first must choose which technology to install. In fact, firm 1 has a choice of four technologies. The first technology has fixed costs of 12 and a constant marginal cost of 1. The second has fixed costs of 8.1 and a constant marginal cost of 2. For the third, fixed costs are 5 and marginal costs are 3. For the fourth, fixed costs are 2.7 and marginal costs are 4. Firm 1 must select which of these four technologies to install; having done so, the two firms will compete in duopolistic fashion.

(a) Imagine that the sequence of events and competitive variables are: Firm 1 chooses one of the four technologies to install; firm 2 sees which technology firm 1 has chosen; firms 1 and 2 then simultaneously and independently choose output quantities. (That is, they engage in Cournot competition.) What are the Nash equilibria of this game, including the technology selection stage? If you find more than one equilibrium, do you find any one of them particularly persuasive, and why?

(b) Suppose that the sequence of events and competitive variables are: Firm 1 chooses one of the four technologies to install; firm 2 sees which technology firm 1 has chosen; firms 1 and 2 then simultaneously and independently choose prices that they quote. (That is, they engage in Bertrand competition. Please recall that these products are differentiated, so it isn't the case that Bertrand competition must lead to zero profits.) What are the Nash equilibria of this game, including the technology selection stage? If you find more than one equilibrium, do you find any one of them particularly persuasive, and why?

(c) Imagine that the sequence of events and competitive variables are: Firm 1 chooses one of the four technologies to install; firm 2 *does not* see which technology firm 1 has chosen; firms 1 and 2 then simultaneously and independently choose quantities to output. (That is, they engage in Cournot competition.) What are the Nash equilibria of this game, including the technology selection stage? If you find more than one equilibrium, do you find any one of them particularly persuasive, and why?

(d) Suppose that the sequence of events and competitive variables are:

Firm 1 chooses one of the four technologies to install; firm 2 *does not* see which technology firm 1 has chosen; firms 1 and 2 then simultaneously and independently choose prices that they quote. (That is, they engage in Bertrand competition.) What are the Nash equilibria of this game, including the technology selection stage? If you find more than one equilibrium, do you find any one of them particularly persuasive, and why?

Throughout this problem, restrict yourself to pure strategy Nash equilibria only. You will probably find the results a bit surprising. Can you explain what is going on here?

■ 21. Imagine an N firm oligopoly for "nominally differentiated" goods. That is, each of the N firms produces a product that "looks" different from the products of its competitors, but that "really" isn't any different. However, each firm is able to fool some of the buying public. Specifically, each of the N firms (which are identical and have zero marginal cost of production) has a captive market — consumers who will buy only from that firm. The demand generated by each of these captive markets is given by the demand function $P_n = A - x_n$, where x_n is the amount supplied to this captive market and P_n is the price of the production of firm n . There is also a group of intelligent consumers who realize that the products are really undifferentiated. These consumers are represented by the demand curve $P = A - X/B$, where P is the price of goods sold to these consumers and X is their demand. (If $x_n > A$ or $X/B > A$, then the prices in the respective markets are zero. Prices do not become negative.)

Firms compete Cournot style. Each firm n supplies a total quantity X_n , which is divided between its loyal customers and the customers who are willing to buy from any firm. If we let x_n be the part of X_n that goes to loyal customers, then the price of good n is necessarily $P_n = A - x_n$. The price that goes to the "shoppers" is $P = A - (\sum_n (X_n - x_n))/B$. In an equilibrium, $P_n \geq P$, and if $X_n > x_n$, then $P_n = P$. (That is, a firm can charge a "higher than market" price for its own output, but then it will sell only to its own loyal customers.) By Cournot competition, we mean that each firm n assumes that the other firms will hold fixed their total amounts of output.

(a) For a given vector of output quantities by the firms, $(X_1, \dots, X_N)$, is there a unique set of equilibrium prices (and a division of the output of each firm) that meets the conditions given above for equilibrium? If so, how do you find those prices? If not, can you characterize all the market equilibria?

(b) Find as many Cournot equilibria as you can for this model, as a function of the parameters N and B . (WARNING: Do not assume that solutions of first-order conditions are necessarily global maxima.)

chapter thirteen

Incomplete information and irrationality

In this chapter we present a methodological innovation, the notion of a *game of incomplete information*, which has been extremely useful in the modeling of all sorts of economic phenomena. As an example of this sort of application, we discuss again the problem of entry deterrence in monopoly. We use games of incomplete information to model situations in which one or more of the players is suspected of being "irrational." And with this tool at our disposal, we return a final time to questions concerning refinements of Nash equilibrium.

13.1. Games of incomplete information

The following story provides an excellent example of what we are after. Imagine two individuals, whom we will call (without much imagination) player 1 and player 2. Player 2 unhappily is something of a bully; he likes to pick fights with cowards. But he is equally averse to picking a fight with someone with courage; he will pick a fight with someone only if he assesses probability .5 or more that this individual is a coward. (He is indifferent between picking and not picking a fight at the assessment .5.) He considers picking a fight with player 1, but he is uncertain whether player 1 is a coward or is brave. He initially assesses probability .8 that player 1 is brave and .2 that player 1 is a coward (these being the only possibilities), and so it seems that he will not pick a fight with this individual.

Before deciding whether to pick a fight with player 1, player 2 gets a signal about player 1's disposition that may be of value, namely what player 1 will have for breakfast. Player 1 will have one of two breakfasts: beer or quiche. If player 1 is brave, then she prefers a breakfast of beer to a breakfast of quiche. If player 1 is a coward, she prefers quiche to beer. So, it might seem, player 2 should observe what player 1 has for breakfast and conclude: If 1 has quiche, she is certainly a coward; pick a fight. But if

she has beer, she is certainly brave; avoid a fight. But complicating this is the fact that whether brave or a coward player 1 prefers that player 2 not pick a fight with her, and she cares more about this than she does about the type of breakfast she has. And player 1 realizes that player 2 is lurking at the street corner, watching carefully what player 1 has for breakfast.

We wish to create a game theoretic model of this situation to analyze what player 1 "should" have for breakfast and how player 2 "should" react to this signal. The difficulty we encounter is that player 2 is uncertain about some of the characteristics of player 1: What are player 1's dispositions in terms of a fight? Or, equivalently, what are player 1's preferences concerning breakfast? And we haven't said much about how player 1 regards all this. Does player 1 know that player 2 is uncertain about player 1's tastes? The last line of the previous paragraph seems to indicate that this, at least, is so. But this leads to more questions. For example, does player 1 know precisely what odds player 2 assesses that player 1 is brave?

In answer to these questions, consider the models of this situation given by the games in figures 13.1(a) and 13.1(b). Start with 13.1(a). Here we have modeled the situation as follows. Nature determines whether player 1 is brave or is a coward, with probability .8 that 1 is brave. Player 1 knows her own temperament, and player 1 must choose (perhaps contingent on her temperament) which breakfast to have. Player 2 observes this breakfast choice but not the initial choice of nature and then must decide whether to pick a fight with player 1 or not. Then payoffs are made. Note that player 2's payoffs are designed so that player 2 gets one unit of utility if he picks a fight with a coward or avoids a fight with a brave opponent, and player 2 gets zero if he does the "wrong" thing. Thus, as we said before, player 2 will pick the fight if he assesses probability more than .5 that his opponent is a coward; he avoids the fight if this assessment is below .5; and he is indifferent at the assessment .5. Player 1 gets two units of utility if she avoids the fight, to which is added one unit of utility if she has her preferred breakfast (depending on her disposition, which is determined by nature).

In this model, we have used a move by nature to capture the idea that player 2 is uncertain about the characteristics of player 1. Nature begins the game by selecting those characteristics. This selection of characteristics is known to player 1, and (*please note carefully*) player 1 is aware as well of player 2's initial uncertainty and assessment. Moreover, player 2 is aware of the fact that player 1 is aware of these things, and so on.

Compare this model with the extensive form game in figure 13.1(b). Here are four initial nodes, which we label bo, bp, co, and cp. The first letter in these labels refers to the type of player 1 — either brave or a

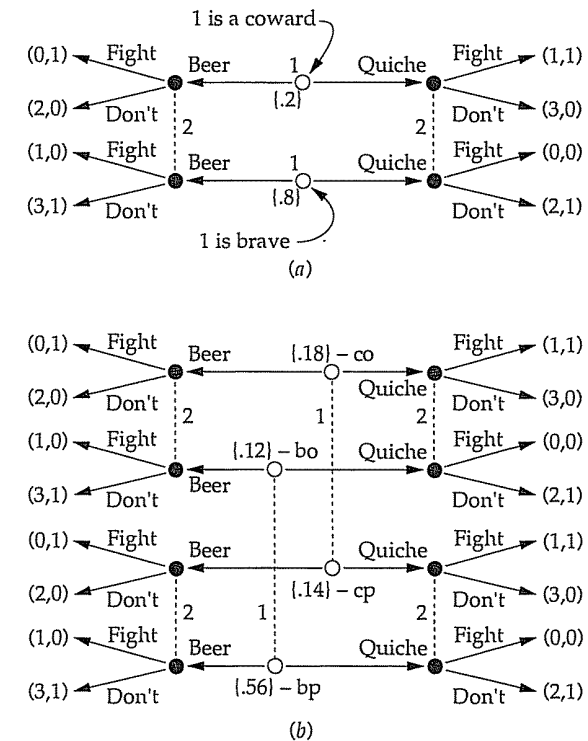


Figure 13.1. Two versions of the beer-quiche game with incomplete information.

coward. Now in this game, we have player 1 choosing her breakfast knowing the first letter of the initial node but not the second. That is, {bo, bp} is one information set for player 1, and {co, cp} is another. And then player 2 chooses whether to pick a fight based on two pieces of information — whether player 1 had beer or quiche for breakfast, and whether the second letter of the initial node is o or p. That is, player 2 has four information sets: {bo-beer, co-beer}; {bp-beer, cp-beer}; {bo-quiche, co-quiche}; and {bp-quiche, cp-quiche}.

This second extensive form game is a model of the following situation. Player 2 has precisely the sorts of doubts that he has in the first model — whether player 1 is brave or a coward. But now player 1 has doubts too. She knows that player 2 is uncertain about her character.

But we have modeled things so that player 1 is unsure about player 2's assessment. Player 1 thinks that player 2 may initially be pessimistic, assessing probability .8 that player 1 is brave (which is, in fact, the situation), or that player 2 may initially be optimistic, assessing probability .4 that player 1 is brave. Player 1 assesses probability .7 that player 2 has the first, pessimistic assessment, and .3 that player 2 has the second, optimistic assessment. The letters o and p in the initial nodes are mnemonics for optimistic and pessimistic. So the initial node bo is: nature begins the game with a brave player 1 and an optimistic player 2. This node, note, has probability .12, obtained as the product of the probability that player 2 is optimistic as assessed by player 1 and the probability that player 1 is brave as assessed by player 2, if player 2 is, in fact, optimistic. Note that in the model player 1 always knows (whenever it is her turn to move) whether the game started at a b-class initial node or a c-class initial node; player 1 always knows her disposition. And player 2, whenever he is called upon to move, knows whether the game started at an o-class initial node or a p-class node; he knows his own initial assessment. Finally, the model "closes" here; player 1 knows what sorts of assessment player 2 might be making, and player 2 knows that player 1 knows this. Player 2 knows the odds that player 1 assigns to the chances that player 2 makes various assessments, and player 1 knows that player 2 knows these odds, and so on.

These are examples of what are called *games of (or with) incomplete information*. These games are used to model situations in which some players have more or less information at the outset than do others. One player will know her own utility function. A second will know what options he has available. A third will have assessments about those two things, and the first two players may have assessments about those assessments. And, if we wanted something really complex, we could imagine that player 3 has an assessment concerning player 2's assessment of player 3's assessment of player 1's utility function. (If you think this is painful to read, imagine having to draw the corresponding extensive form.)

The form of the model is: Nature acts first to determine all these things, and then players' "initial" states of information are set by giving them varying amounts of information about nature's initial choices. These models are not completely general. In particular, they presume:

(a) There is a single, consistent "base model" in the minds of all the players. Everyone agrees on the possible initial states among which nature chose. No player is completely ignorant about something another player knows, although one player may assess very low (conditional) probability

to certain states that other players know pertain for sure. (We will see later how you can model your way around this restriction.)

(b) Players agree on the prior distribution on initial states. This is implicit in the model; it is planted when we put at the initial nodes a single probability distribution representing the (unanimous) assessment of how nature starts things off. There is no reason to make this assumption in theory. We could, for example, assume that different players have different priors over nature's initial move, although we would still assume that each player is aware of the priors assessed by all others. (Note well: This isn't to say that each is aware of the posteriors assessed by others. We saw how one player can be unsure what assessment a second player makes, after nature selectively reveals some of what was done in getting the game underway.) But it is conventionally assumed that players have a common prior — more of the religious doctrine about which we spoke in chapter 2 and again in chapter 11.

(c) The model "closes" at a certain point. By this we mean the following. Suppose that player 2 is unaware of some characteristic of player 1. Player 2 will then have a probability assessment over this characteristic. We see in figure 13.1(b) how to model the notion that player 1 is uncertain about player 2's assessment. And we could further complicate the model and have player 2 uncertain about player 1's assessment concerning player 2's assessment. And we could have player 1 uncertain about 2's assessment about 1's assessment about 2's assessment. (Such a model would be quite intractable, so you needn't worry that we will ever look at such a thing.) But, in the end, this chain ends — say with player 1 knowing 2's assessment of 1's assessment of 2's assessment of 1's assessment of 2's assessment of 1's unknown characteristic. In figure 13.1(a) the model closes rather sooner than that; player 2 had an assessment over player 1's character, and player 1 knew 2's assessment. In figure 13.1(b) the model closes a step further along.

This sort of closure is theoretically unnecessary. With some very high-powered mathematics, one can show that an infinite regress of assessments is modelable. But, speaking practically, a model that doesn't close (and quite quickly) would be useless in applications, so the models that people write down and analyze almost always close quickly.

Despite these limitations, this way of modeling situations of incomplete information has proven very powerful in generating insights into behavior in competitive situations. The proof of that pudding will only

come in the eating, however, so without too much further ado we proceed to an application of this sort of model.

But before doing so, we make one short remark. The model given in figure 13.1(a) may remind you of a game you have seen earlier; it is the "truth game" described in problem 3 in chapter 11 and discussed and depicted in subsection 12.7.4. The story has changed, but the two games, as extensive form games, are entirely isomorphic. This provides a good opportunity to introduce some conventional language. In the story about the truth game, the randomization by nature (the flipping of the bent coin) was entirely within the game. Because player 1 was told the outcome of the flip and player 2 wasn't, the two players were put on different informational footings. This sort of situation is called, in the literature, a *game with imperfect information*. In contrast, in the beer-quiche story nature is invoked as an external artifice to get the players from some never-relevant point of common information to a "starting point" where one player knows something about which another is uncertain. This sort of situation is conventionally called a *game with incomplete information*. As mathematical models, the two are the same; the different names refer (somewhat vaguely, to my mind) to the sort of story that is being told.

One further point on this: It is commonplace to regard the truth game as a two-player game, with player 1's expected payoffs computed by averaging over the two possible outcomes of the coin flip. No one ever quibbles with this. But in beer-quiche, things are not so straightforward. Is this a two-player game, or should the two different manifestations of player 1 be treated as different players, so that this is in some sense a three-player game? Making interpersonal comparisons of utility between a cowardly player 1 and a brave player 1, which is what we would do if we took expectations over player 1's disposition, is suspicious at least. For certain game-theoretic considerations this would make a difference; cf. problem 1. There is no set conventional wisdom on this matter; for what little it is worth, my prejudices are to regard this as a three-player game.

13.2. An application: Entry deterrence

To illustrate the uses to which this sort of model has been put in the literature, we briefly consider the problem of entry deterrence by a monopolist that was discussed in section 9.2.¹ This is not meant to be anything like a complete treatment of entry deterrence; consult the relevant papers or, say, Tirole (1988) for that. But, at the same time, to introduce

¹ Since there will be both a monopoly firm and an entrant, it is convenient to use personal pronouns of different genders. Hence we will refer to the monopolist as *she* and to the potential entrant as *he*.

the reason for this application we will have to wander a bit far afield of the main subject of this chapter. Your forbearance is, therefore, requested.

To begin, recall the simple story of section 9.2. A monopolist faces demand given by $D(p) = 9 - P$ and has a constant marginal cost production technology with marginal cost of production equal to \$1. The monopolist must also pay a fixed cost of \$2.25. The standard theory of monopoly holds that this monopolist will equate marginal revenue, or $9 - 2X$ at output level X , with marginal cost; this gives $X = 4$, $P = \$5$, and profits of \$13.75.

Despite this, the story goes, we find that this monopolist sets $X = 5$, which leads to $P = \$4$ and profits of \$12.75. When pressed to explain this, the monopolist says:

A single potential competitor is sitting out there who is considering entry into this market. This competitor has exactly the same production technology as I do. If this entrant believes that I will produce four units, the entrant faces residual demand of $5 - X_2 = P$, which leads the entrant to produce $X_2 = 2$ if he enters, depressing price to \$3, and leaving him with net profits of $\$4 - \$2.25 = \$1.75$. Hence this competitor will enter. And if he enters, I will make net profits of $\$8 - \$2.25 = \$5.75$.

But if I produce five units and the entrant believes that I will continue to do this, then the entrant faces residual demand of $4 - X_2$, which leads him to produce $X_2 = 1.5$ if he enters. This pushes price down to \$2.50, giving him a gross profit of \$2.25 (gross of fixed costs), and a net profit of \$0. Hence this entrant will not enter, and I will continue to make profits of \$12.75. So I'm really much better off with my production level of 5.

This, as we noted in chapter 9, is a curious story on several grounds, chief among which is, Why does the potential entrant suppose that, if he enters, the monopolist (now a duopolist) will persist in the quantity that she produces preentry? It is not impossible to tell stories to justify this. One can imagine, for example, a world in which the incumbent monopolist must commit to a certain level of production that is observed by the potential entrant, and the entrant can then decide whether to enter and how much to produce. That is, we imagine a structure of competition like the von Stackelberg extensive form game. But it isn't clear that this is a very realistic model of the details of competition, and so one may wish to look further, particularly at models where the incumbent monopolist and the entrant interact competitively only after the incumbent decides to enter.

This sort of model presents a problem for the story of entry deterrence. If the entrant decides whether to enter or not, the incumbent monopolist sees this, and the two compete, it follows that the postentry competition can be thought of as a proper subgame. And there is no reason (yet) to imagine that the monopolist's preentry pricing or quantity decisions have any effect at all on the entrant's calculations. To take the simplest example once again, imagine the following details of competition: The monopolist, in a first period, exercises however she wishes her monopoly power. The entrant observes this and decides whether to enter. The monopolist observes what the entrant does, and then a second period of production/sales takes place, either in a monopoly (if the entrant chooses not to enter) or in a duopoly (if the entrant does enter). If the entrant does enter, the form of competition in the second period, for the sake of concreteness, is assumed to be Cournot style. The two firms simultaneously and independently choose quantities to supply to the market. (If you have any doubt about it at all, give the extensive form for the game just described.)

The point is that in this game, what happens subsequently if the entrant does enter is a proper subgame. It is a Cournot-style duopoly game, and we know the Cournot equilibrium for this subgame; each firm produces $8/3$ units of the good, price is $\$11/3$, and each firm's net profits are approximately $\$4.86$. (You should be working all this out as we go along.)

Now the entire game has many Nash equilibria. For example, it is an equilibrium for the incumbent monopolist to threaten that she will produce 5.01 units in the second period if the entrant enters; this causes the entrant to choose not to enter. Since the entrant doesn't enter, the threat is off the path of play and is okay as part of a Nash equilibrium. But this relies on behavior that is subgame imperfect; the game has a unique subgame perfect equilibrium. If the entrant enters, the incumbent monopolist accommodates herself to this fait accompli and produces $8/3$; the entrant nets $\$4.86$; and so the entrant enters. And, going back to the first stage, the incumbent monopolist realizes that nothing she does in the first period will influence this, so she (optimally) chooses to produce four units, knowing that her days as a monopolist are numbered but knowing as well she can do nothing about that.²

So how do we resurrect the story of the entry-detering monopolist? At least three ways are suggested in the literature. First, one can move to a formulation in which the second period is followed by a third, a fourth,

² If you like irrelevant logical puzzles, try to build a (subgame imperfect) Nash equilibrium for the game in which the incumbent monopolist produces five units in the first period. In this equilibrium, how much will the monopolist necessarily produce in the second period (assuming it is the last)?

ad infinitum, and where profits in different periods are evaluated using discounted present value. This sort of technique, which is the subject of chapter 14, does give us enough leverage to resurrect the story. We leave it to the next chapter to see how.

Second, one can invent ways in which decisions made by the monopolist in the first period affect (physically) what happens in the second. There are a large number of variations on this theme, but the general idea is the same: The incumbent monopolist by something she does in the first period that is observed by the entrant is able to shift around her own second period incentives in a way that makes her "more aggressive" in the second period in the event of entry — aggressive enough so that entry is forestalled. I will not work out an example of this here, but refer you instead to problem 2.

Third (and the point of all this here) is that we can link the two periods by supposing that the potential entrant learns something about the incumbent monopolist from the monopolist's actions in the first period. To illustrate this, we change the numbers around a bit. Suppose that the entrant is uncertain about the marginal cost of the monopolist. The entrant faces a constant marginal cost of 3; the monopolist either has a constant marginal cost of 3 or a marginal cost of only 1. And suppose that the entrant's (and the incumbent's) fixed costs are $\$3$, which is known to both parties.

These numbers are selected so that the entrant will enter if he is sure that she has marginal costs of $\$3$; but he won't enter if he is sure her marginal costs equal $\$1$. Specifically, if the entrant knows that the monopolist has costs of $\$3$, Cournot competition leads each to produce two units, gross profits for each are $\$4$, and so (net of the $\$3$ fixed cost) the entrant makes a $\$1$ profit and will enter. Whereas if the entrant knows that the monopolist has costs of $\$1$, the Cournot equilibrium if the entrant enters will be for the entrant to produce $4/3$ and the monopolist to produce $10/3$. Price will be $13/3$, and the entrant's gross profits will be $16/9$ for net profits of $\$ - 11/9$. The entrant doesn't enter.

Suppose the monopolist, in the first period, chooses her single-period optimal level of quantity, as suggested by the standard theory of monopoly. If her costs are 1 per unit, she prices the good at $\$5$. If her costs are 3 per unit, she prices the good at $\$6$. Hence if the entrant anticipates that the monopolist will act in this fashion in the first period, the entrant can learn from the monopolist's first period decisions whether to enter or not. But then the monopolist, if her costs are in fact 3, might wish to price the good at $\$5$ to mislead the entrant into thinking that entry is a bad idea. (This would then be our phenomenon of entry-detering pricing in the

first period.) But then perhaps the entrant would not pay attention to the monopolist's first period decision.

How do we model and analyze this? Needless to say, the appropriate model depends on what (and how much) the incumbent monopolist knows about the uncertainties of the entrant and how much the entrant knows about what the monopolist knows. We can use games of incomplete information to model a number of situations along these lines, of which we will give the simplest example here.

Suppose the entrant assesses probability ρ that the incumbent has marginal costs of \$3; suppose the incumbent monopolist knows this; the entrant knows that the monopolist knows this, and so on. Consider the game of incomplete information where Nature first chooses the production function (marginal costs) of the incumbent monopolist, with ρ giving the probability that nature chooses \$3, and reveals this to the monopolist. Depending on this, the monopolist chooses her first period levels of output and price. The entrant observes this (but not the monopolist's cost structure) and decides whether to enter. If the entrant enters, the entrant and monopolist play a simultaneous move Cournot "subgame." Since the entrant may be uncertain about what the monopolist's costs are, he may face uncertainty about his profits; we will assume he seeks to maximize the expected level of profits he attains. And since the monopolist will be receiving profits in each of two periods, we have to know how she trades off a dollar today against a dollar tomorrow; we will assume for simplicity that she maximizes the undiscounted sum of her profits. (Accommodating a discount factor would not be difficult.)

We depict this extensive form game symbolically in figure 13.2. Note that we show six information sets for the incumbent — three each for each of her "types." She must decide how much to produce in the first period, how much to produce in the second if the entrant doesn't enter, and how much to produce in the second period if the entrant does enter. (Actually, she has many information sets of the latter two types, as she can condition her second period decisions on her level of output in the first period.) We show two information sets for the entrant: whether to enter or not, and if he enters, how much to produce. This is very deceptive. The entrant has a pair of these information sets for each first-period output decision of the incumbent, and the whole point of the exercise is, How will the first-period output decision of the incumbent affect the decisions taken by the entrant?

We will now undertake to analyze this extensive form game, using as solution concept sequential equilibrium.^a This analysis will be tough

^a To be very precise, the notion of sequential equilibrium doesn't really apply here, since

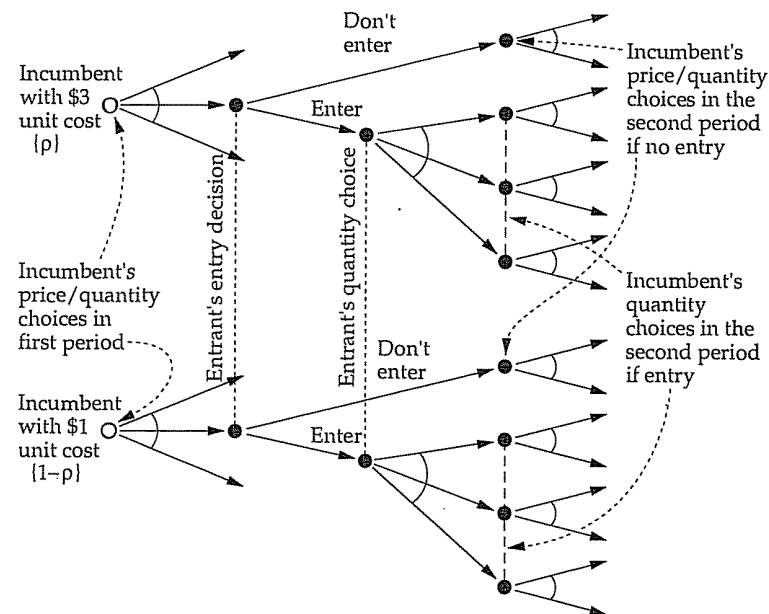


Figure 13.2. An entry deterrence game.

going for many readers; it is the first "serious" problem using noncooperative game theory we've undertaken to examine. We will take things slowly and in steps, but still this will not be easy to follow. You should persevere; many recent developments in microeconomics are based on arguments of this sort. If, after struggling with this analysis, you are still at sea, it may be worthwhile to move immediately to chapter 17 and then return to this analysis.

To begin, we note why we use the sequential equilibrium concept. We have already seen in our discussion of the entry game without incomplete information that we could get "entry deterrence" on the cheap, using Nash equilibria that are held together by the monopolist's threats to produce very large quantities in the event of entry. To avoid these "threat" equilibria, we needed to invoke subgame perfection. We are unable to

sequential equilibrium is defined for finite game trees only. Our analysis is really only in the "spirit" of sequential equilibrium.

invoke subgame perfection here; because of the incomplete information, this game has no proper subgames. By using the sequential equilibrium concept, we invoke sequential rationality for the monopolist, which means that she can't threaten to meet entry with an incredibly large amount of output; she must respond optimally to what she supposes the entrant will do if he enters.

Step 1. The Cournot subgame given the entrant's beliefs

Of course, the sequential equilibrium notion does more than avoid threats by the monopolist. It also ties the actions of the entrant to his beliefs, after seeing the price/quantity choice of the monopolist in the first round, concerning the monopolist's costs. More precisely, suppose that we write $\mu(p)$ for the beliefs held by the entrant that the incumbent monopolist has marginal cost \$3, as a function of the price p that the monopolist charges in the first round. Given those beliefs, what ensues if the entrant decides to enter is completely fixed by $\mu(p)$ in the fashion of problem 17 in the last chapter.

More precisely, in that problem we asked what would be the outcome of Cournot-style competition between two firms, where the first had marginal costs of c , but the first was uncertain about the marginal costs of the second. In terms of the numbers here, we imagine that the entrant chooses output level $x_e(p)$, while the monopolist chooses $x_m(p, 1)$ if her costs are \$1 and $x_m(p, 3)$ if her costs are \$3. Note that all three of these are taken as functions of p , the price the monopolist charged in the first period, since that may influence the incumbent's beliefs about the monopolist's costs.

In an equilibrium, the monopolist's quantities must be best responses to the entrant's quantity, or

$$x_m(p, 1) = \frac{9 - 1 - x_e(p)}{2} \text{ and } x_m(p, 3) = \frac{9 - 3 - x_e(p)}{2}.$$

(If you aren't sure where these came from, compute the monopolist's profits as a function of her output quantity and optimize.) The entrant's quantity must at the same time be a best response to the (from his perspective) random level of output selected by the monopolist. His expected profits, given his beliefs and given his output level x , are

$$\begin{aligned} & [\mu(p)][(9 - 3 - x_m(p, 3) - x)x] + [1 - \mu(p)][(9 - 3 - x_m(p, 1) - x)x] \\ & = (6 - [\mu(p)x_m(p, 3) + (1 - \mu(p))x_m(p, 1)] - x)x, \end{aligned}$$

which is maximized at

$$x_e(p) = \frac{6 - [\mu(p)x_m(p, 3) + (1 - \mu(p))x_m(p, 1)]}{2}.$$

Note well where this expression for his profits comes from; he assesses probability $\mu(p)$ that the monopolist has costs \$3 and so will produce at level $x_m(p, 3)$ and probability $1 - \mu(p)$ that she will produce at level $x_m(p, 1)$ because her costs are \$1.

This gives us three equations in three unknowns, and if you solve them, you will find that

$$\begin{aligned} x_e(p) &= \frac{2(6 - 3\mu(p) - 4(1 - \mu(p)))}{3} = \frac{2(2 + \mu(p))}{3}; \\ x_m(p, 1) &= \frac{6 + 3\mu(p) + 4(1 - \mu(p))}{3} = \frac{10 - \mu(p)}{3}; \text{ and} \\ x_m(p, 3) &= \frac{3 + 3\mu(p) + 4(1 - \mu(p))}{3} = \frac{7 - \mu(p)}{3}. \end{aligned}$$

And if you work at the algebra a bit more, you'll find that the profits of the entrant, not including his fixed cost, are $4(2 + \mu(p))^2/9$.

This is an algebraic swamp, but what emerges is sensible: The higher $\mu(p)$ is, the more "optimistic" the entrant is. Higher $\mu(p)$ means he assesses a higher probability that the monopolist has higher costs. This emboldens him; his level of output rises with $\mu(p)$. And this rise in his level of output causes the equilibrium levels of output of the monopolist to fall. All of which raises his expected gross profits.

When seeing this sort of construction for the first time, readers will sometimes latch onto what was just said and conclude that the entrant should "choose to have optimistic beliefs." It should be stressed that players don't choose their beliefs; they have them based on their experiences with life, similar situations, and so on. Players choose their actions optimally given their beliefs, but we don't think of players choosing to be optimistic when being optimistic is to their benefit. Choosing advantageous beliefs would be like choosing to like broccoli this year because the price of broccoli is low.

Let us take stock of where we are. We are looking for sequential equilibria to the extensive form game we have described, a game with which we hope to capture some aspects of entry deterrence. We have shown what will ensue after the monopolist charges price p at the first date, if the entrant, therefore, assesses probability $\mu(p)$ that the entrant

has cost \$3 and enters. Namely, the entrant and monopolist will act (in equilibrium) in a fashion that leads the entrant to gross profits of $4(2 + \mu(p))^2/9$.

One skeleton is buried in what was just asserted, which should be exhumed if not fully explained. In any sequential equilibrium of this game, the beliefs of the entrant at an information set where he is deciding whether to enter will be identical with his beliefs at the subsequent information set if he enters, whether he is supposed to enter or not in equilibrium. This is a consequence of the consistency criterion of beliefs and strategies in a sequential equilibrium. Besides, it is fairly sensible. Why should something the entrant does or doesn't do change his beliefs about the costs of the monopolist?

Step 2. The entry decision given the entrant's beliefs

So should the entrant enter, if he believes that the incumbent has costs of \$3 with probability $\mu(p)$? He will enter if his expected gross profits are enough to cover his fixed costs of \$3. If you compare $4(2 + \mu)^2/9$ with \$3, you will find that the critical value of $\mu(p)$ is approximately .598.^b If the entrant believes that the incumbent monopolist has marginal costs of \$3 with probability exceeding .598, he enters. If he assesses probability less than .598, he stays out. (If he assesses precisely .598, he is indifferent between entering and staying out.)

So, with all this as prelude, we can ask, What are the sequential equilibria of this game? There are, it turns out, many of them. We will not attempt to characterize them all. Instead, we will discuss one set of strategies that do not form an equilibrium, and we will give one sequential equilibrium for the case $\rho < .598$ and another sequential equilibrium for the case $\rho > .598$. For now, be content to understand why the nonequilibrium is not an equilibrium, why the two types of equilibria are equilibria, and what these two equilibria entail for the story of entry deterrence. Later, after completing chapter 17, you will be told to return and conduct a more complete analysis of this problem.

Step 3. A nonequilibrium

No matter what the value of ρ is, it won't be an equilibrium for the monopolist to charge \$5 in the first period if her costs are \$1 and \$6 if her costs are \$3. For if she does this, then the entrant will infer that her costs are \$3 if she charges \$6 and \$1 if she charges \$5. That is, $\mu(6) = 1$ and $\mu(5) = 0$. These beliefs are forced by Bayes' rule; if the monopolist charges

^b Precisely, $3\sqrt{3}/2 - 2$.

\$5 or \$6 in the first period, then these observations are along the path of play, and Bayes' rule must be applied.^c

But given these beliefs, we know that the entrant won't enter if he sees \$5 charged in the first period, and he will enter if \$6 is charged. Consider the plight of the monopolist if this is the entrant's strategy and she has marginal costs of \$3. If she follows the strategy outlined and charges \$6, she will net \$6 in the first period (net of fixed costs), but then the entrant will enter, and she will make only \$1 net in the second period. If, on the other hand, she charges \$5 in the first period, mimicking the strategy of the \$1 monopolist, her net profits in the first period will fall to \$5 in the first period (from \$6). But then the entrant will not enter, and she can charge \$6 in the second period and make net profits of \$6. This is far better than following the strategy outlined, which is, therefore, not part of an equilibrium strategy profile.

Step 4. A pooling equilibrium

Now suppose $\rho < .598$. We claim the following is a sequential equilibrium: The monopolist charges \$5 in the first period, no matter what her costs are. The entrant enters if the monopolist charges anything more than \$5 in the first period, based on beliefs $\mu(p) = 1$ for $p > \$5$. The entrant does not enter if the monopolist charges anything at or below \$5, based on beliefs that $\mu(p) = \rho$ for $p \leq \$5$. (If there is entry, the two play according to the scheme outlined in step 1. If there is no entry, the monopolist charges her monopoly price in the second period.)

Why is this a sequential equilibrium? First note that the entrant is playing optimally given his presumed beliefs. He is entering when he should and staying out when he should, according to the analysis in step 2. Note second that his beliefs are consistent with the supposed equilibrium strategies, at least along the path of play. The only price that is used in the first period is \$5 (in these strategies), and since both types of monopolist charge this price, the posterior assessment of the monopolist's type given this price should be the prior. All other prices are off the path, so beliefs can be arbitrary.^d But they even make some sense: The higher the price the monopolist charges, the greater should be our suspicion that she has high costs. The beliefs we have mandated have this monotonicity character, at least.³

^c This is always true in a sequential equilibrium.

^d They should be consistent in the sense of sequential equilibria, and these are (allowing for the fact that we aren't working with a finite game and so must fudge definitions a bit).

³ "Aha," the reader is saying, "what was all this stuff about not choosing beliefs? Seems like beliefs are being chosen here!" At the risk of confusing the reader who is not confused,

And finally, the monopolist is optimizing given the entrant's strategy. If the monopolist's costs are \$1, she is getting to charge her favorite price in the first period *and* forestall entry. Nothing could be better than that. While if her costs are \$3, she would prefer to charge a higher price in the first period, but that will bring on entry. On balance, it is better to accept the smaller profits in the first period and keep the larger profits in the second.

The "intuition" of this equilibrium resides in the last assertion. Because ρ is low, the entrant, if he doesn't learn anything from the first period price charged by the monopoly, will choose to forego entry. Hence the monopolist, if her costs are \$3, wishes to hide that fact. And in this equilibrium she hides that fact by charging a \$5 price in the first period, just "like" in the story of section 9.2!

This is called a *pooling equilibrium* because the two types of monopolist do the same thing in equilibrium.

Step 5. A screening equilibrium

This pooling equilibrium doesn't work if $\rho > .598$. In this case a pool would be ineffective; entry would not be forestalled at all. And what's the point of the \$3 monopolist charging a short-run suboptimal price in the first period if she is going to see entry anyway?

When $\rho > .598$, we assert the following is an equilibrium: If the monopolist has costs \$3, she charges \$6 in the first period. If she has costs \$1, she charges \$3.76 (!) in the first period. The entrant enters if the monopolist charges anything above \$3.76, based on beliefs $\mu(p) = 1$ for $p > \$3.76$. The entrant stays out if the monopolist charges anything

let us be very clear about this. A sequential equilibrium describes what people do and what they believe, both in and out of equilibrium. We aren't saying why they act as they do, although we assume as part of the justification that they all are pretty sure how each will act, and they all choose their actions optimally given their beliefs. Similarly, we aren't saying why they believe what they do (out of equilibrium). We are only seeing whether beliefs and intended actions are in equilibrium, which means that the intended actions are optimal. If we are to have faith in a particular equilibrium, then we need to have some reason to believe that individuals will see the prescribed actions as "obvious ways to play the game," and when we employ sequential equilibria, we essentially are supposing there is reason to think the specified out-of-equilibrium beliefs describe what the players will find the obvious beliefs to have. When we remark, as we did at the end of the previous paragraph, that the beliefs conform to a sensible monotonicity property, we are imposing a *partial* test on the reasonableness of supposed beliefs. But this is a pretty weak test, and if we are going to have faith in this equilibrium, we will need either empirical evidence or some convincing story as to why people will think and act as we are postulating. In this sense, what is demonstrated here is the *possibility* or *coherence* of various sorts of entry deterrence; we would need some other argument to assert that this is what *will* happen.

at or below \$3.76, based on beliefs $\mu(p) = 0$ for $p \leq \$3.76$. (If there is entry, the two play according to the scheme outlined in step 1. If there is no entry, the monopolist charges her monopoly price in the second period.)

Why is this an equilibrium? First, the entrant is behaving optimally given his beliefs, according to step 2. And those beliefs make sense given the strategy of the monopolist; a first period price of \$6 is a sure indication that the monopolist has costs \$3, while a first period price of \$3.76 is a sure indication that the monopolist has costs \$1. The other prices are off the path of play and so any beliefs will do; the ones we've selected even have the appropriate monotonicity property.

So is the monopolist behaving optimally? Consider first the monopolist if her costs are \$1. She would prefer to charge a higher price than \$3.76 in the first period, but that would attract entry. We leave to you the task of showing that given the beliefs of the entrant for prices above \$3.76 she prefers to keep the entrant out and sacrifice a bit of her first period profits.

The more interesting case is the monopolist if her costs are \$3. If she follows the prescriptions above, she makes \$6 (net of fixed costs) in the first period and, after entry, \$1 (net of fixed costs) in the second. She can only forestall entry by charging a price of \$3.76 or less, which forestalls entry completely, and then she would make \$6 in the second period. But what does she make if she charges \$3.76 in the first period? She makes \$.76 per unit on sales of 5.24 units, or \$3.98 gross and \$.98 net of fixed costs. This is just short of the \$1 threshold needed to make forestalling entry worthwhile, and we have an equilibrium. N.B., the level \$3.76 is chosen so that, if the monopolist's marginal costs are \$3, she just fails to prefer to mimic the low-cost monopoly strategy of charging \$3.76 and forestall entry.

Intuitively, in this equilibrium the \$1 monopolist to demonstrate convincingly that she has \$1 costs must charge a very low price in the first period — a price so low that the \$3 monopolist would rather give up the benefits of forestalling entry than take such low profits in the first period. This is called a *screening equilibrium* because the two types of monopolist are screened or separated by their first period actions. Note that in this equilibrium entry occurs whenever it would have had there been complete information about the monopolist's costs. The only thing that incomplete information brings is a better deal for consumers in the first period, if the monopolist has low costs — the low-cost monopolist charges a low price in order to "prove" she has low costs and (thereby) forestall entry

that would be forestalled if only she could reveal her cost structure to the potential entrant.⁴

The objective of this example is not to analyze entry deterrence. The model given is frightfully simple, and there is an enormous amount more to the story. The point instead is the form of the model, where we use a game of incomplete information, and then the style of analysis. In particular, note what is presumed by this simple model: The incumbent monopolist is aware of the uncertainty in the mind of the entrant, and (at least in the case $\rho < .598$ and the monopolist's costs are \$3) the monopolist uses this uncertainty to her own advantage. If we thought that the monopolist was uncertain what the entrant assessed, we would need a more complex model. We might enrich the model as well by supposing that the monopolist is uncertain about the entrant's fixed or marginal costs: Then the monopolist would be unsure whether the price she charges will deter entry, and she would update her assessment of the entrant's cost structure if the entrant does enter. (And a higher-cost entrant would have a bit of an incentive to enter insofar as this would mislead the monopolist into believing that he is a lower-cost entrant.)

There are other ways we could think of enriching the model. An obvious one is to suppose that the monopolist may be a bit of a dope — unaware of the entrant's uncertainty about her cost structure, or unaware that he will be watching her first period price to uncover that cost structure. Such an unaware monopolist is not fully rational, at least in the sense that she doesn't quite understand the full implications of her actions. So we turn next to the question How would we model "irrationality" of this sort and of others?

13.3. Modeling irrationality

Games of incomplete information can be used to model situations in which one player assesses significant probability that another player is "irrational."

The key comes from identifying what we mean by "irrational." To take the simplest case first, we would call irrational any behavior in the game that flies in the face of the player's own "best interests" as determined by the player's payoffs. For example, consider the extensive form game given in figure 13.3(a). Solved by backwards induction, we see that player 1

⁴ We will discuss this in chapter 17, but you might think through how things would change if the incumbent monopolist had the ability to reveal at no cost to herself the details of her cost structure.

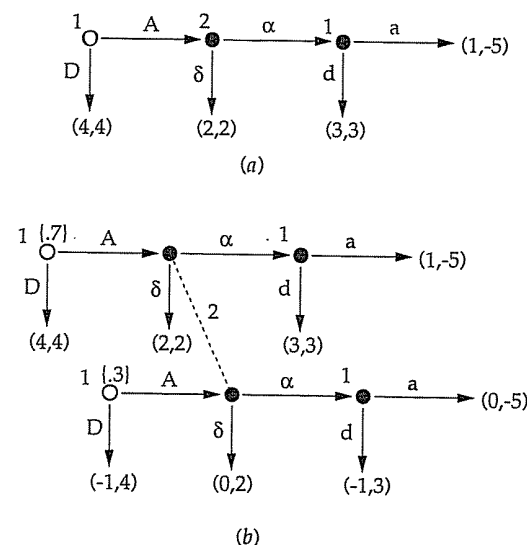


Figure 13.3. Modeling irrationality.

should play d at her final node, hence 2 should play α at his node, and hence player 1 should play D. Suppose, however, that player 2 is concerned that player 1 may for some reason be intending to play Aa. This could be because of inattention on the part of player 1, or because player 1 is malicious, or simply because player 1 has an irrational phobia about the letter d. In particular, suppose that player 2 assesses probability .3 that this player 1's behavior will be to begin with A and, if she gets the opportunity, to continue with a despite 1's (apparent) payoffs.

How do we model this? Consider figure 13.3(b). We have created a game of incomplete information containing two "versions" of the original game, between which nature picks at the outset. In both versions, player 2's payoffs are exactly as in the original game. In the top version, player 1's payoffs are exactly as in the original game. But in the bottom version, player 1 is given payoffs that make the play of Aa a strictly dominant strategy. Notice how we do this: Fixing the strategy Aa for player 1, for each terminal node in the tree, we count the number of information sets of player 1 at which player 1 would have to deviate from the strategy Aa to get us to that terminal node; 1's payoff is set at minus that